

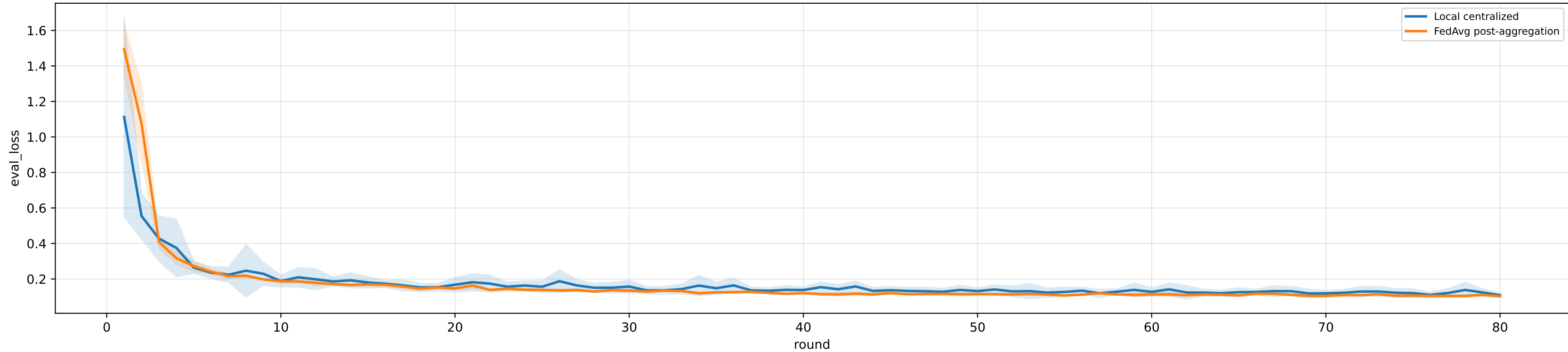
Controlled task-specialization level: Low | FedAvg diagnostics | aggregated / pooled over 5 clients | model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64

family=specialized\_cycle2 | n\_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%  
family=specialized\_cycle3 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%  
family=specialized\_cycle4 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%  
family=specialized\_cycle5 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%  
family=specialized\_cycle6 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%  
specialization=low | specialization\_fraction=0.20 | global\_visible\_support=0.84 | task\_jsd\_mean=0.003 | task\_jsd\_median=0.003 | task\_jsd\_max=0.003 | family\_centroid\_jsd\_mean=0.003

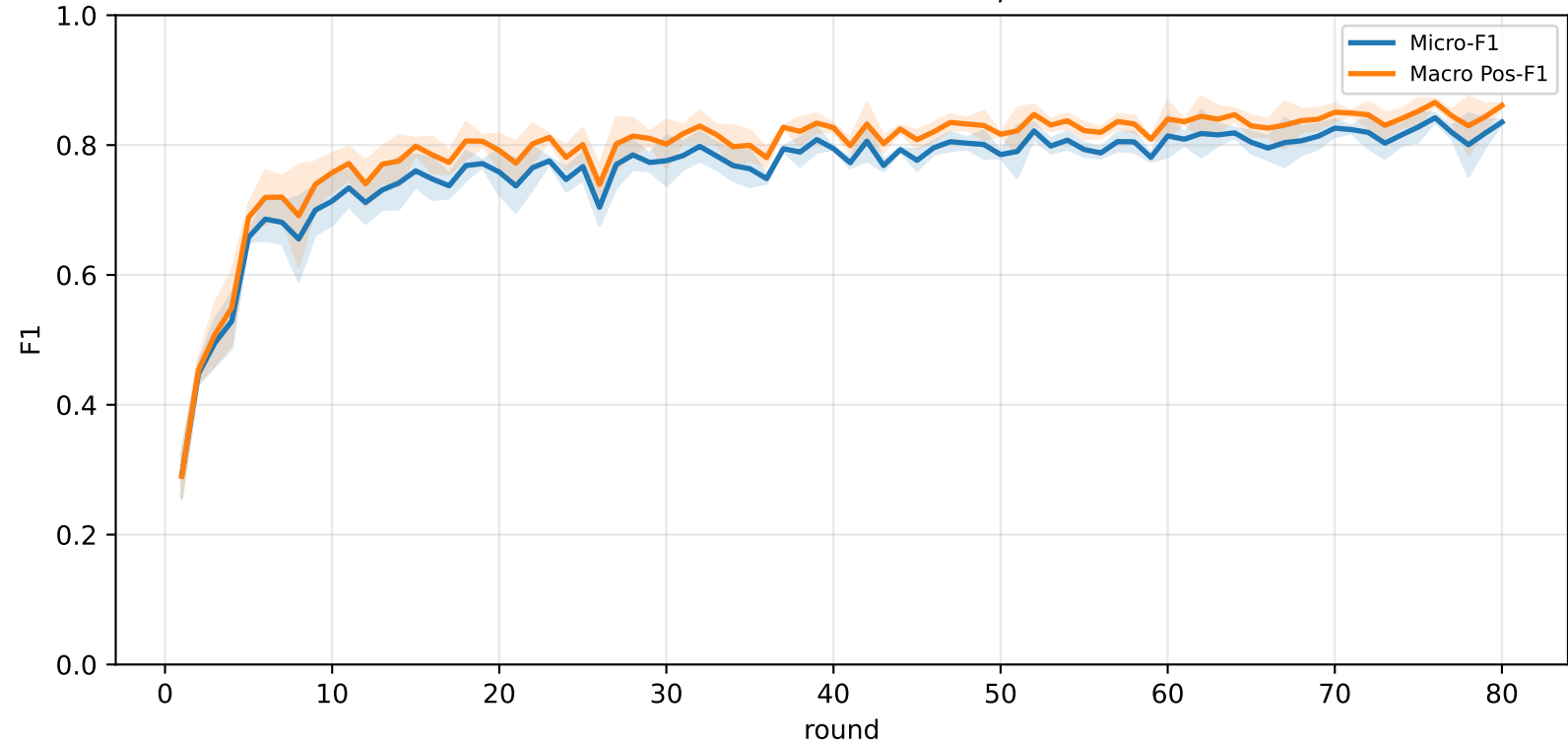
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 83.4% ± 0.3% | 85.7% ± 0.3% | 96.8% ± 0.6%  | 94.9% ± 0.4%  | 86.8% ± 0.2%  | 78.7% ± 0.7%  | 70.4% ± 0.6%  |
| FedAvg            | 81.6% ± 0.3% | 84.0% ± 0.3% | 96.2% ± 0.8%  | 95.4% ± 0.7%  | 82.3% ± 2.8%  | 78.0% ± 0.2%  | 68.4% ± 2.8%  |

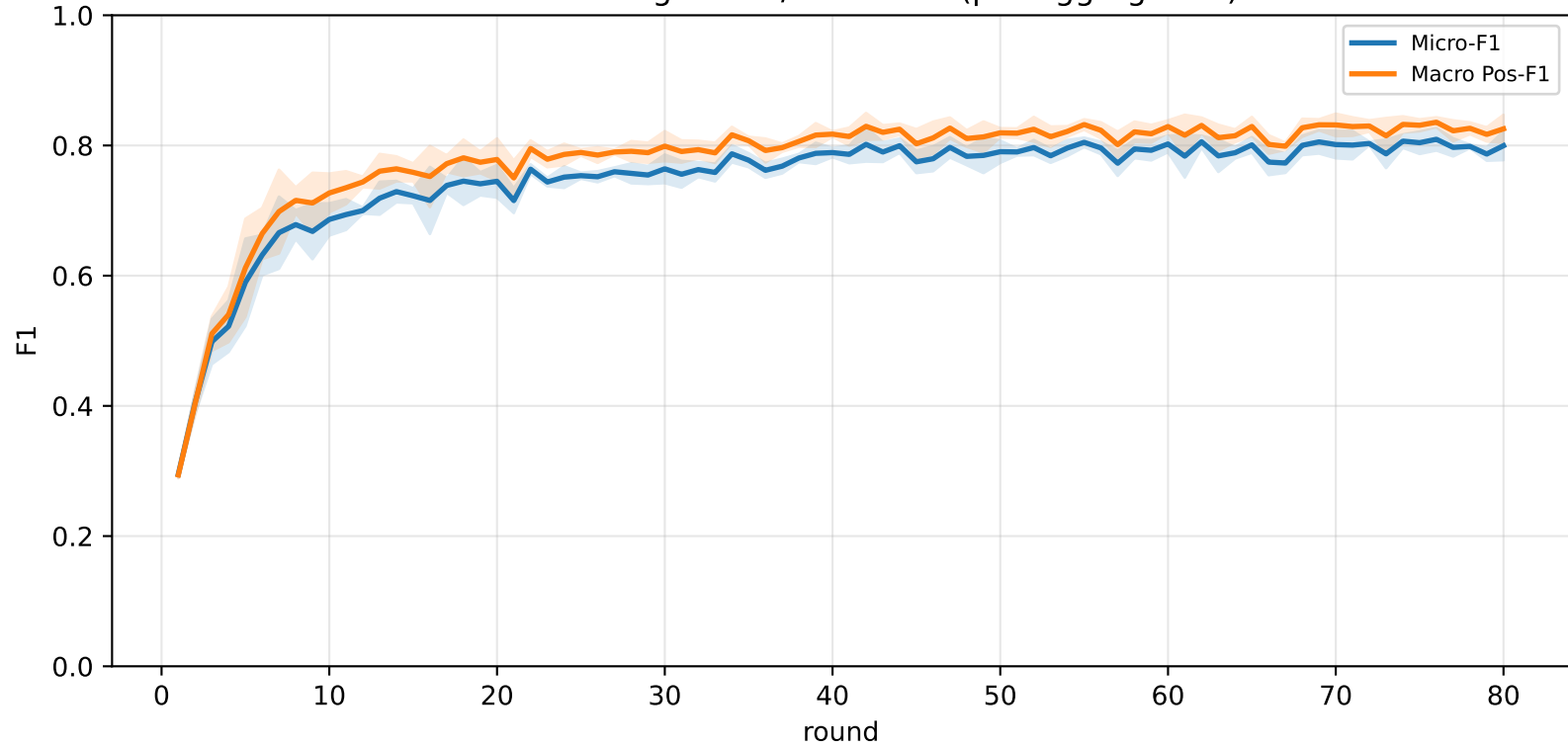
Validation loss comparison



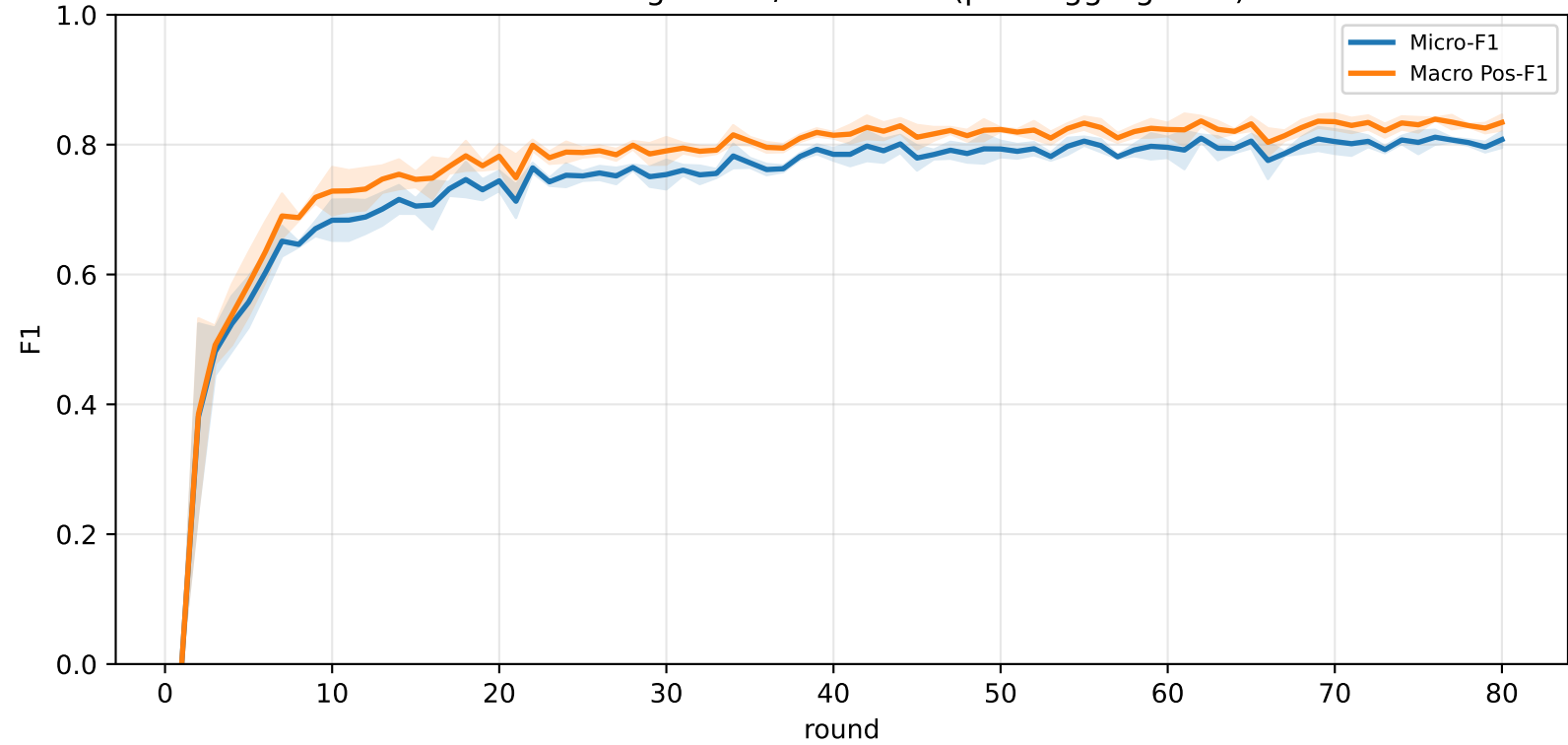
Pooled local centralized micro / macro F1



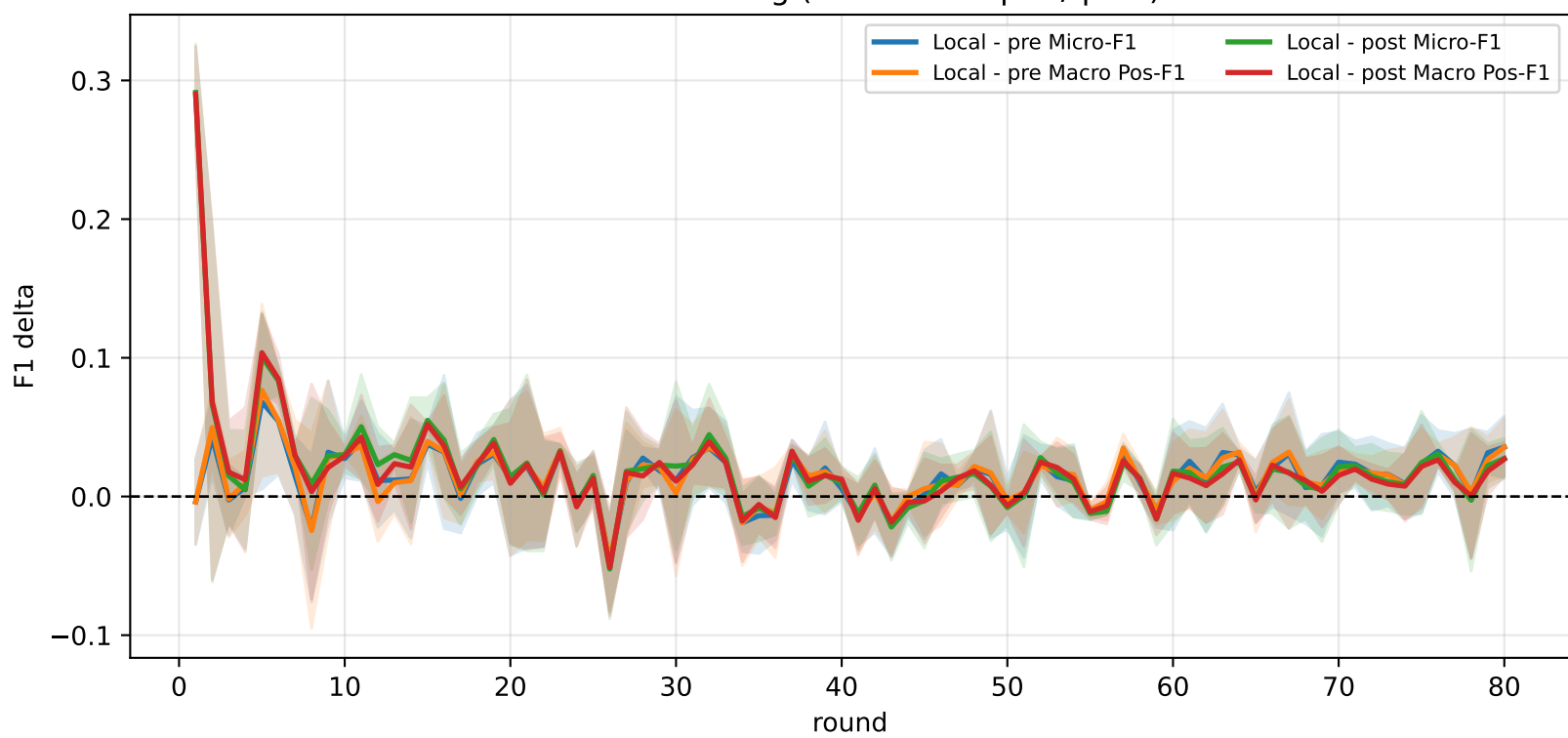
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)

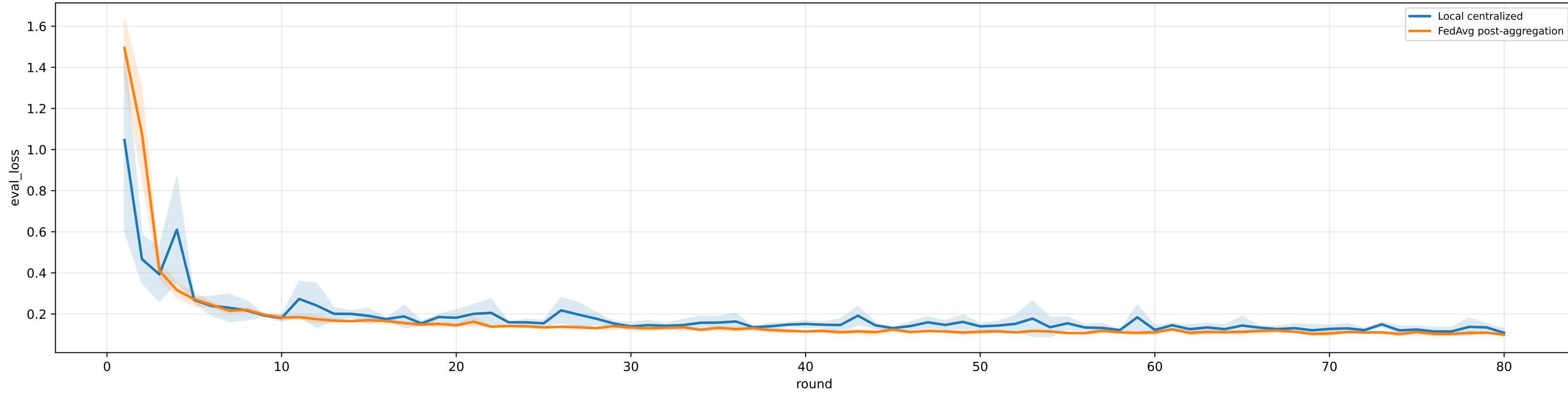


family=specialized\_cycle2 | n\_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% | specialization=low | specialization\_fraction=0.20 | global\_visible\_support=0.84 | task\_jsd\_mean=0.003 | task\_jsd\_median=0.003 | task\_jsd\_max=0.003 | family\_centroid\_jsd\_mean=0.003

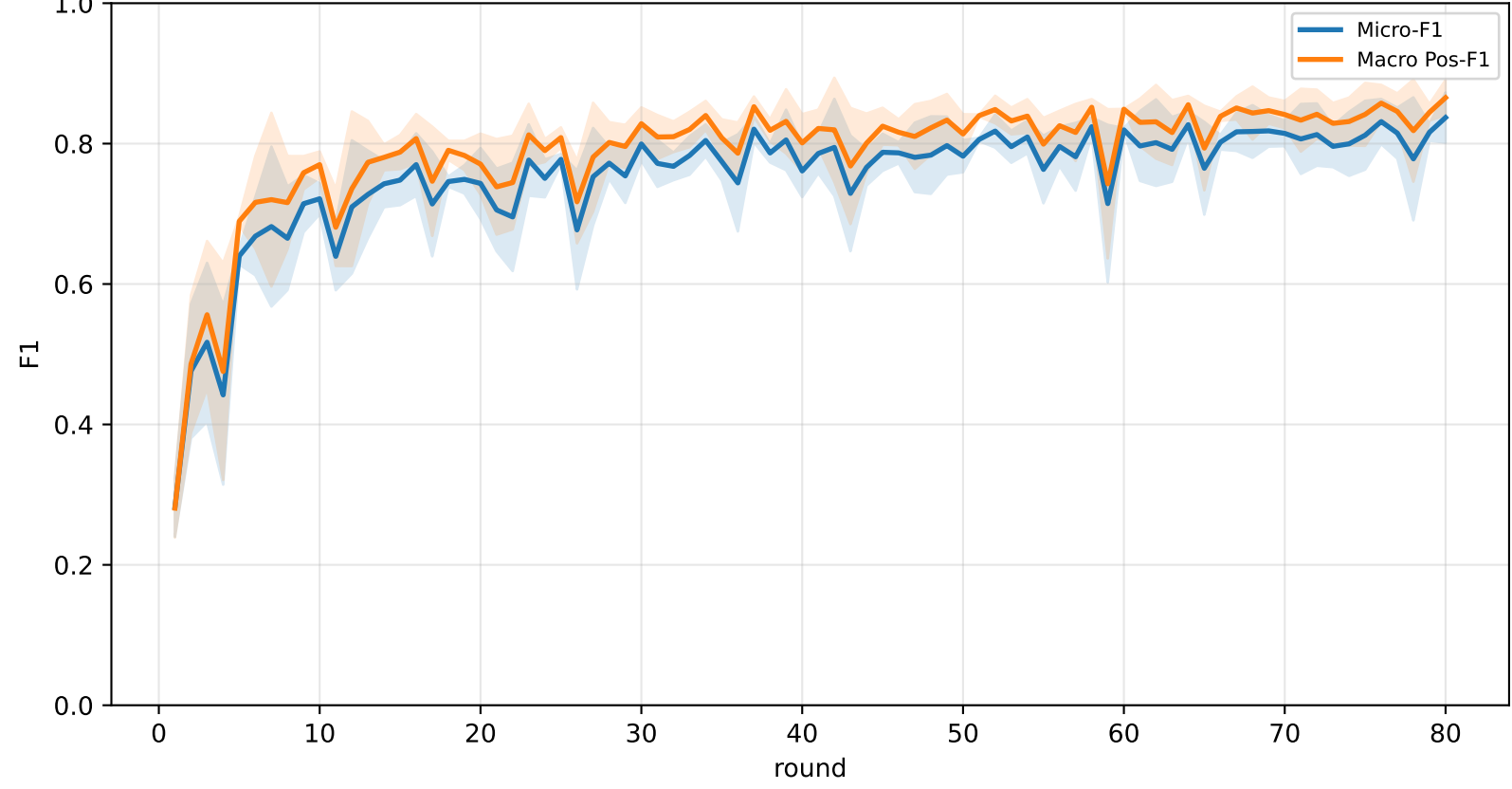
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 81.5% ± 1.5% | 84.0% ± 1.5% | 96.6% ± 1.3%  | 93.0% ± 1.5%  | 85.5% ± 2.0%  | 78.8% ± 2.1%  | 66.4% ± 2.0%  |
| FedAvg            | 81.6% ± 0.2% | 84.2% ± 0.5% | 97.1% ± 1.0%  | 95.7% ± 0.5%  | 81.6% ± 3.0%  | 78.2% ± 0.7%  | 68.3% ± 2.6%  |

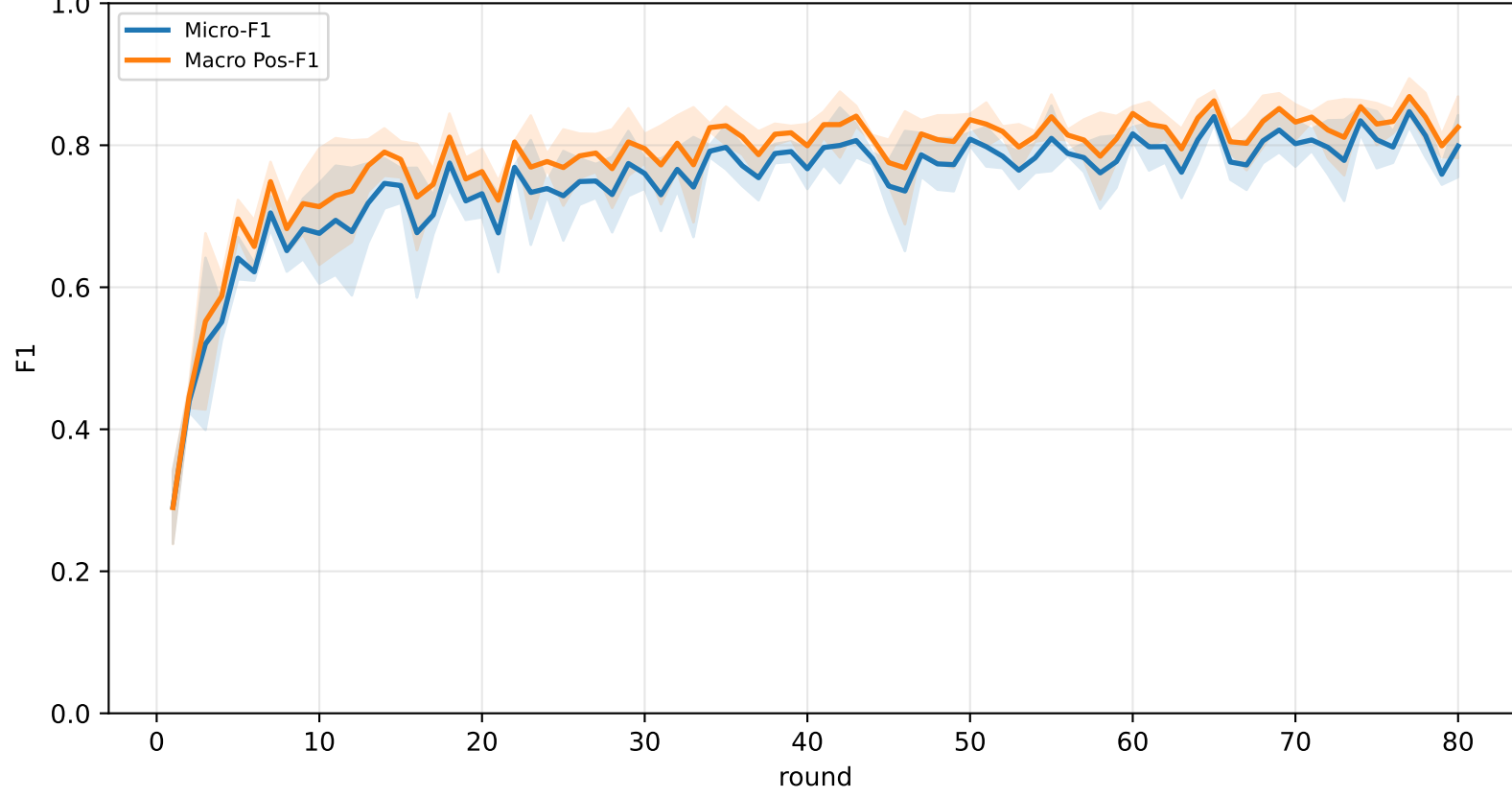
Validation loss comparison



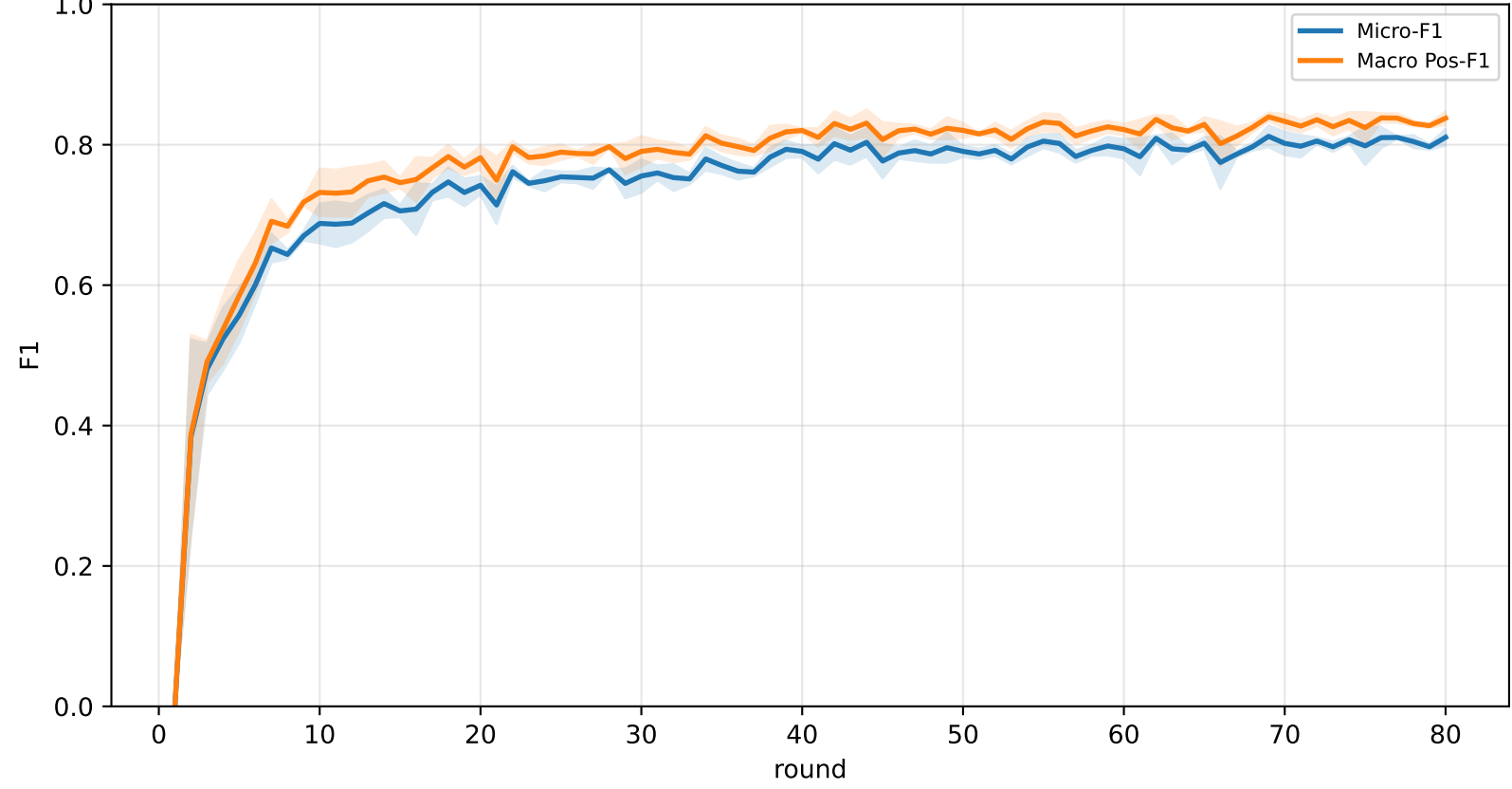
Pooled local centralized micro / macro F1



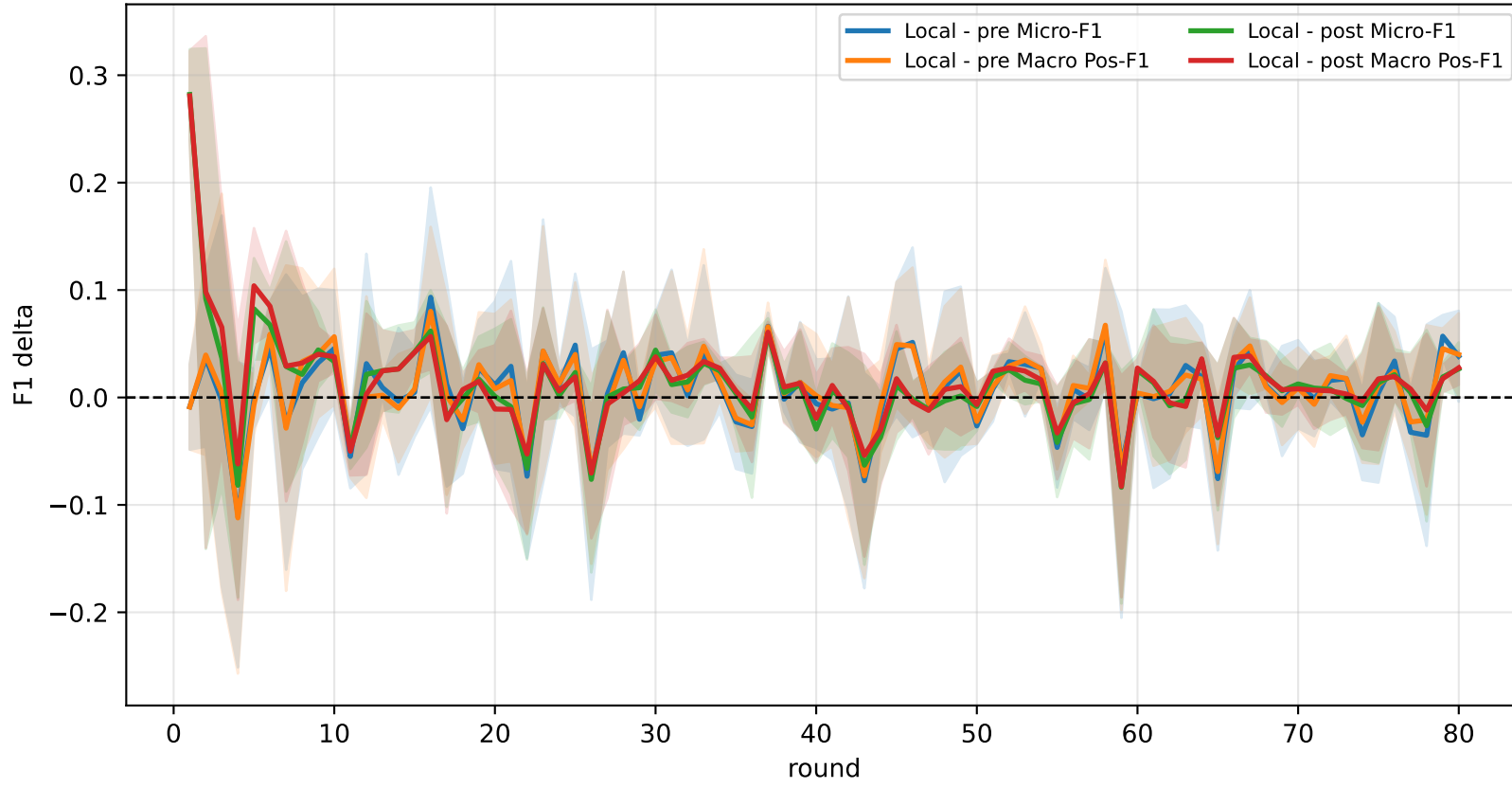
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)



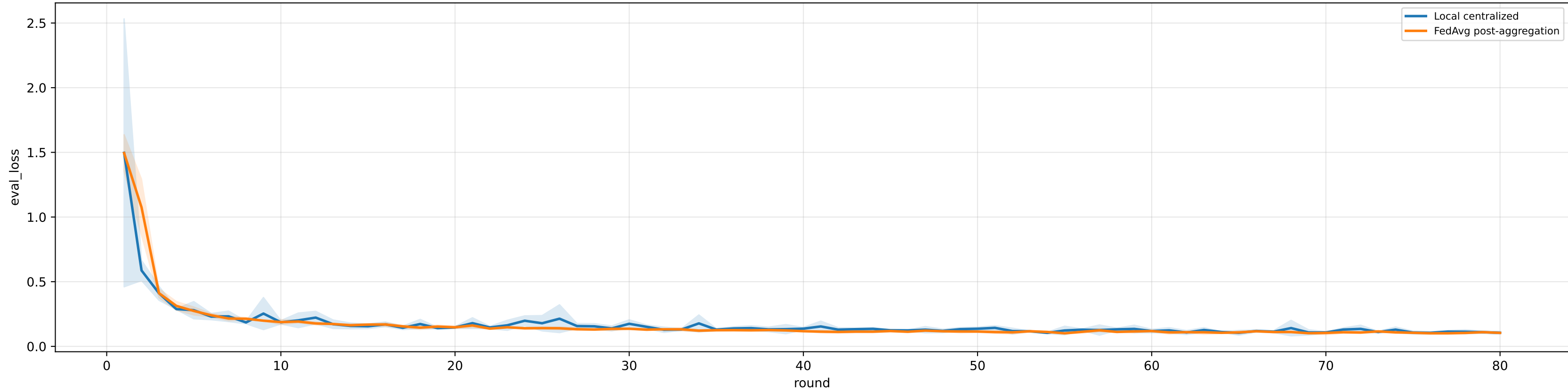
Controlled task-specialization level: Low | FedAvg diagnostics | family=specialized\_cycle3 | n\_clients=1 | model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64

family=specialized\_cycle3 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% | specialization=low | specialization\_fraction=0.20 | global\_visible\_support=0.84 | task\_jsd\_mean=0.003 | task\_jsd\_median=0.003 | task\_jsd\_max=0.003 | family\_centroid\_jsd\_mean=0.003

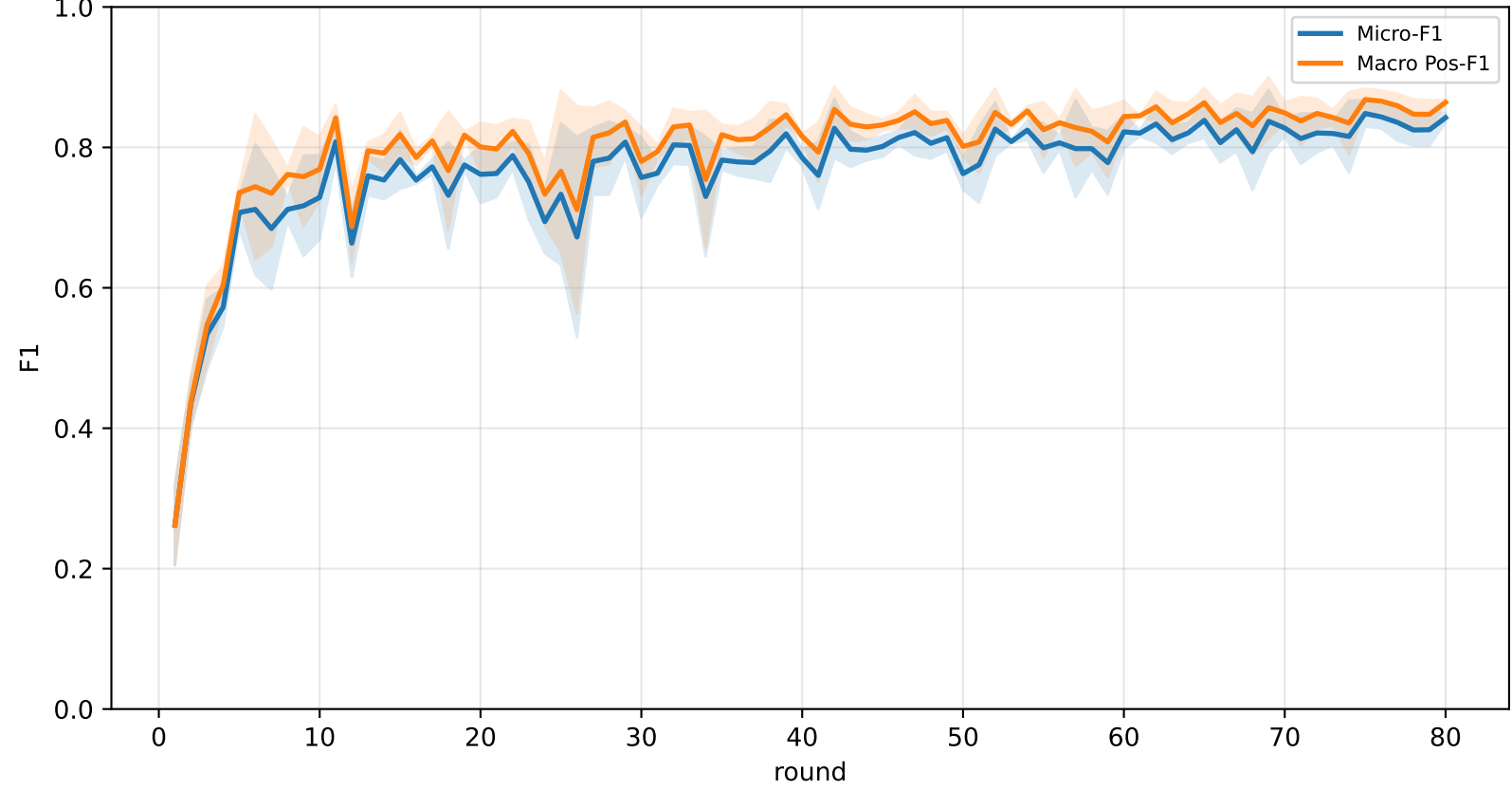
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 84.3% ± 1.4% | 86.1% ± 1.1% | 96.7% ± 0.5%  | 95.8% ± 1.1%  | 86.6% ± 2.6%  | 78.0% ± 2.0%  | 73.7% ± 3.9%  |
| FedAvg            | 81.5% ± 0.2% | 83.9% ± 0.5% | 95.4% ± 0.8%  | 95.8% ± 1.0%  | 82.4% ± 3.4%  | 77.1% ± 0.4%  | 68.7% ± 2.8%  |

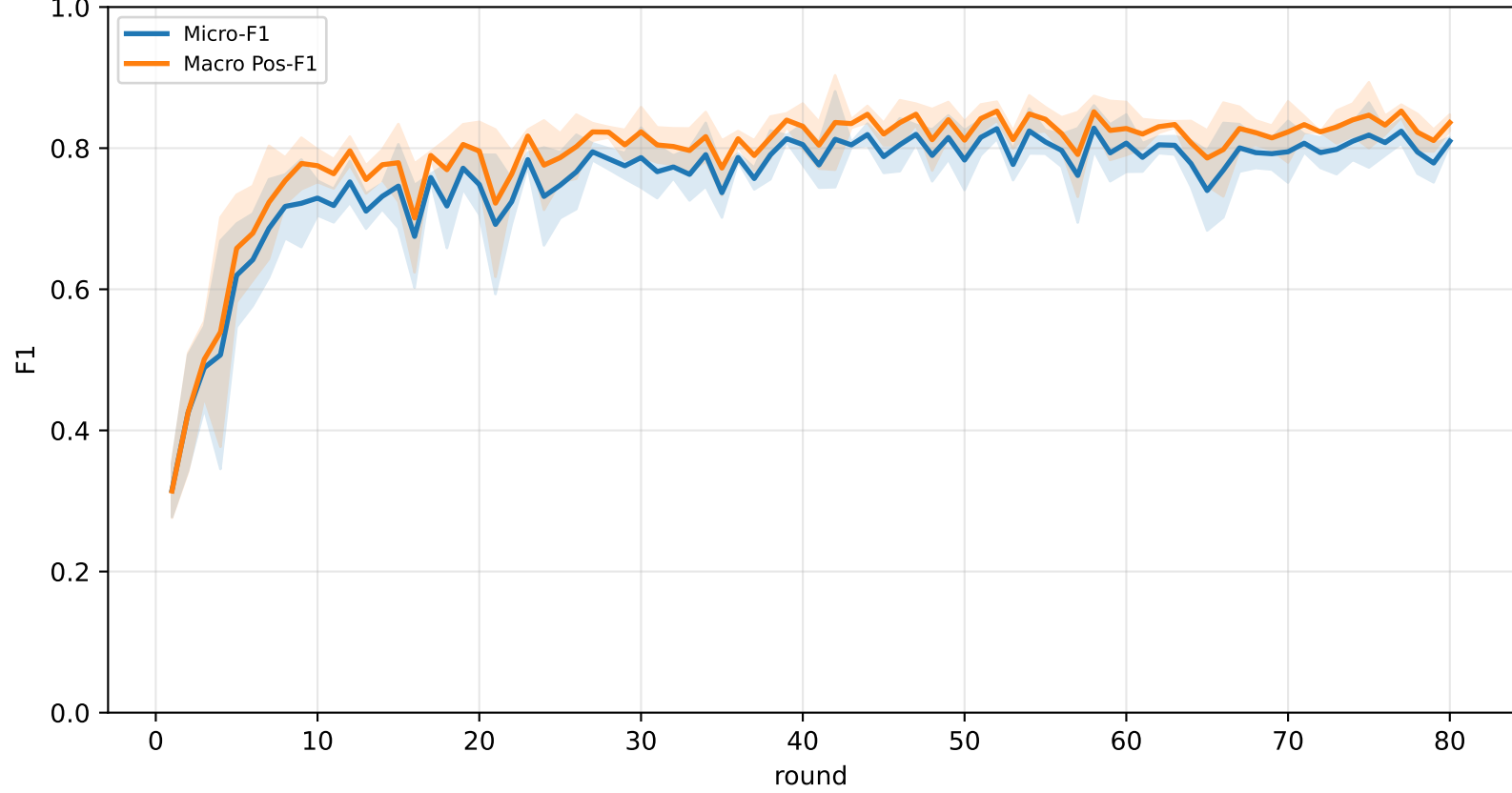
Validation loss comparison



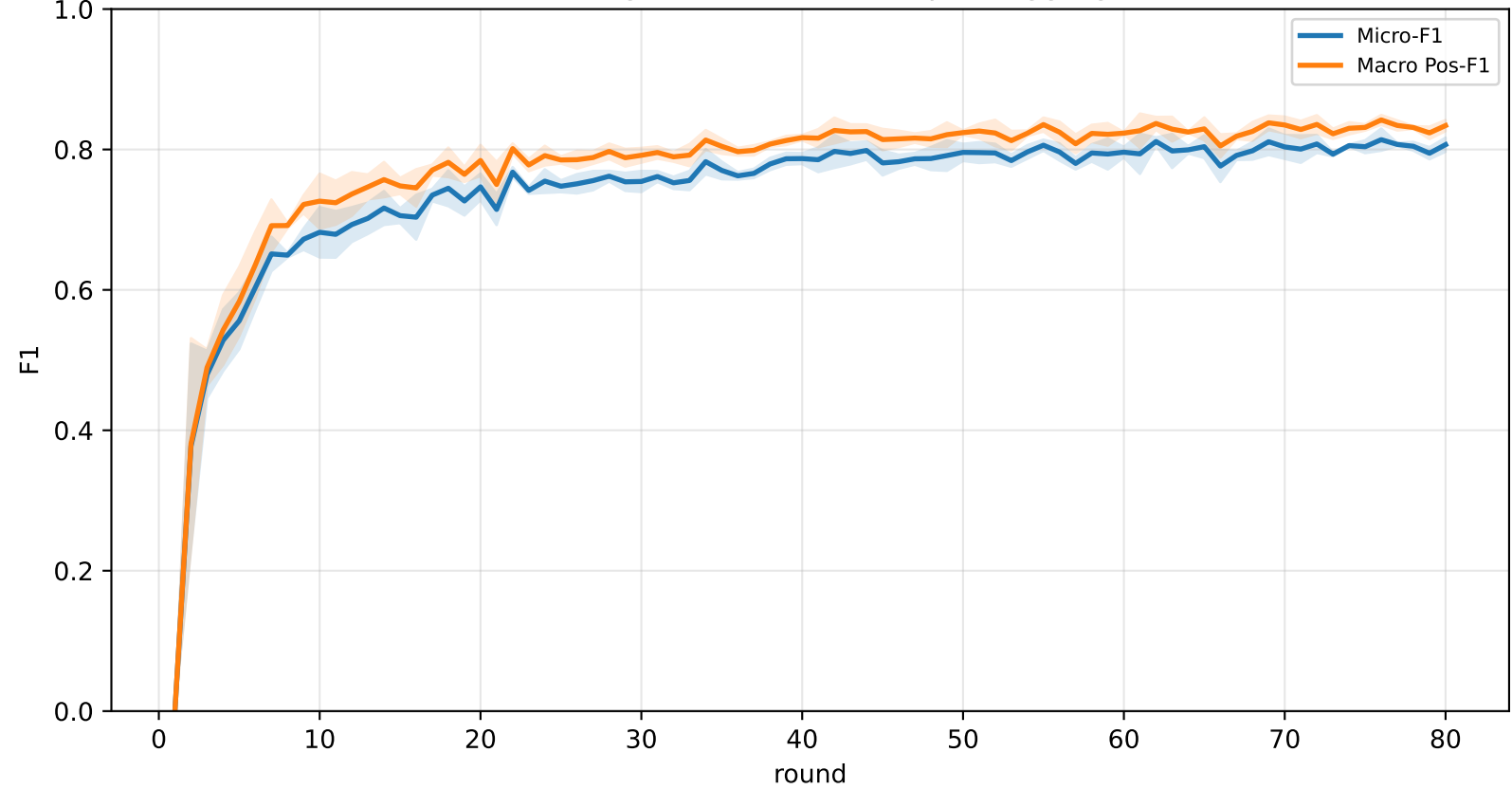
Pooled local centralized micro / macro F1



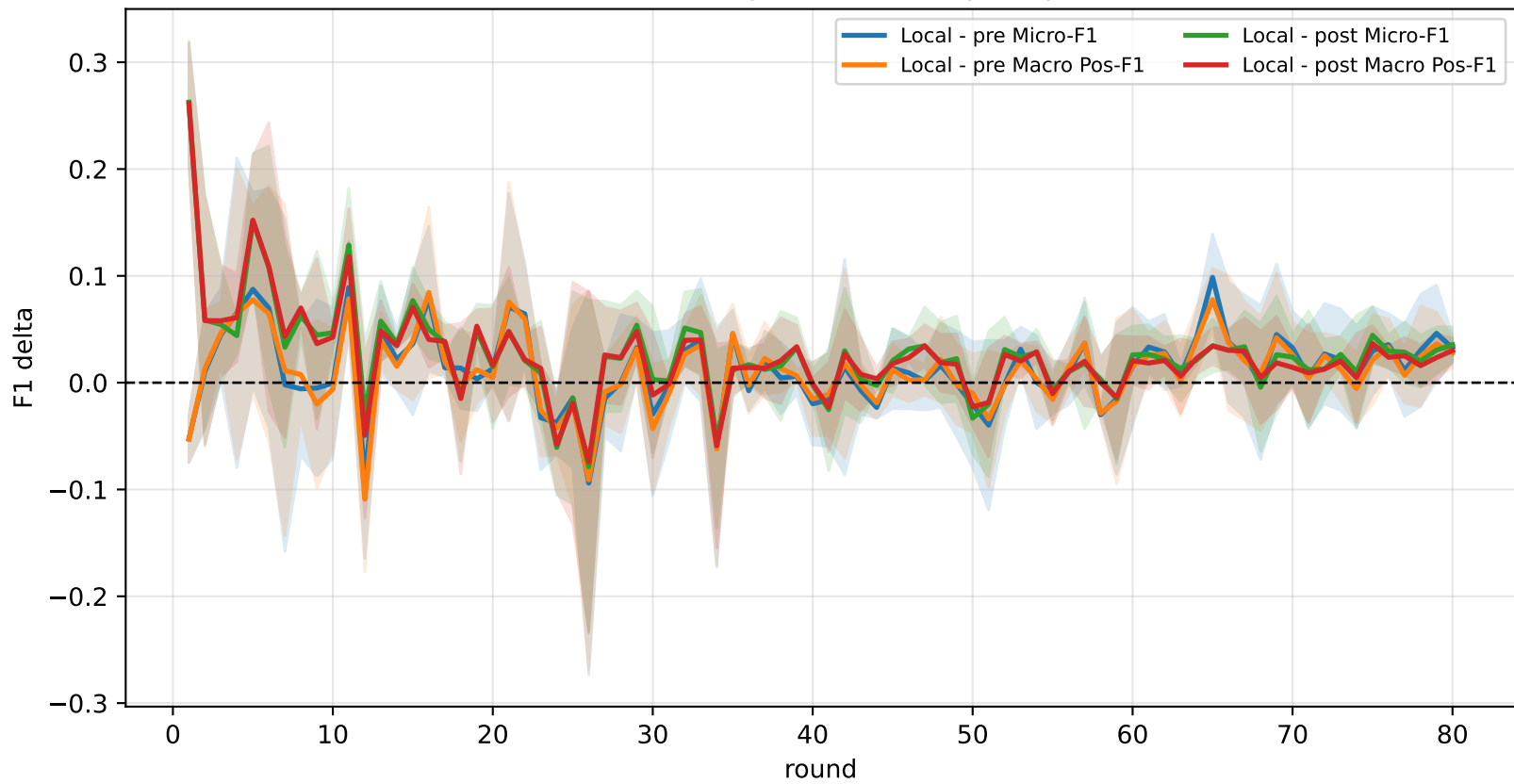
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)



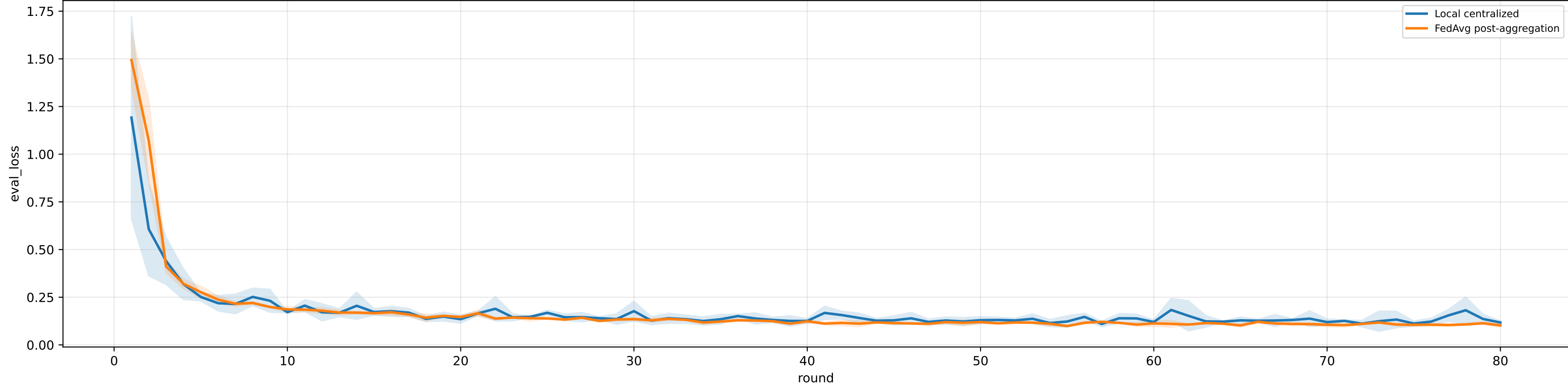


family=specialized\_cycle4 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% | specialization=low | specialization\_fraction=0.20 | global\_visible\_support=0.84 | task\_jsd\_mean=0.003 | task\_jsd\_median=0.003 | task\_jsd\_max=0.003 | family\_centroid\_jsd\_mean=0.003

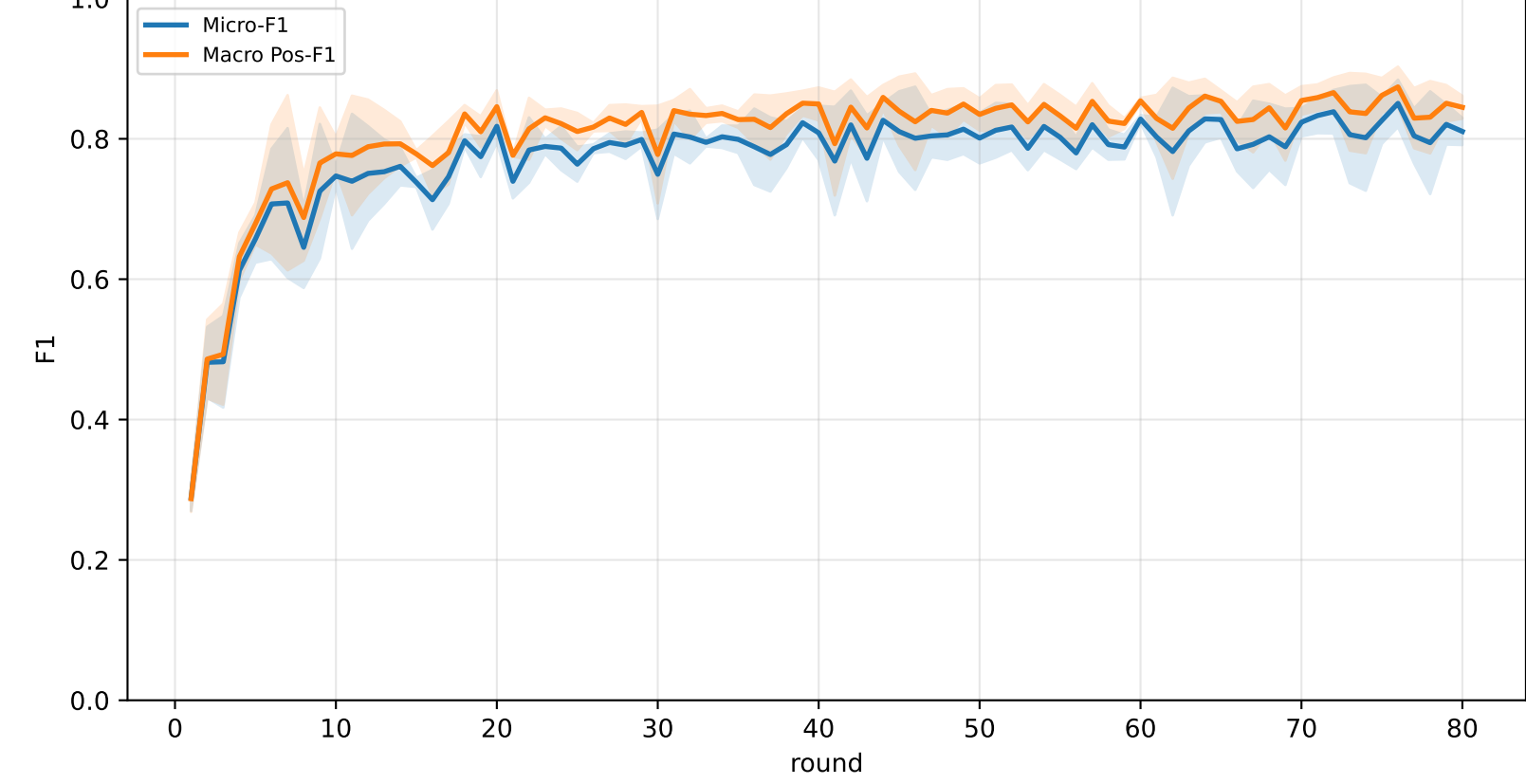
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 81.9% ± 2.9% | 84.6% ± 2.1% | 96.4% ± 1.2%  | 96.1% ± 0.4%  | 85.9% ± 3.7%  | 76.4% ± 0.2%  | 68.1% ± 7.4%  |
| FedAvg            | 81.5% ± 0.6% | 84.0% ± 0.3% | 96.2% ± 1.0%  | 94.5% ± 0.8%  | 82.6% ± 2.1%  | 78.3% ± 0.6%  | 68.1% ± 3.3%  |

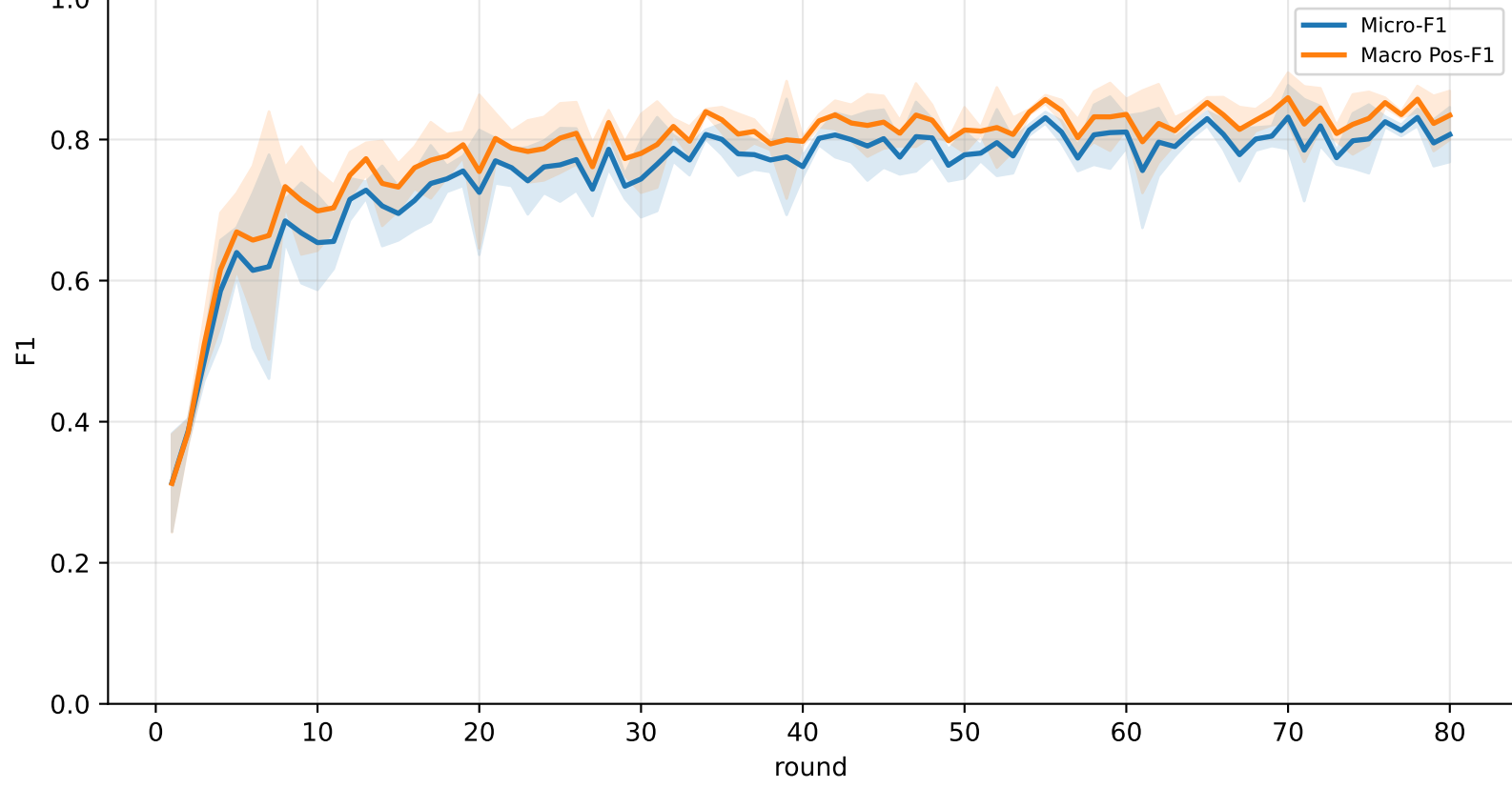
Validation loss comparison



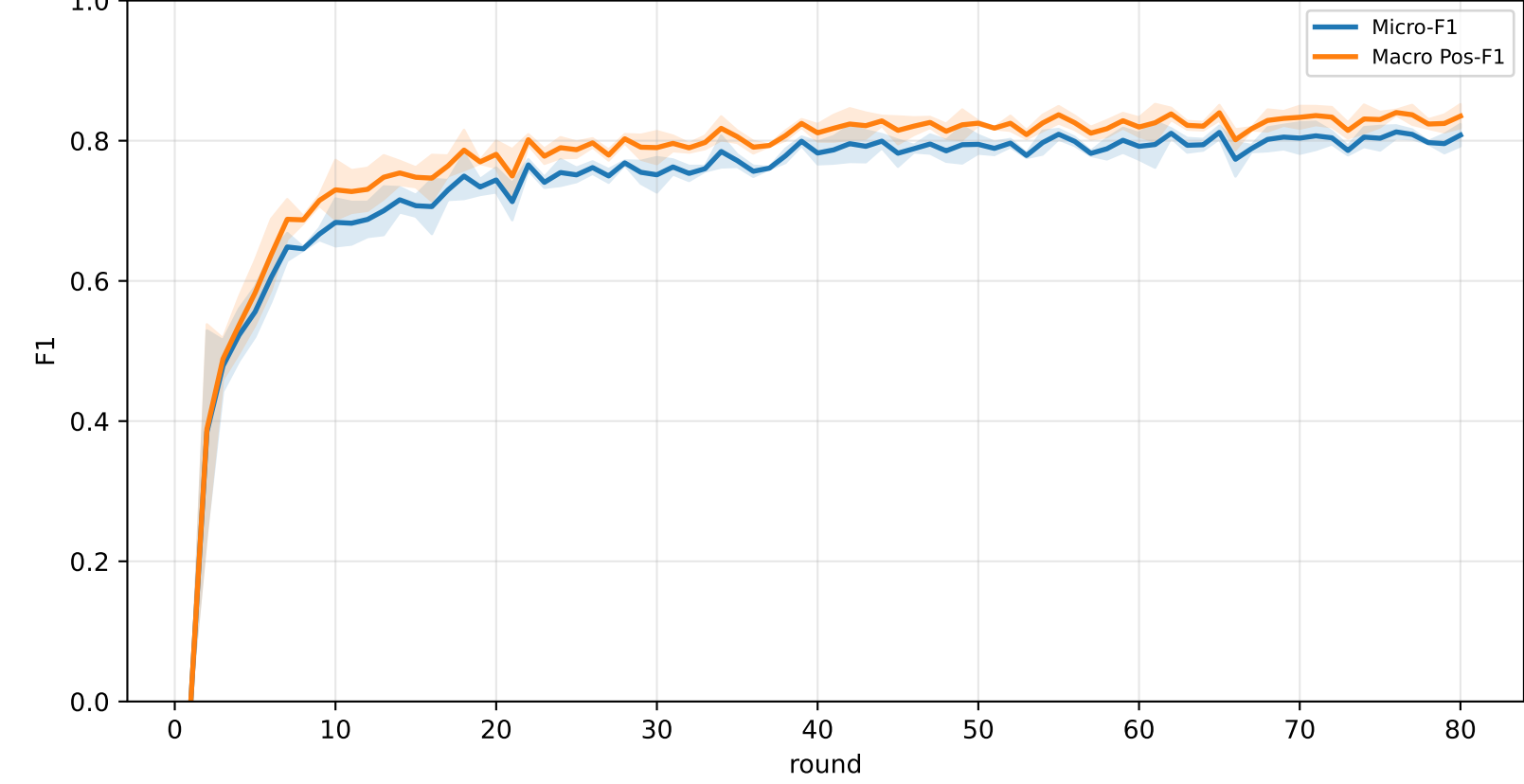
Pooled local centralized micro / macro F1



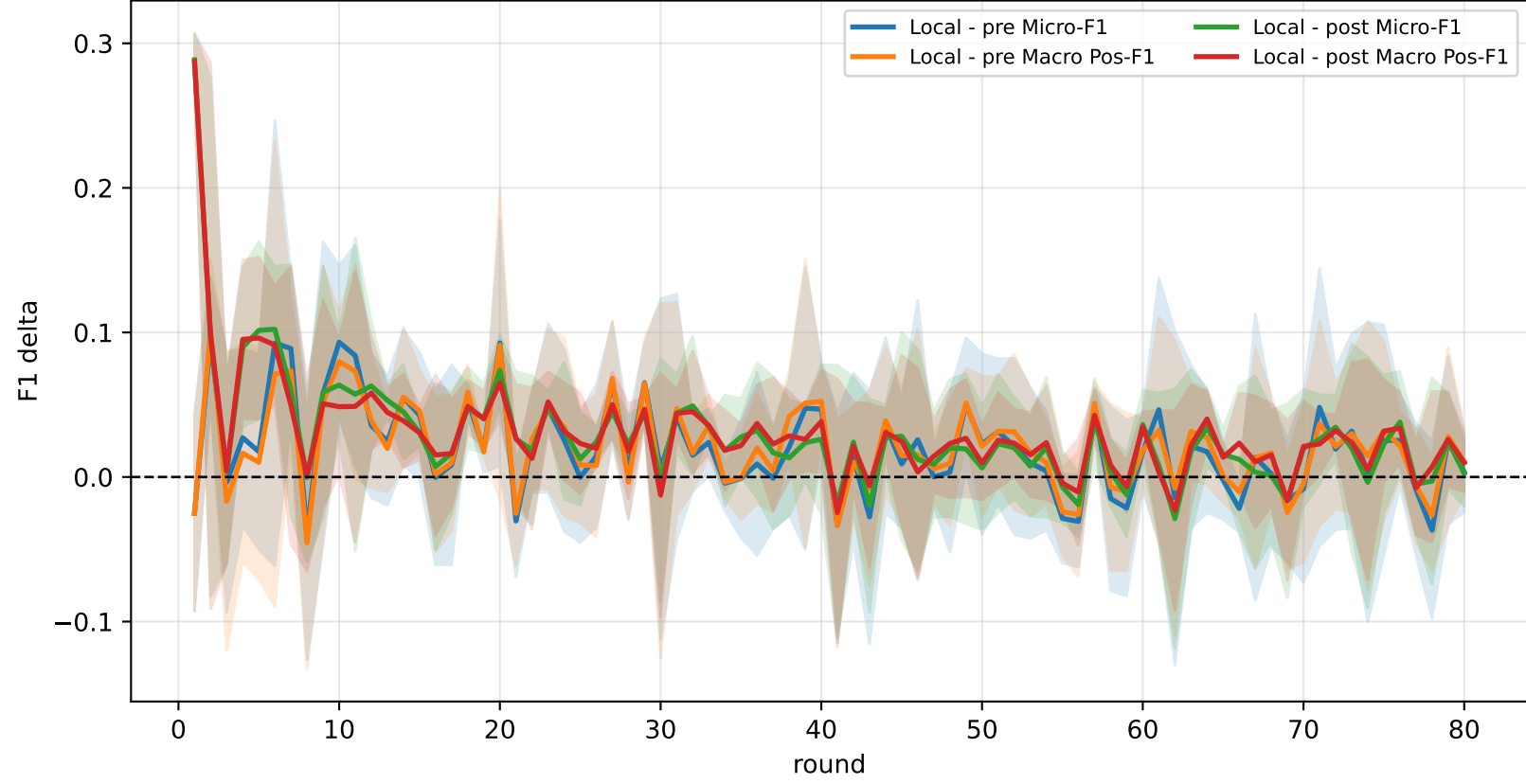
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)



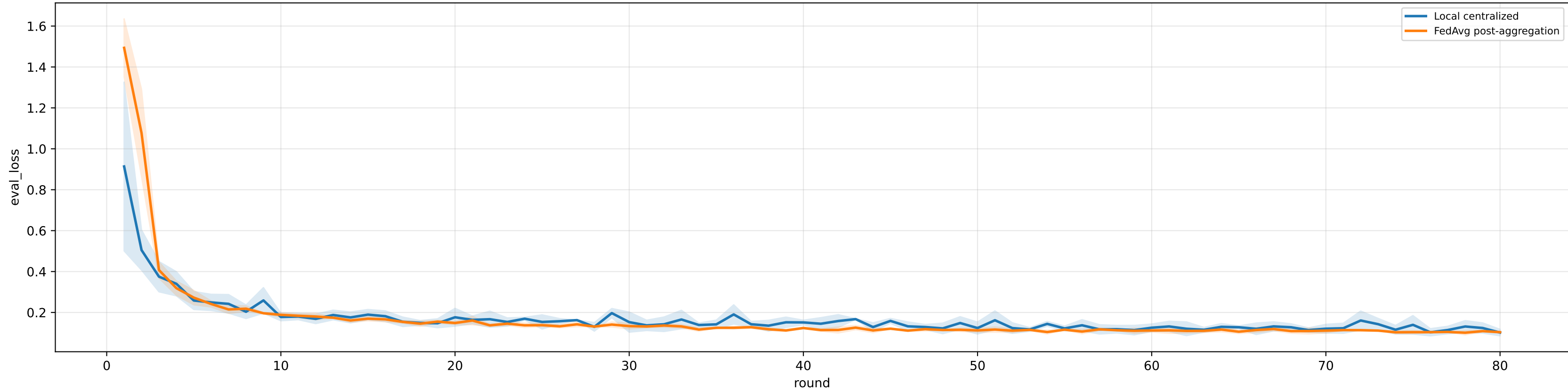


family=specialized\_cycle5 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%  
specialization=low | specialization\_fraction=0.20 | global\_visible\_support=0.84 | task\_jsd\_mean=0.003 | task\_jsd\_median=0.003 | task\_jsd\_max=0.003 | family\_centroid\_jsd\_mean=0.003

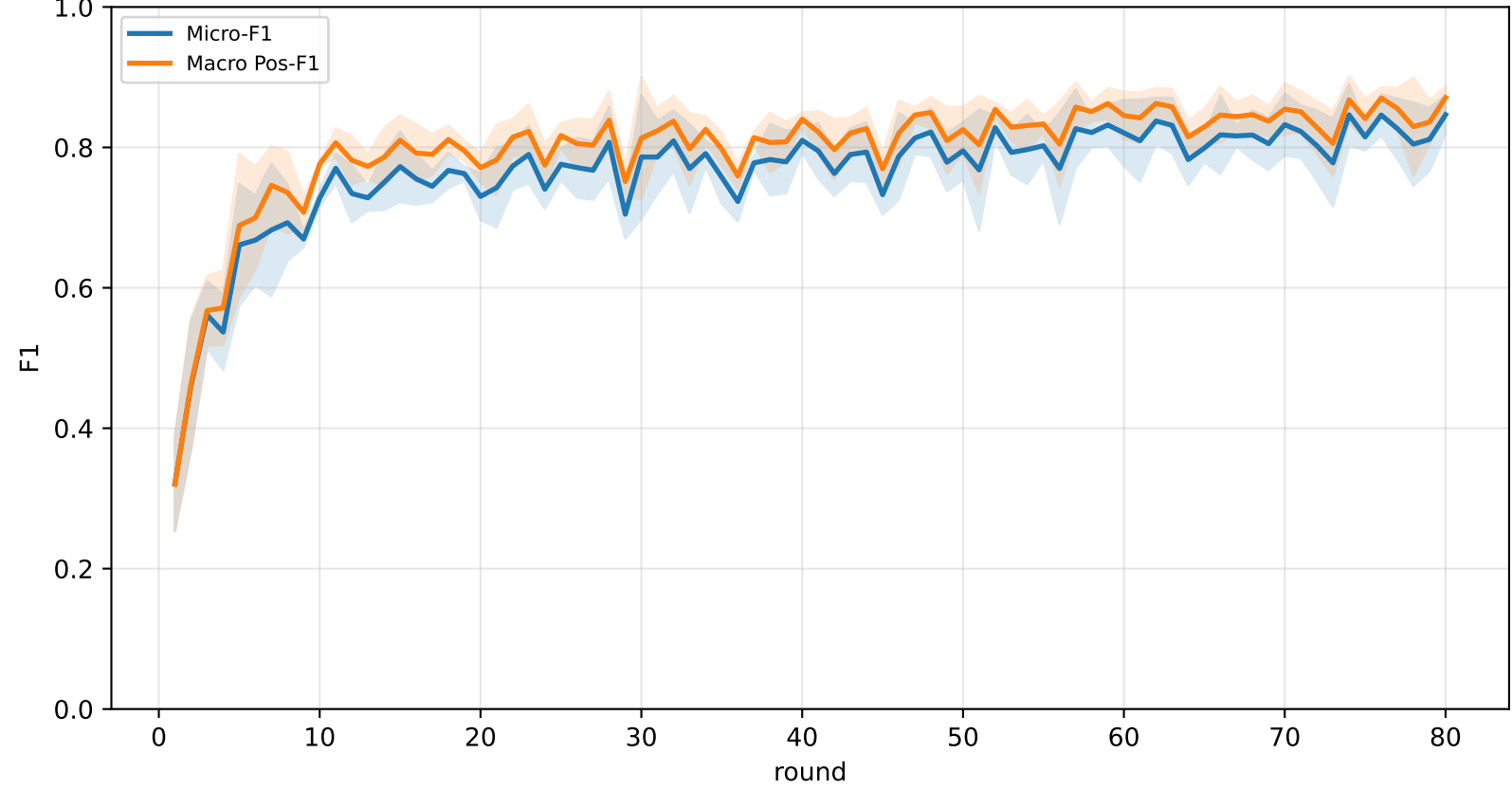
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 85.4% ± 1.5% | 87.2% ± 0.5% | 96.5% ± 1.7%  | 95.7% ± 0.7%  | 88.3% ± 2.6%  | 81.1% ± 0.7%  | 74.6% ± 7.4%  |
| FedAvg            | 81.2% ± 0.3% | 83.5% ± 0.4% | 95.1% ± 1.1%  | 94.9% ± 1.1%  | 81.6% ± 3.1%  | 77.8% ± 0.9%  | 68.3% ± 2.6%  |

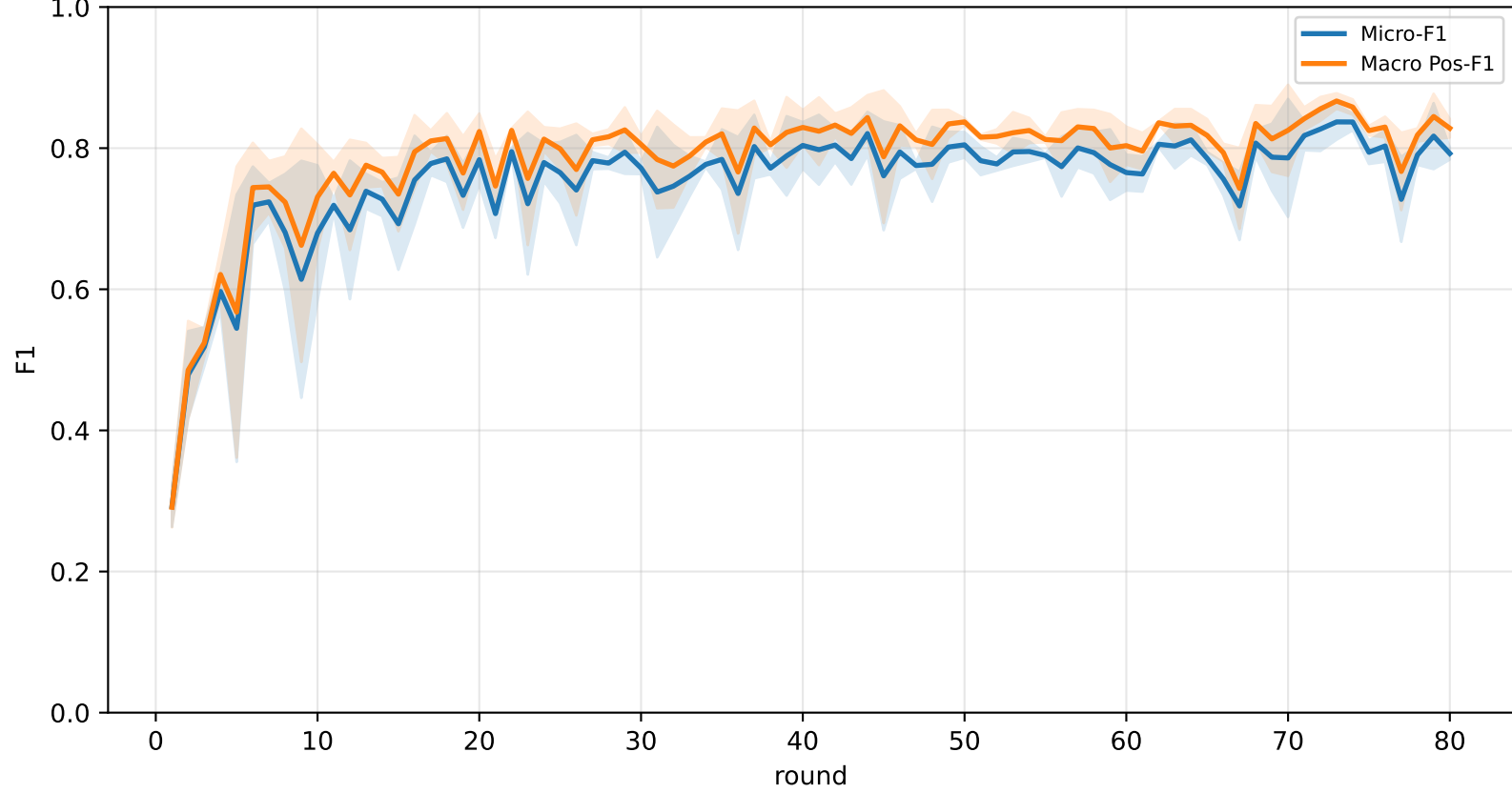
Validation loss comparison



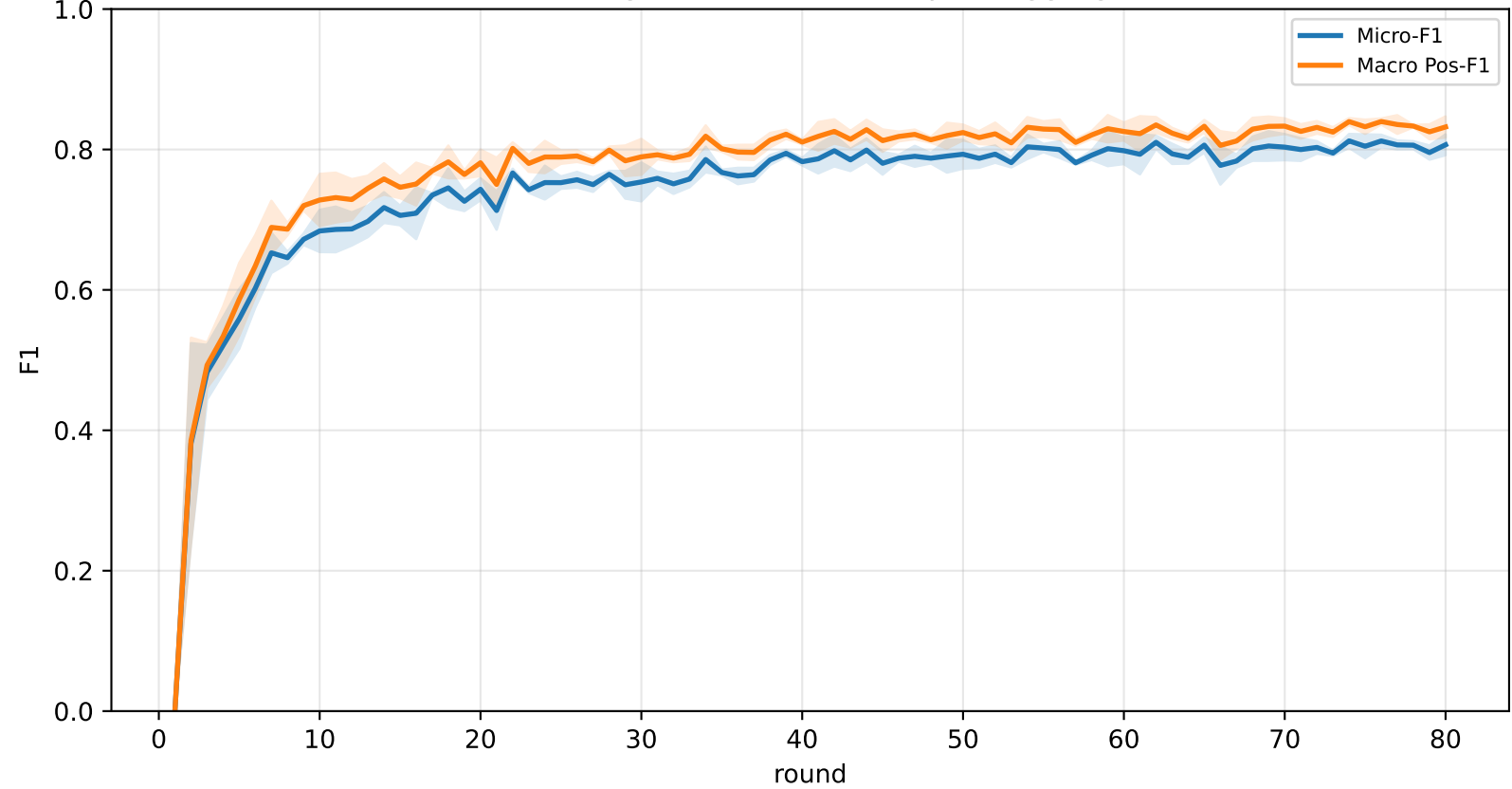
Pooled local centralized micro / macro F1



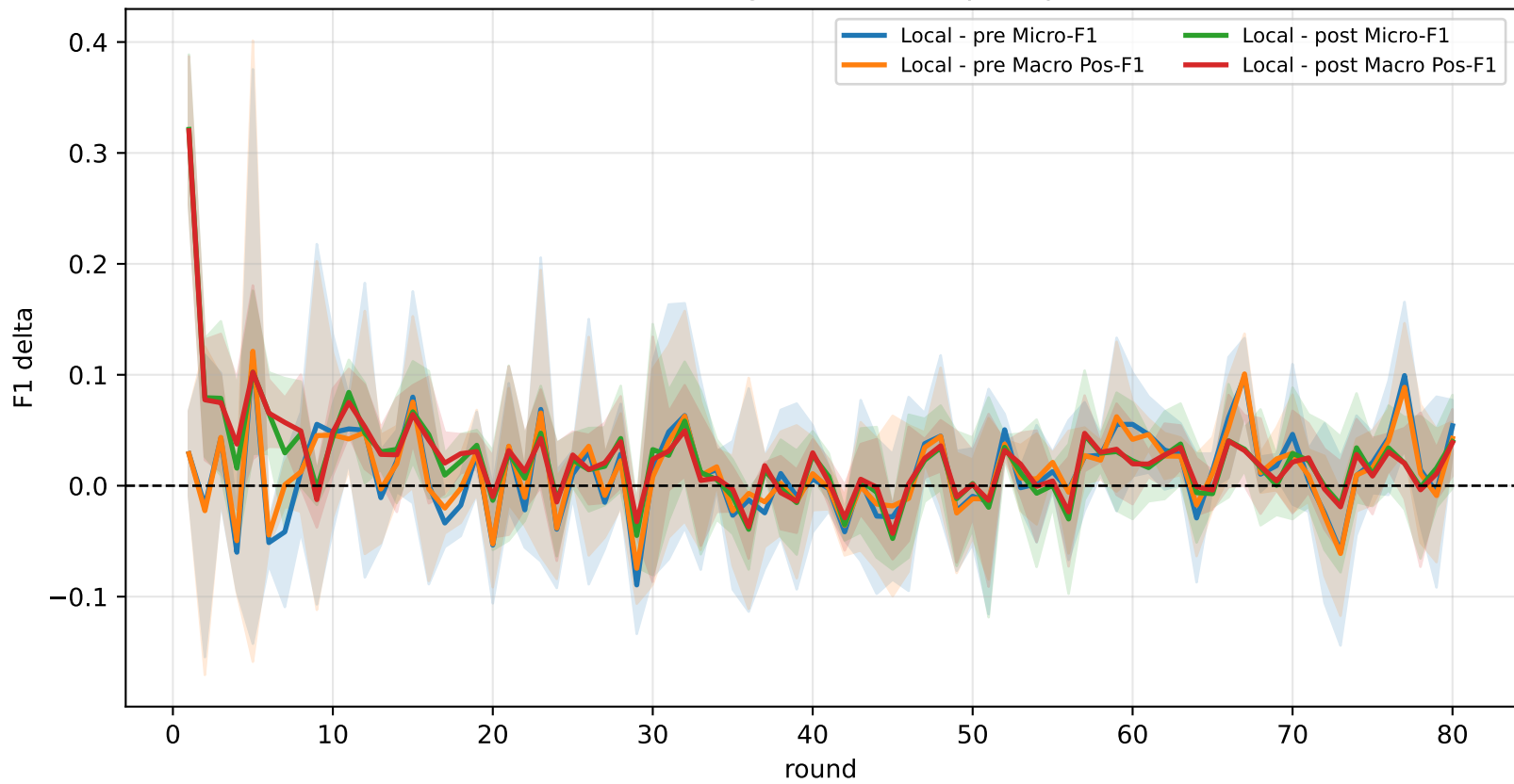
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)

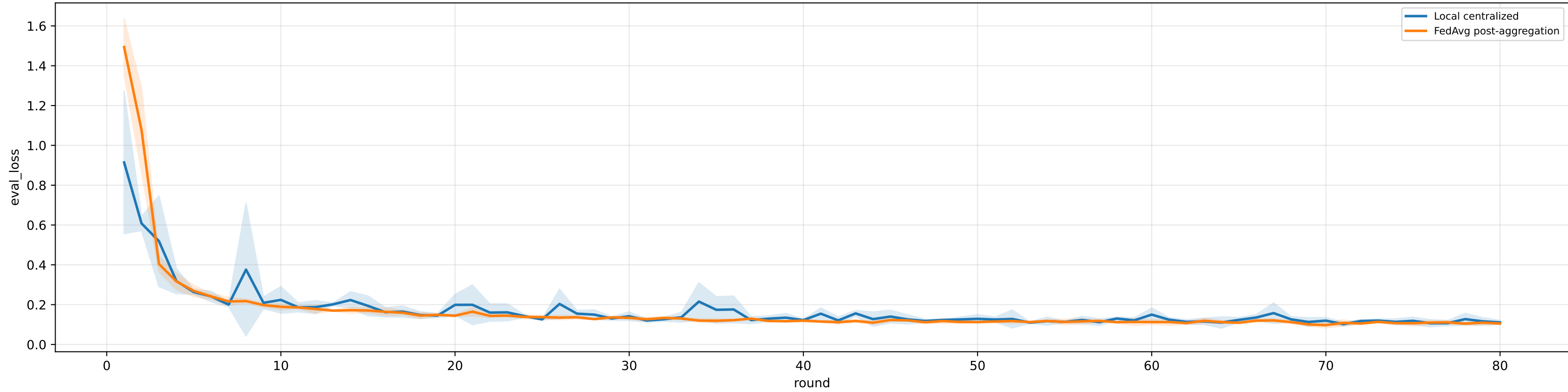


family=specialized\_cycle6 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%  
specialization=low | specialization\_fraction=0.20 | global\_visible\_support=0.84 | task\_jsd\_mean=0.003 | task\_jsd\_median=0.003 | task\_jsd\_max=0.003 | family\_centroid\_jsd\_mean=0.003

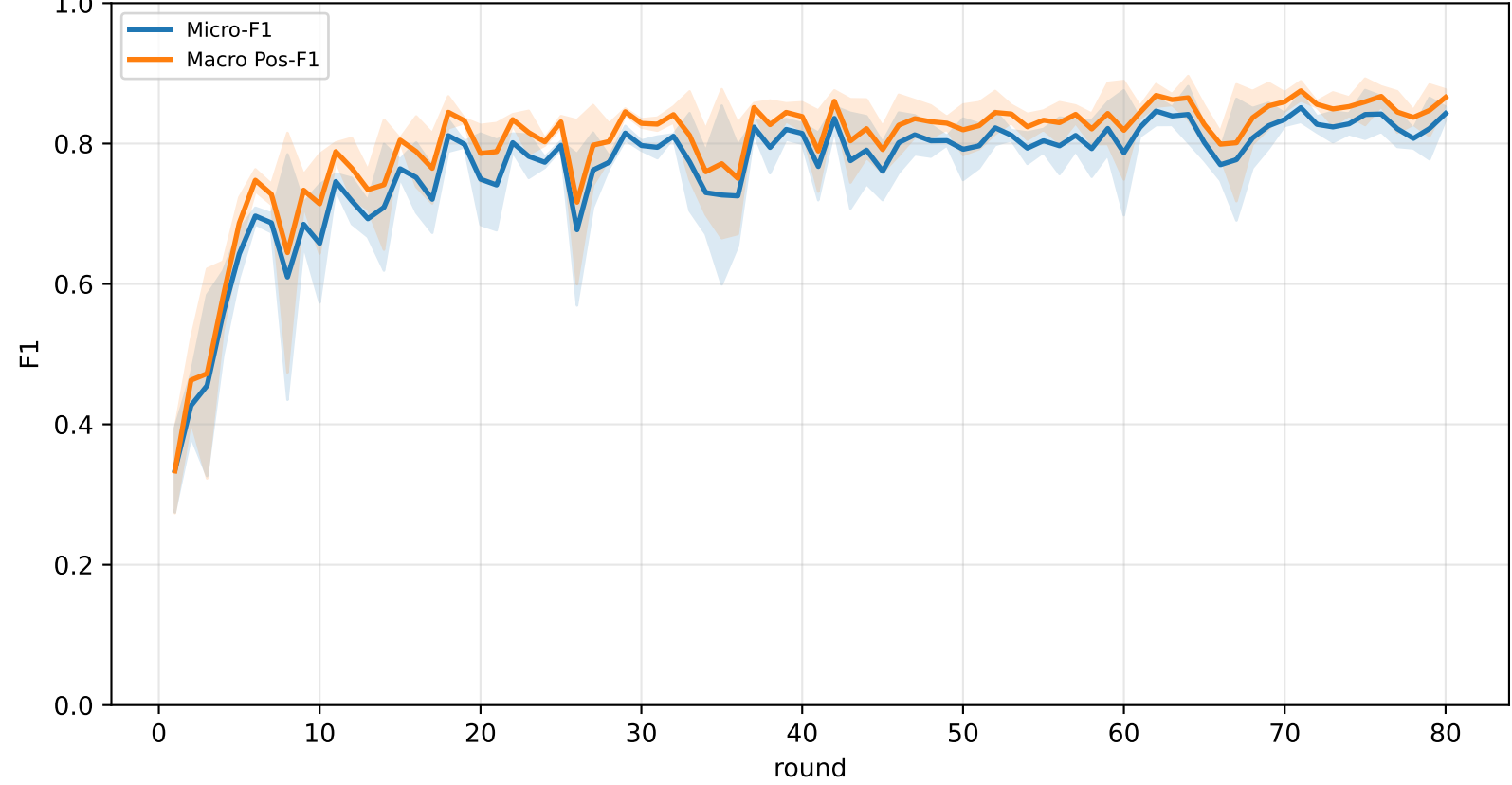
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 84.2% ± 1.5% | 86.2% ± 1.0% | 97.7% ± 0.7%  | 93.8% ± 0.8%  | 88.3% ± 1.2%  | 79.5% ± 0.8%  | 71.9% ± 4.5%  |
| FedAvg            | 82.2% ± 0.7% | 84.7% ± 0.3% | 97.4% ± 0.3%  | 96.0% ± 1.1%  | 83.2% ± 2.3%  | 78.5% ± 0.9%  | 68.6% ± 2.8%  |

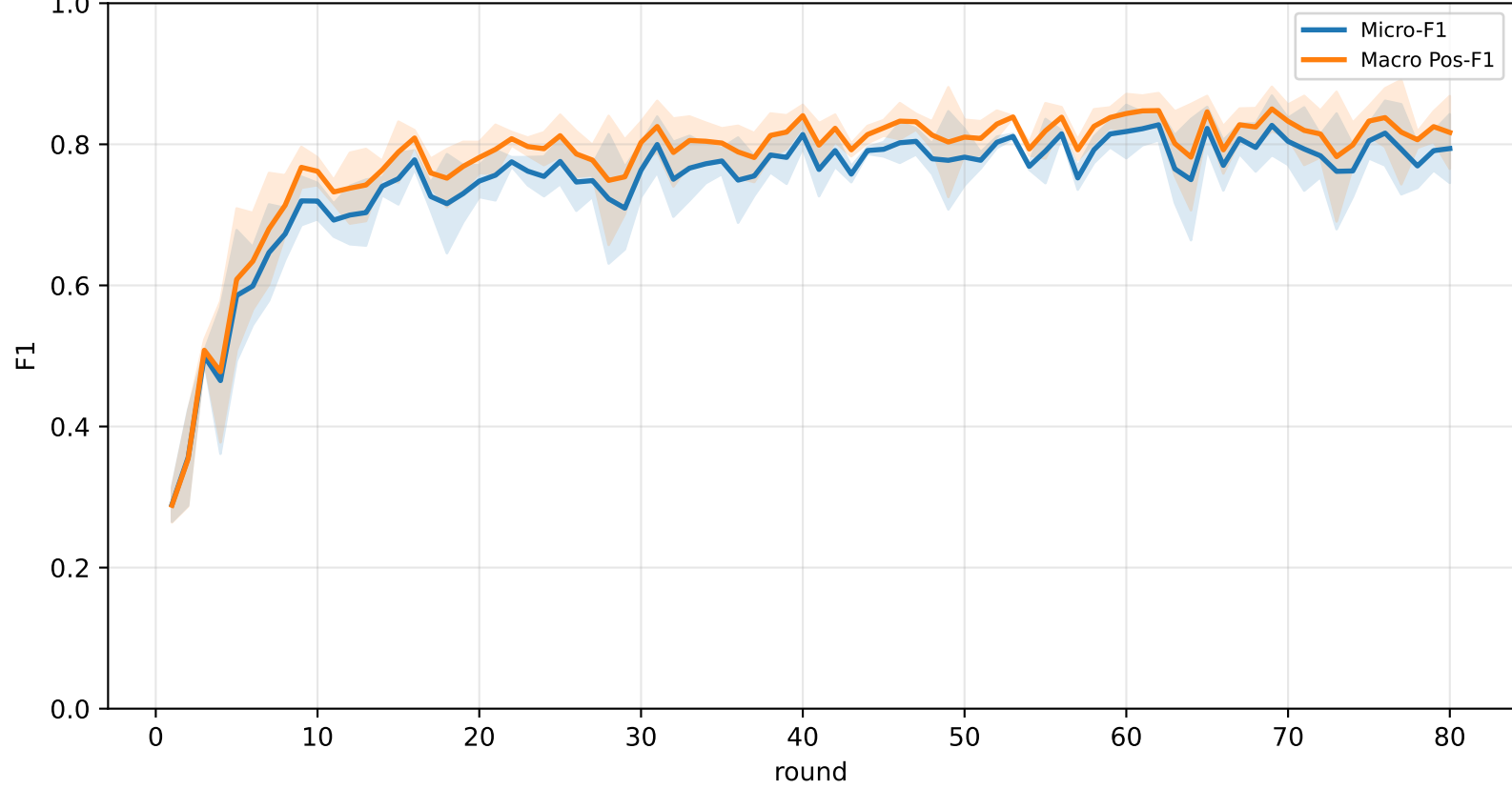
Validation loss comparison



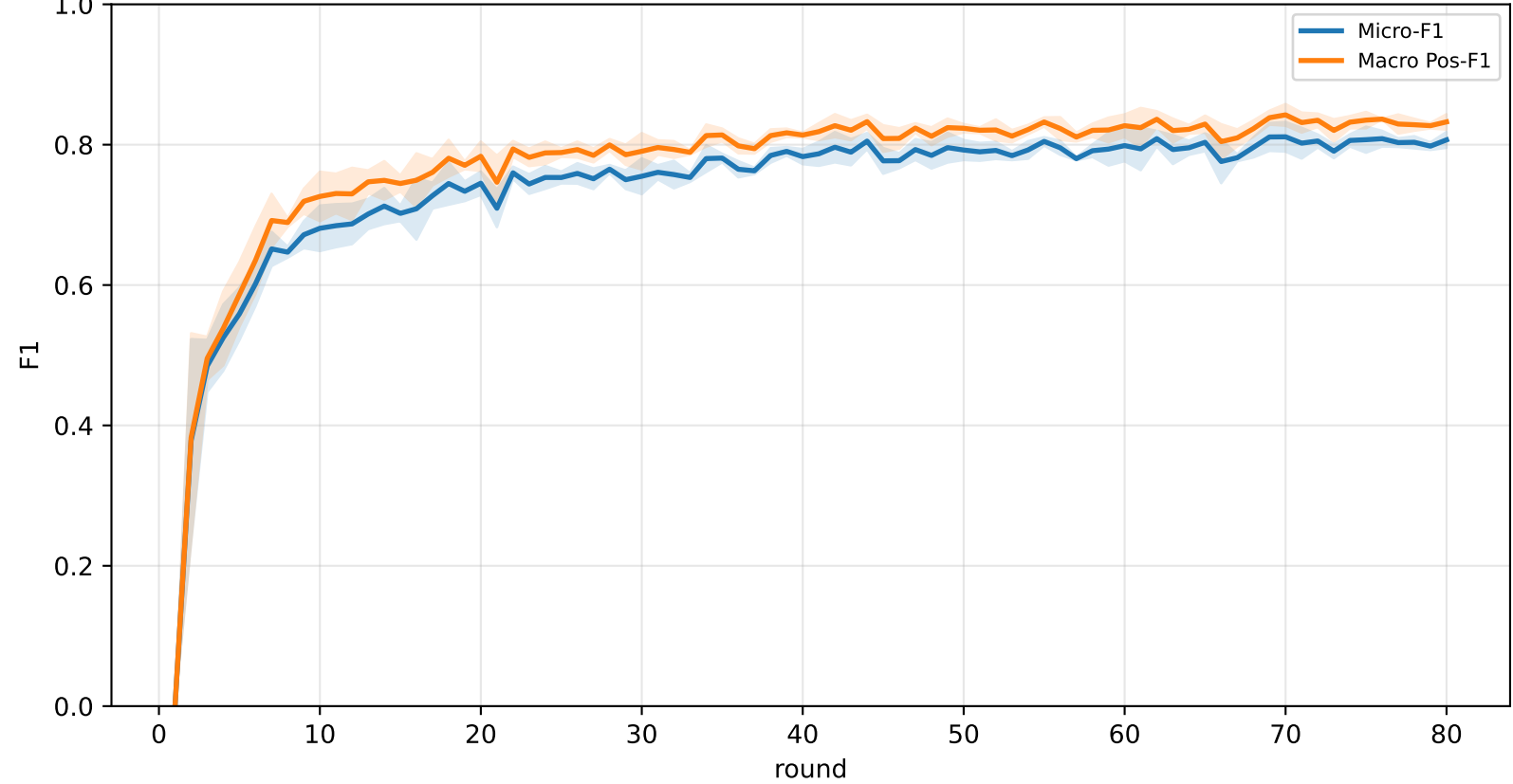
Pooled local centralized micro / macro F1



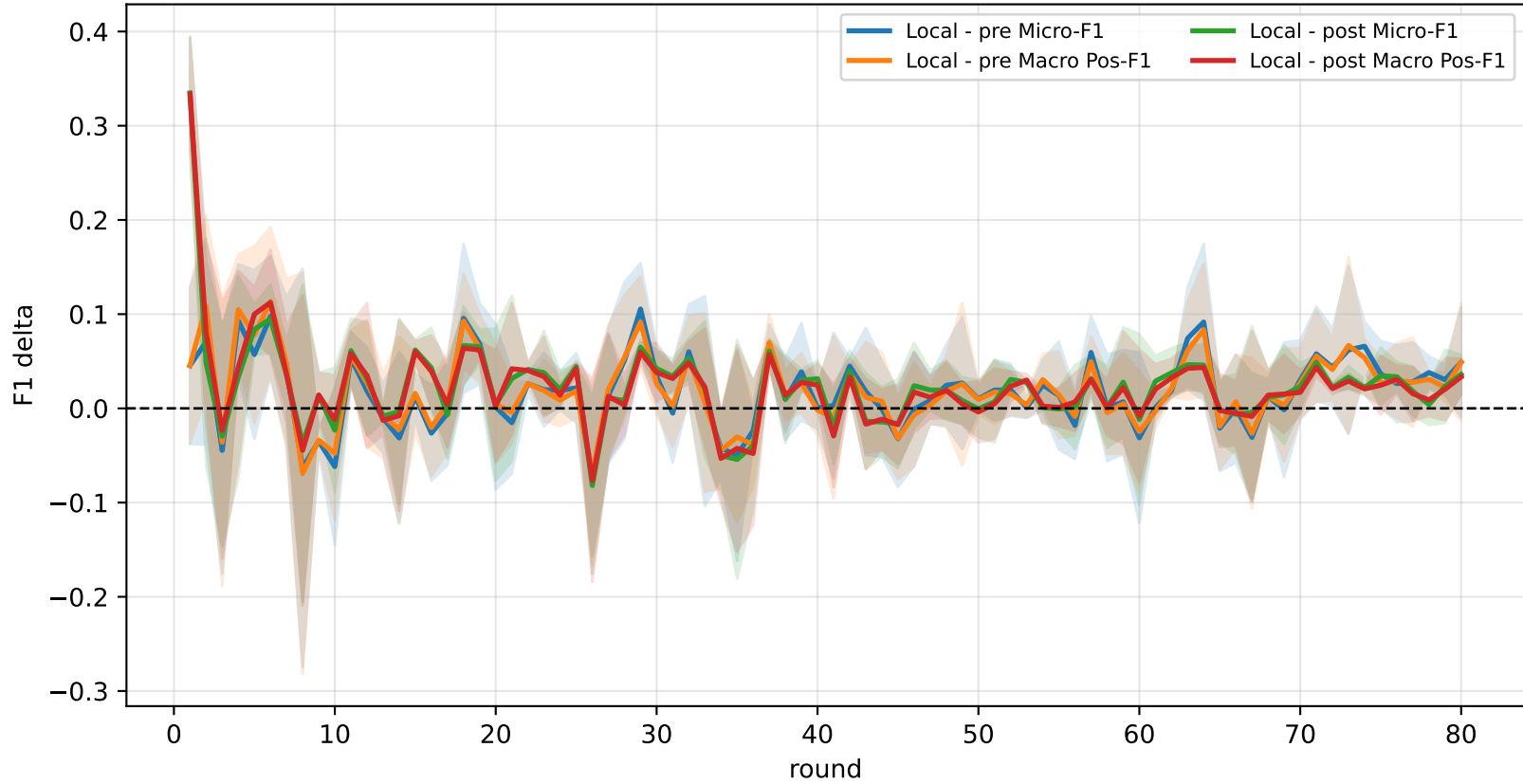
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)



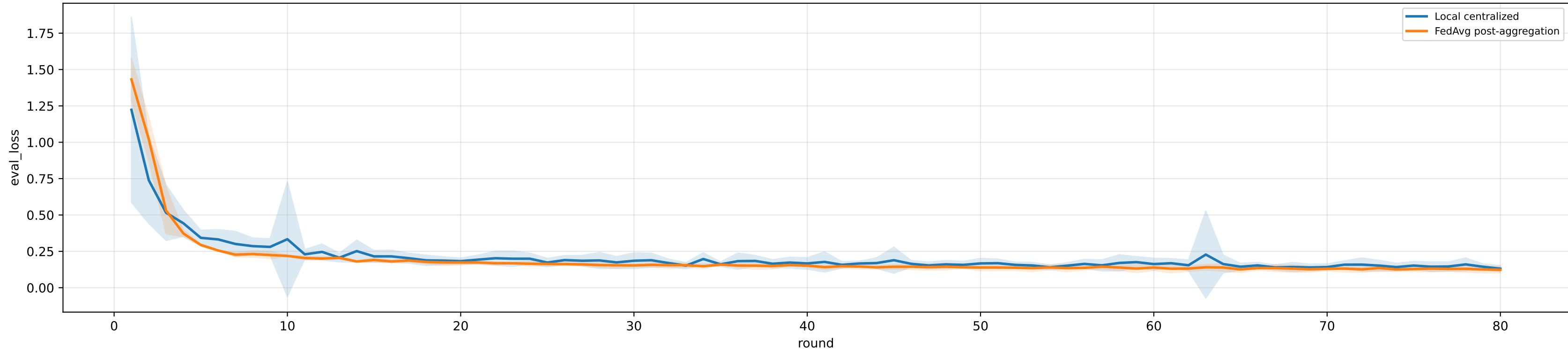
Controlled task-specialization level: Mid | FedAvg diagnostics | aggregated / pooled over 5 clients | model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64

family=specialized\_cycle2 | n\_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%  
family=specialized\_cycle3 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%  
family=specialized\_cycle4 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%  
family=specialized\_cycle5 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%  
family=specialized\_cycle6 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 2.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%  
specialization=mid | specialization\_fraction=0.50 | global\_visible\_support=0.60 | task\_jsd\_mean=0.028 | task\_jsd\_median=0.028 | task\_jsd\_max=0.029 | family\_centroid\_jsd\_mean=0.028

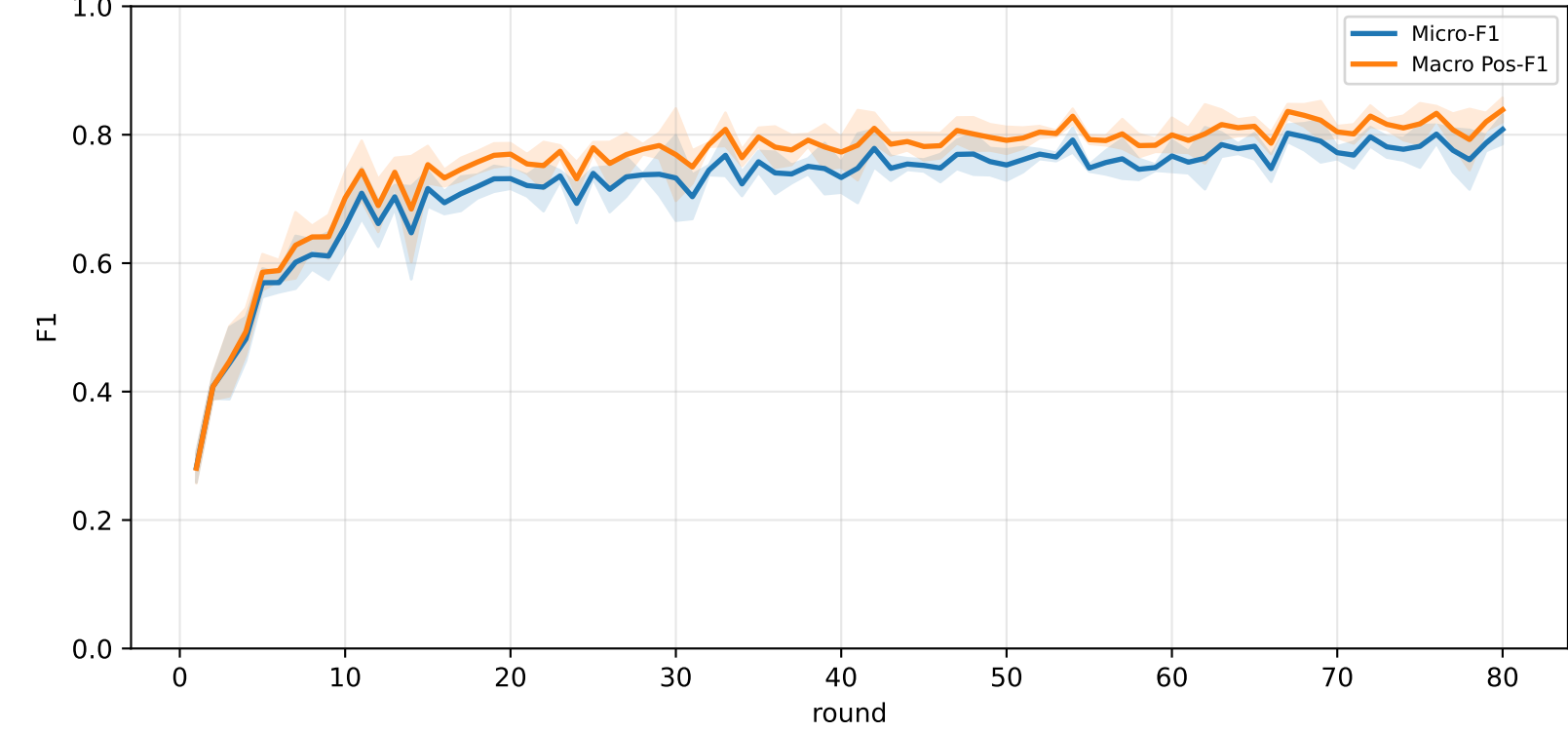
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 81.6% ± 0.7% | 84.6% ± 0.7% | 96.9% ± 0.3%  | 95.3% ± 0.9%  | 87.5% ± 0.4%  | 77.3% ± 1.9%  | 65.6% ± 0.6%  |
| FedAvg            | 78.7% ± 2.2% | 81.3% ± 1.4% | 94.2% ± 1.3%  | 92.5% ± 1.6%  | 79.5% ± 0.1%  | 74.6% ± 3.1%  | 65.8% ± 4.6%  |

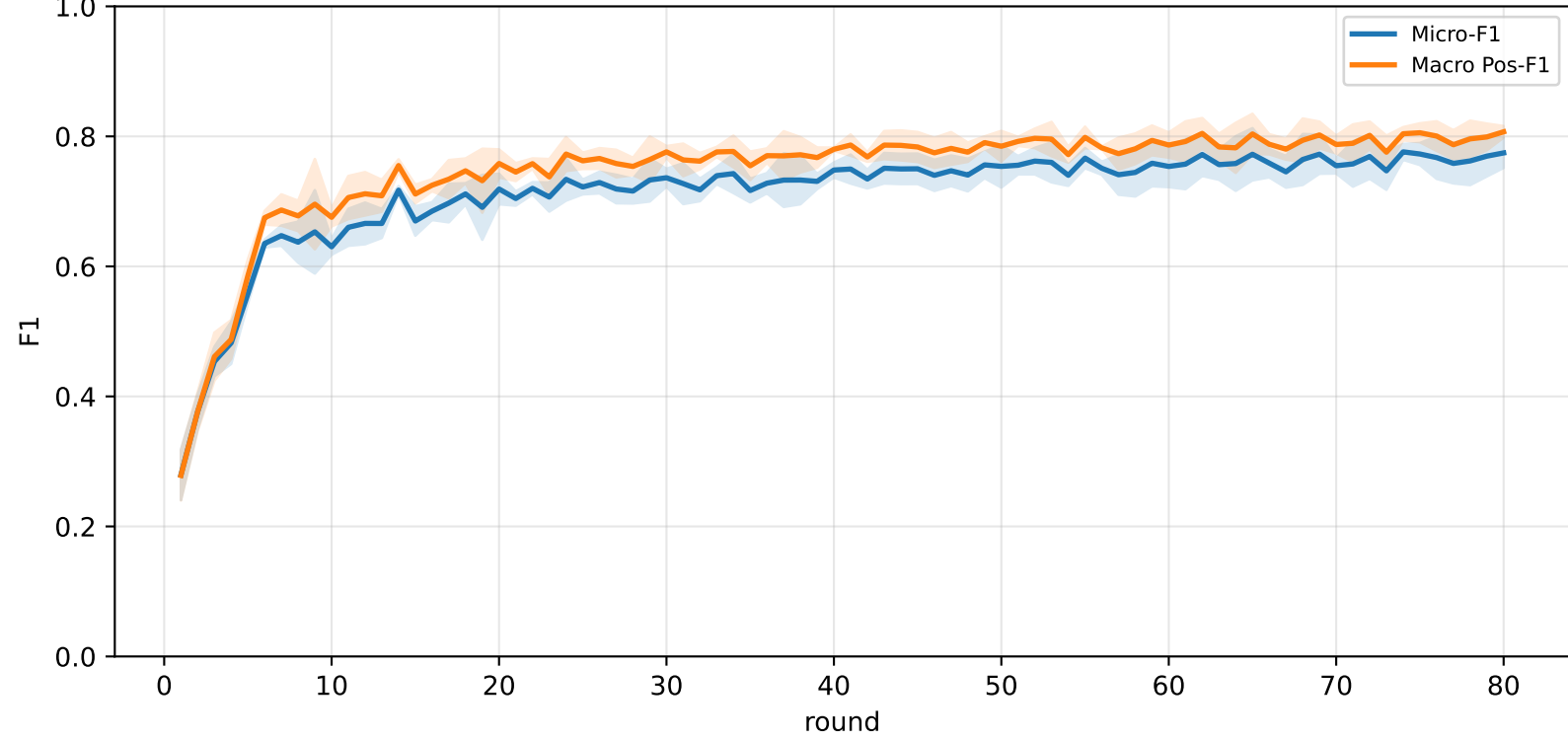
Validation loss comparison



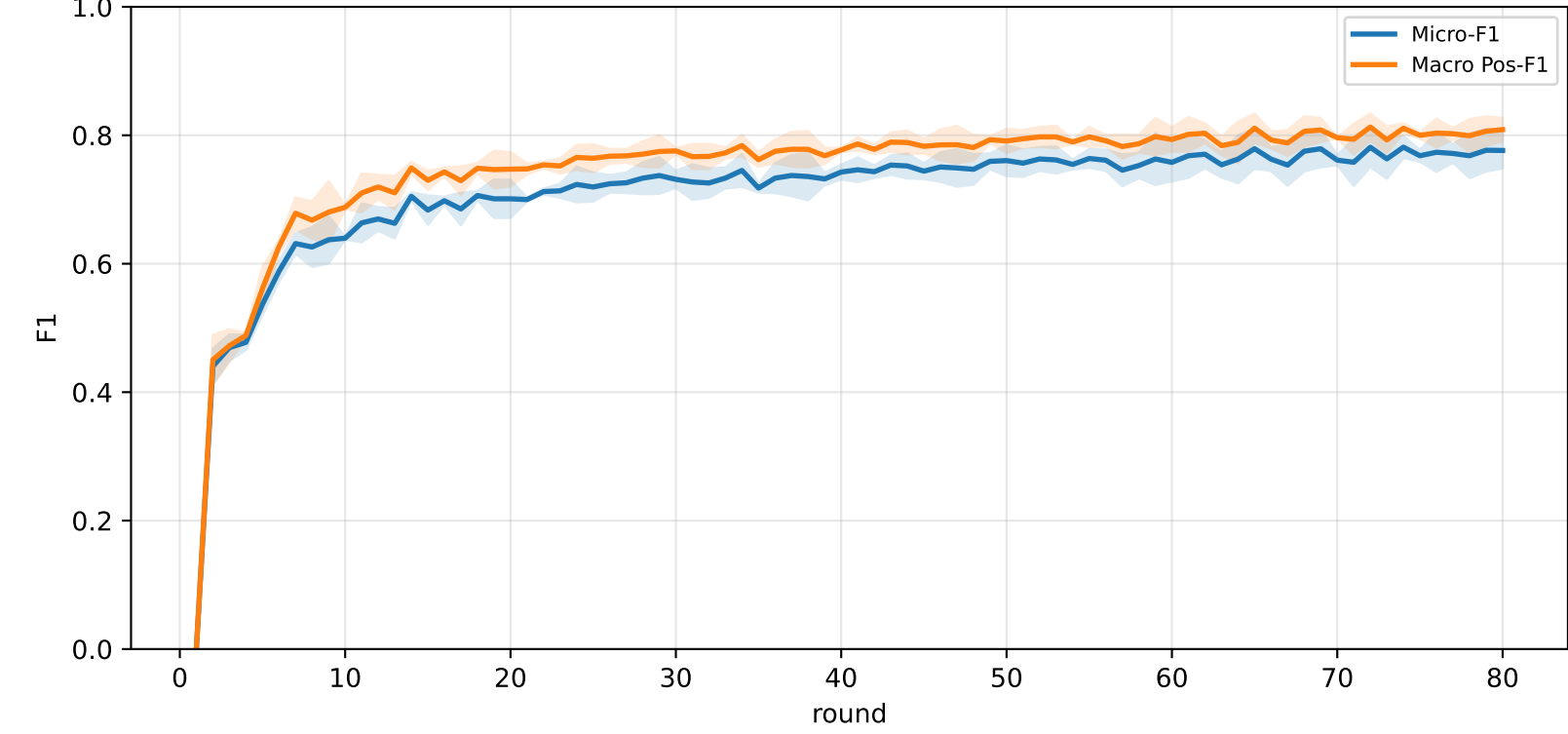
Pooled local centralized micro / macro F1



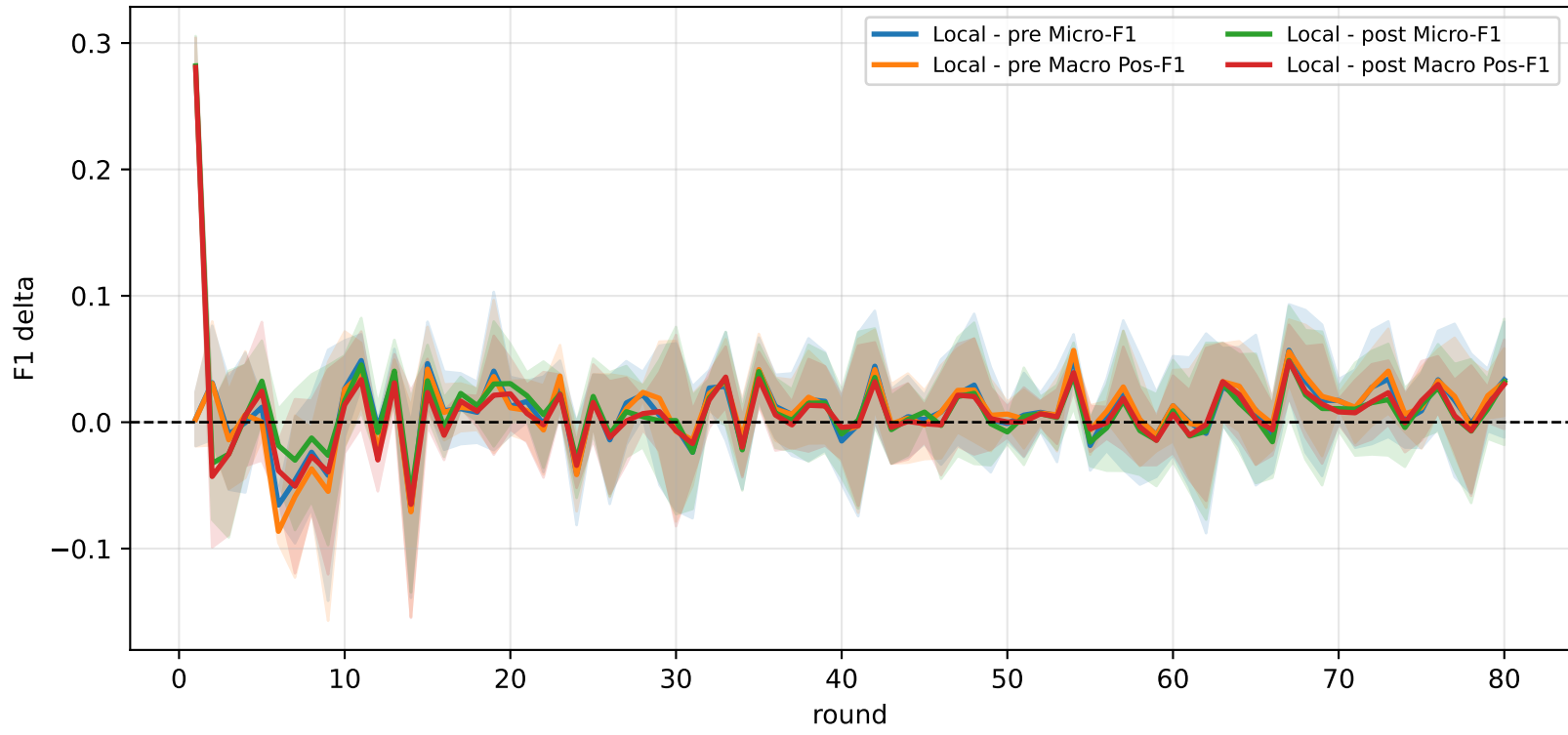
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)



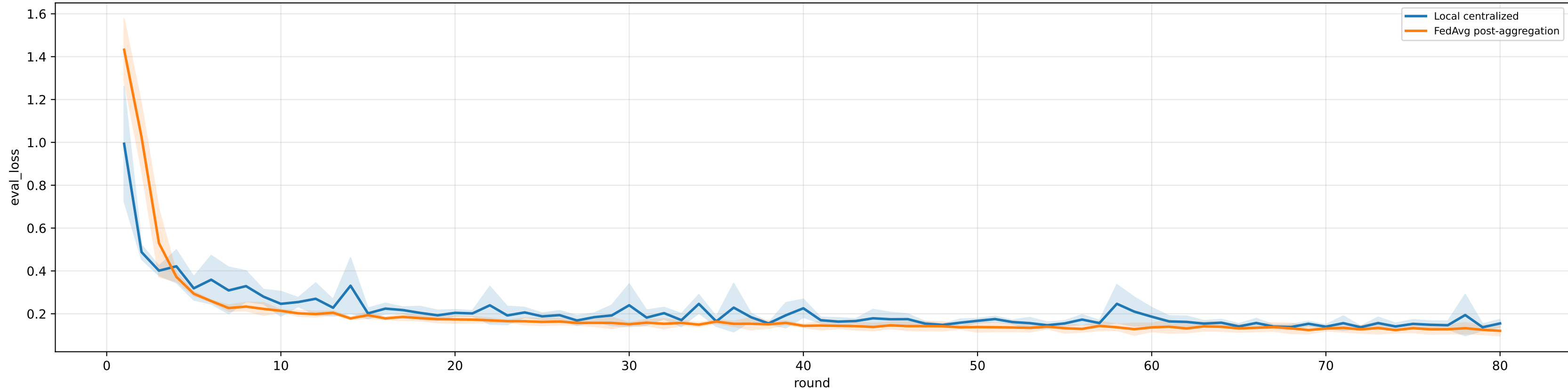


family=specialized\_cycle2 | n\_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% | specialization=mid | specialization\_fraction=0.50 | global\_visible\_support=0.60 | task\_jsd\_mean=0.028 | task\_jsd\_median=0.028 | task\_jsd\_max=0.029 | family\_centroid\_jsd\_mean=0.028

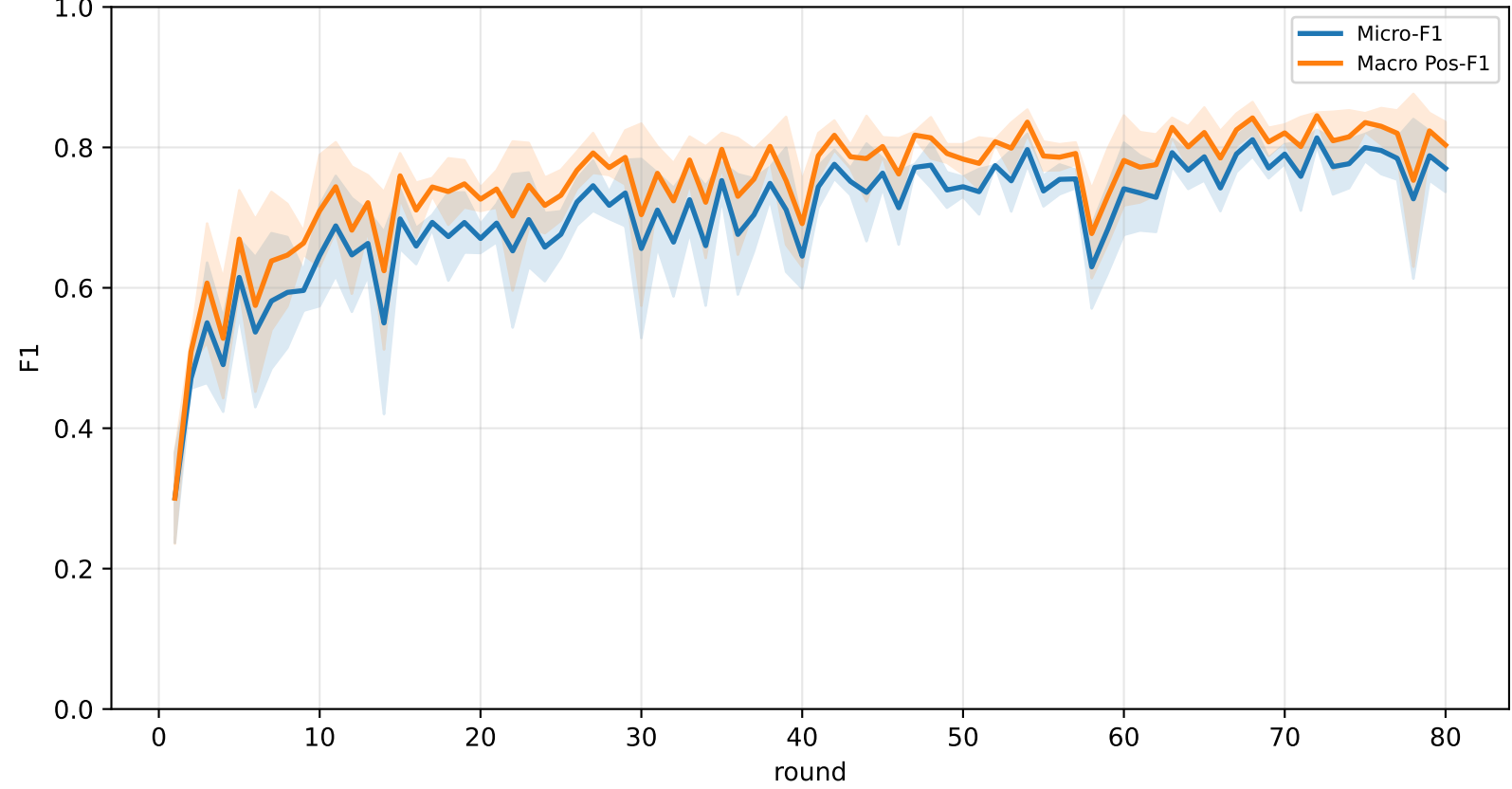
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 81.3% ± 1.7% | 84.2% ± 1.3% | 98.2% ± 0.7%  | 94.8% ± 1.7%  | 86.7% ± 1.3%  | 75.0% ± 3.4%  | 66.4% ± 3.0%  |
| FedAvg            | 78.8% ± 1.9% | 81.4% ± 1.1% | 94.8% ± 1.6%  | 92.9% ± 1.0%  | 79.0% ± 0.6%  | 74.7% ± 2.7%  | 65.7% ± 4.6%  |

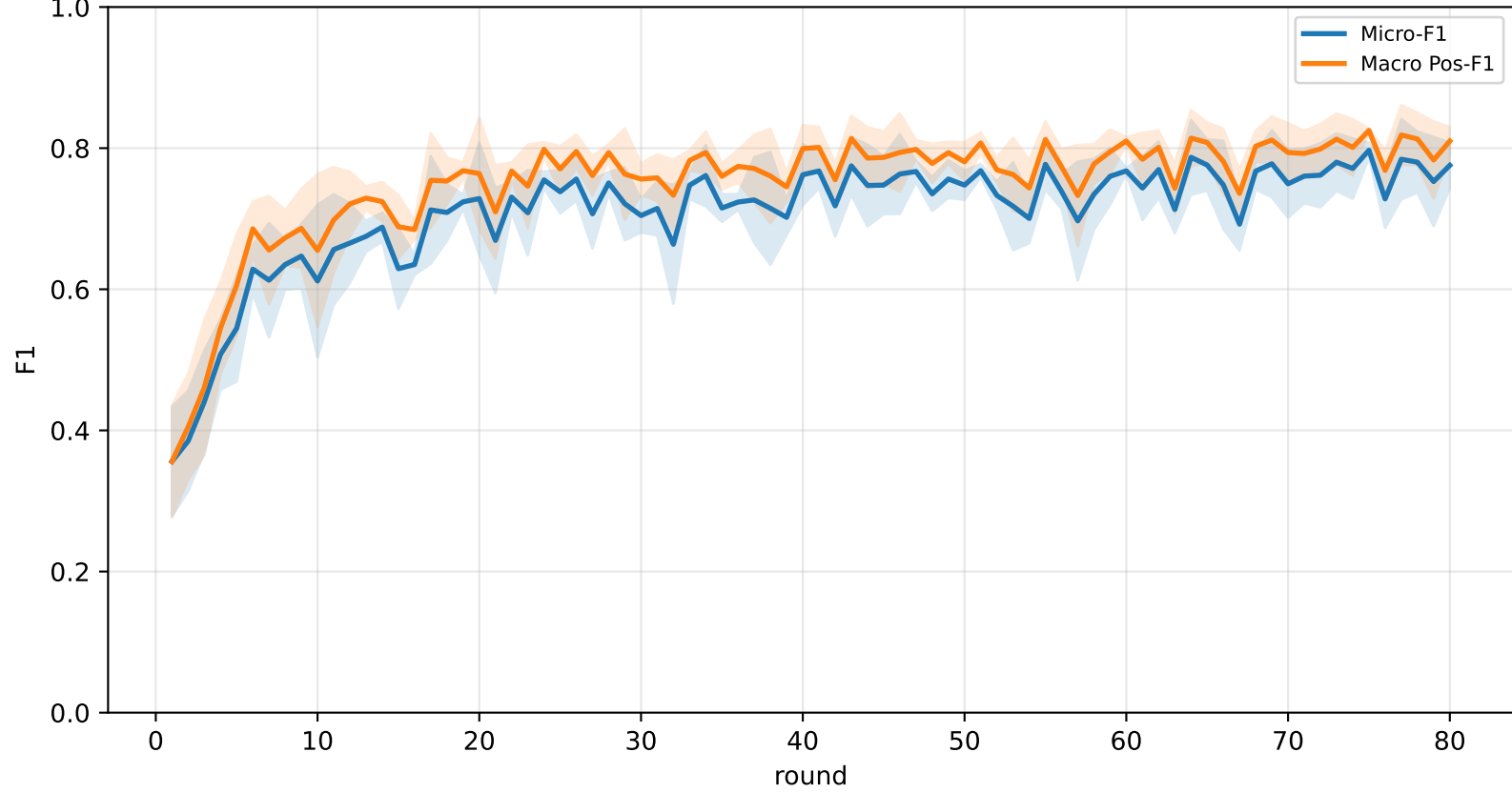
Validation loss comparison



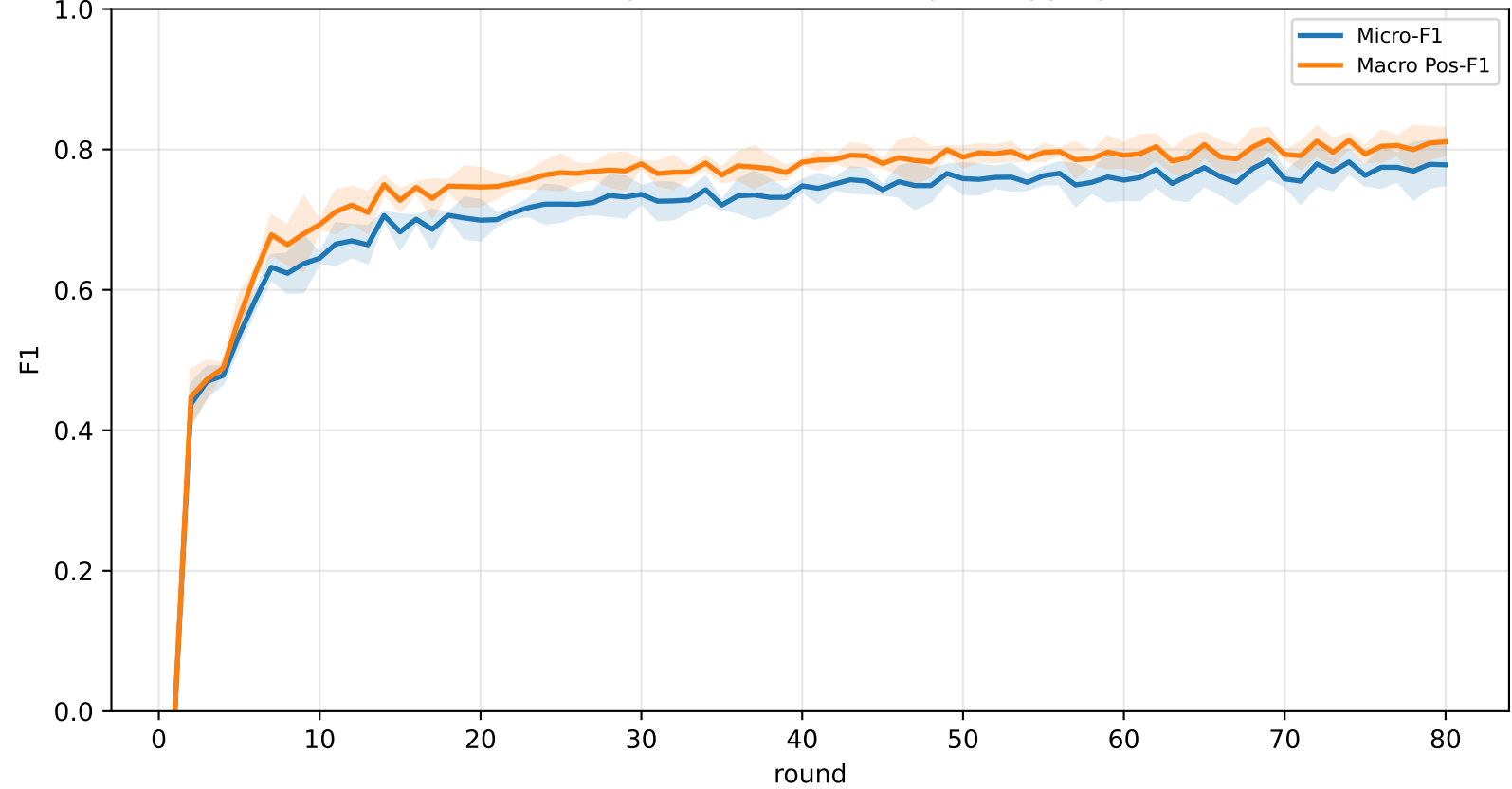
Pooled local centralized micro / macro F1



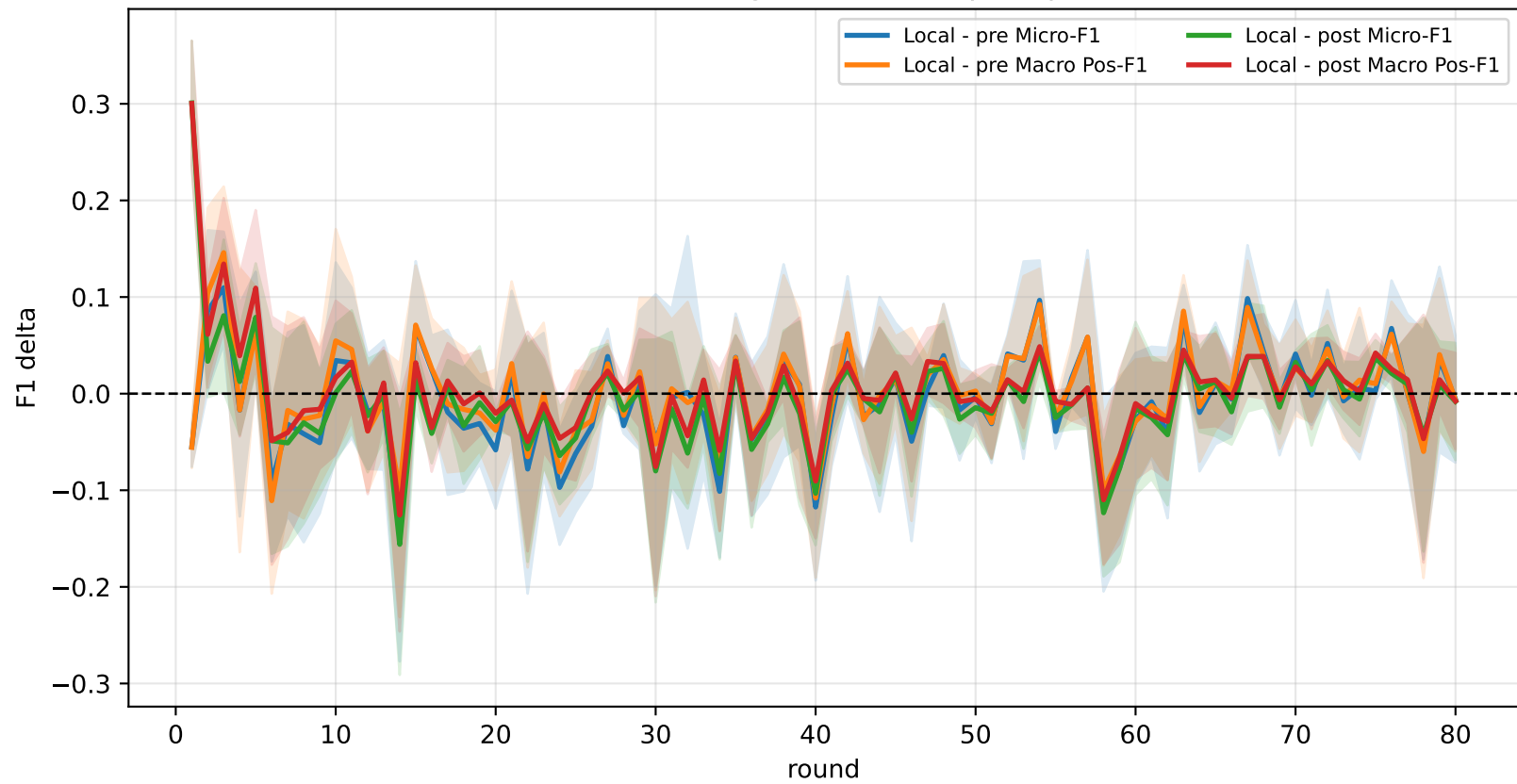
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)



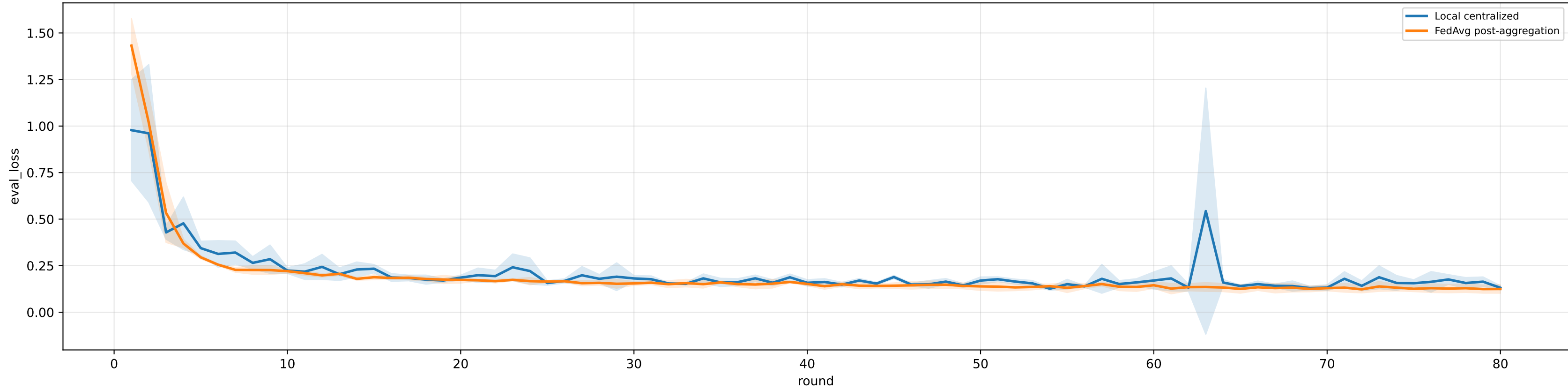


family=specialized\_cycle3 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% | specialization=mid | specialization\_fraction=0.50 | global\_visible\_support=0.60 | task\_jsd\_mean=0.028 | task\_jsd\_median=0.028 | task\_jsd\_max=0.029 | family\_centroid\_jsd\_mean=0.028

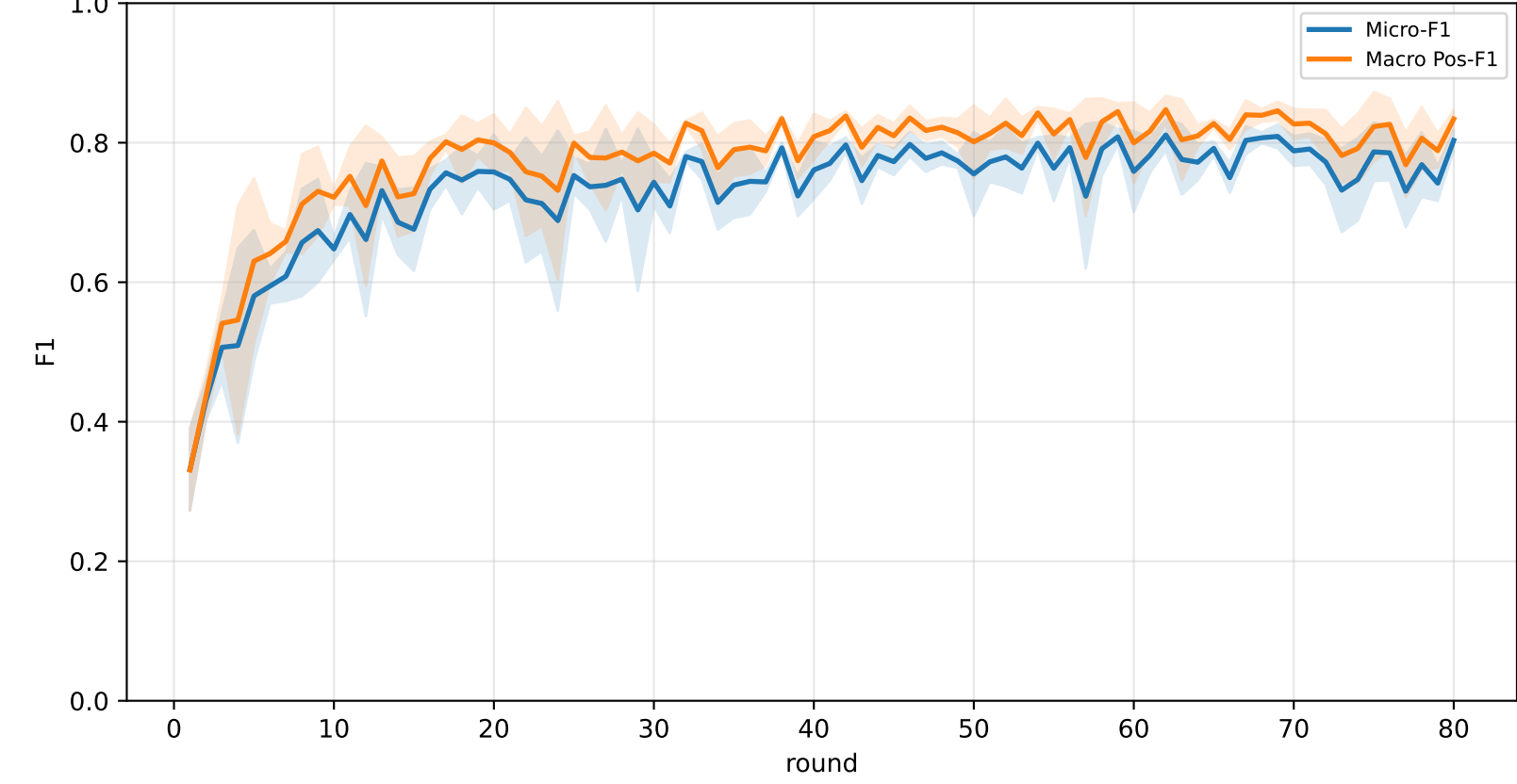
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 81.1% ± 0.2% | 84.7% ± 0.1% | 97.7% ± 0.8%  | 96.9% ± 0.8%  | 90.8% ± 0.3%  | 74.1% ± 1.7%  | 64.0% ± 0.4%  |
| FedAvg            | 78.7% ± 2.1% | 81.3% ± 1.4% | 93.5% ± 0.9%  | 92.5% ± 1.6%  | 80.1% ± 0.7%  | 74.1% ± 2.8%  | 65.9% ± 4.7%  |

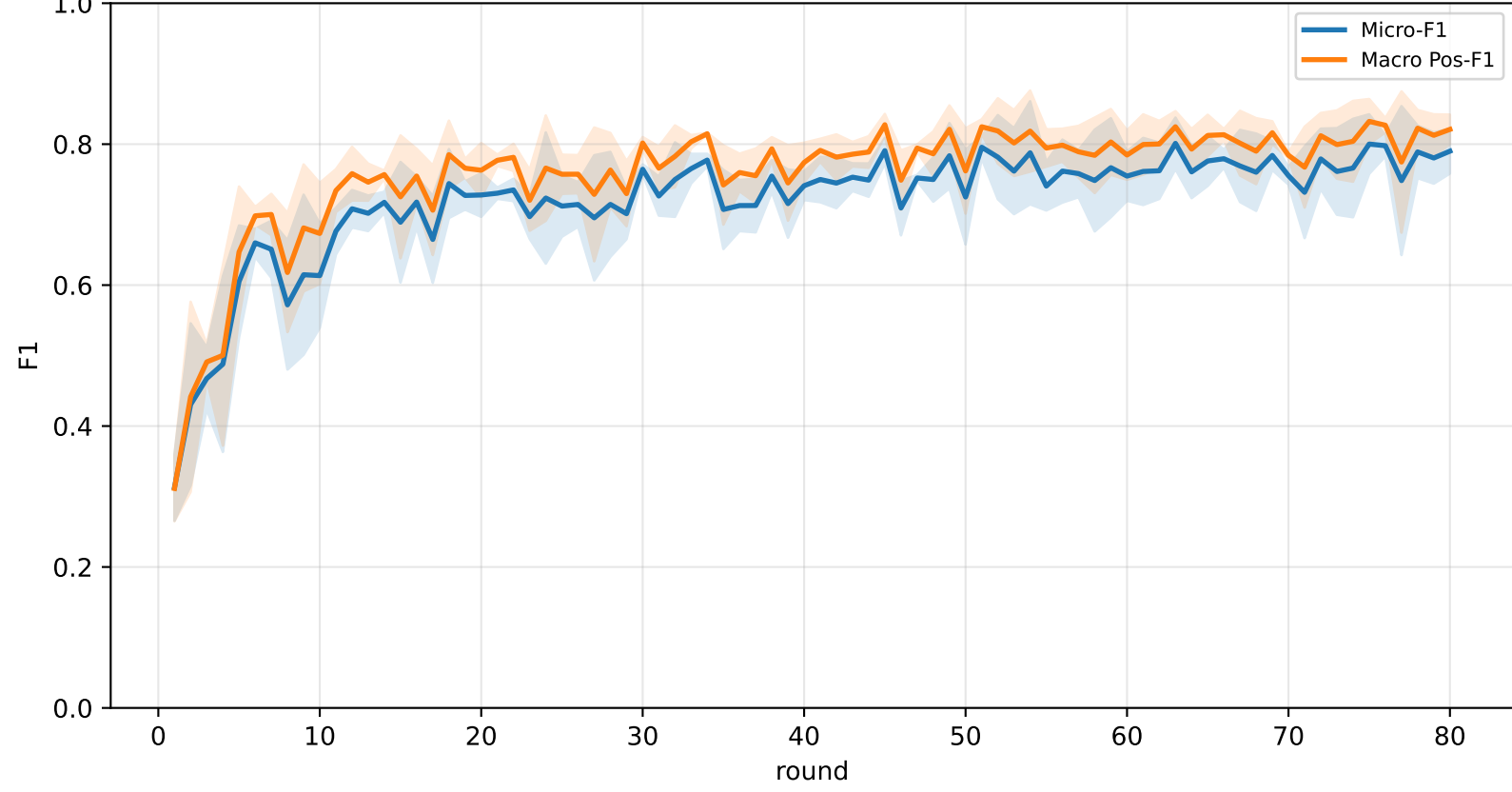
Validation loss comparison



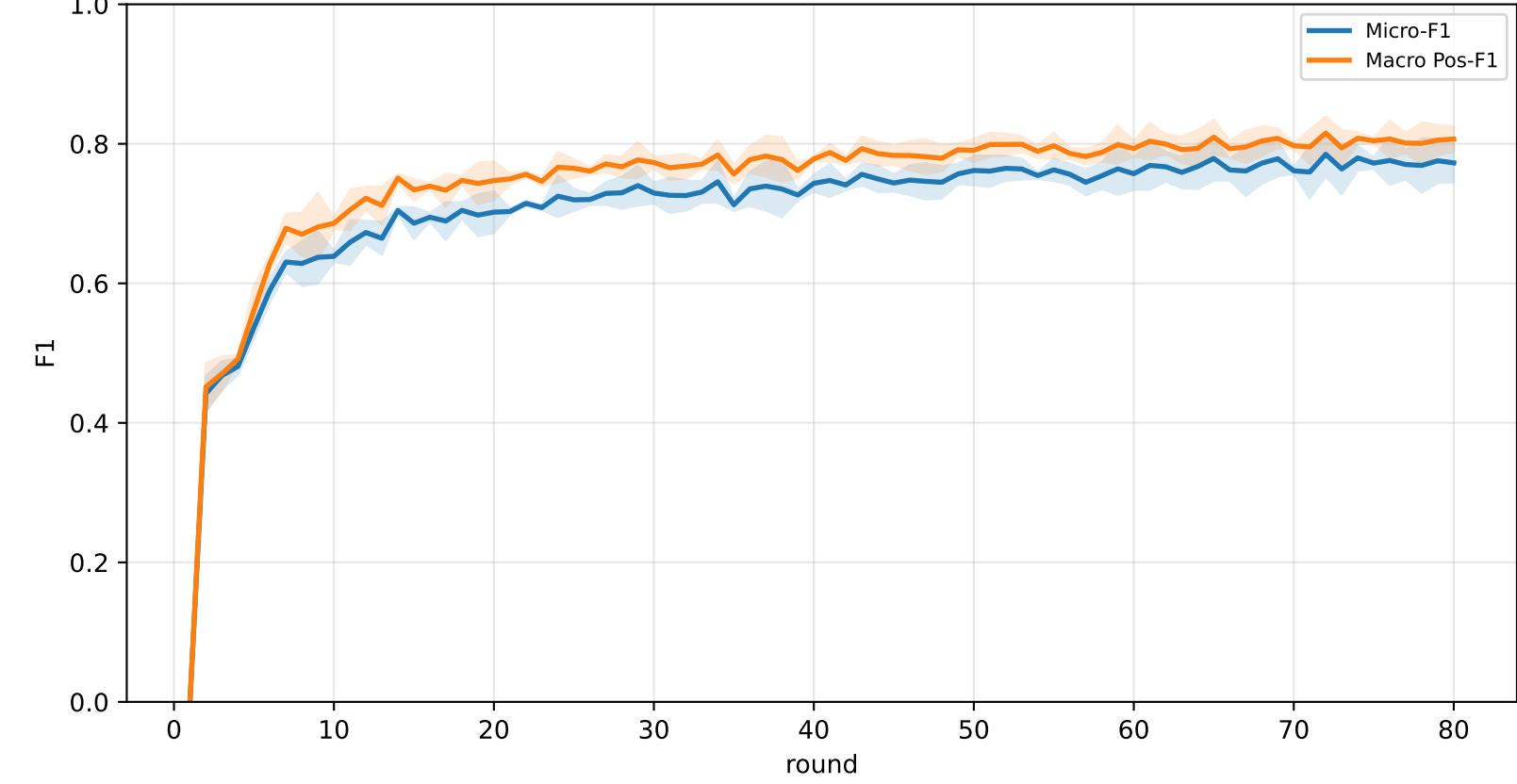
Pooled local centralized micro / macro F1



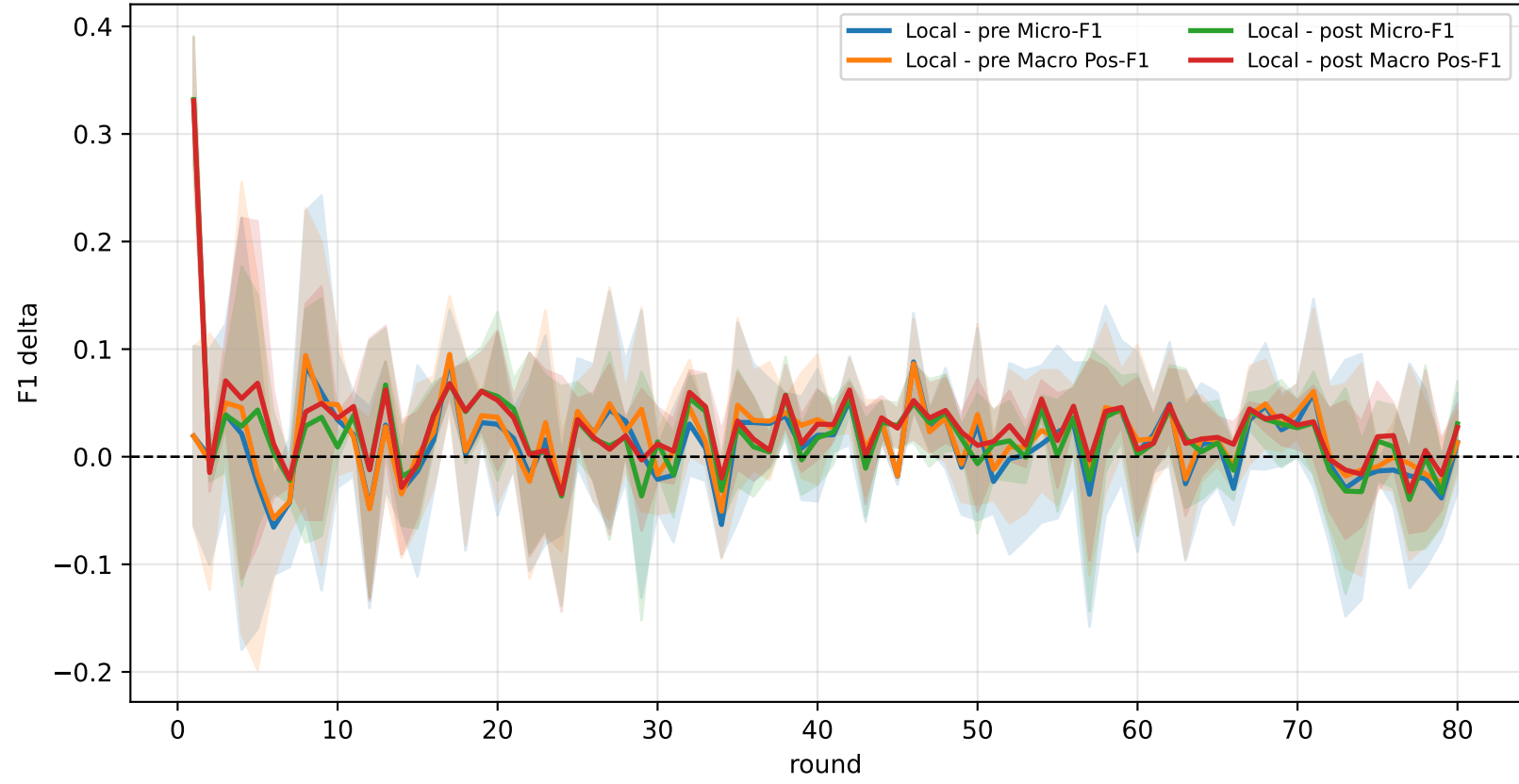
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)

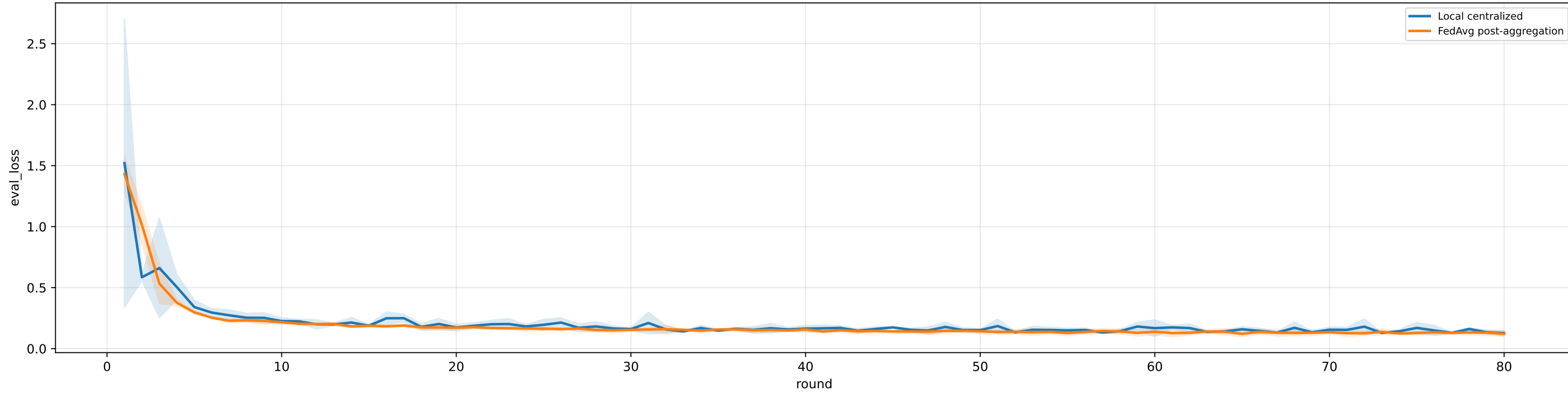


family=specialized\_cycle4 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%  
specialization=mid | specialization\_fraction=0.50 | global\_visible\_support=0.60 | task\_jsd\_mean=0.028 | task\_jsd\_median=0.028 | task\_jsd\_max=0.029 | family\_centroid\_jsd\_mean=0.028

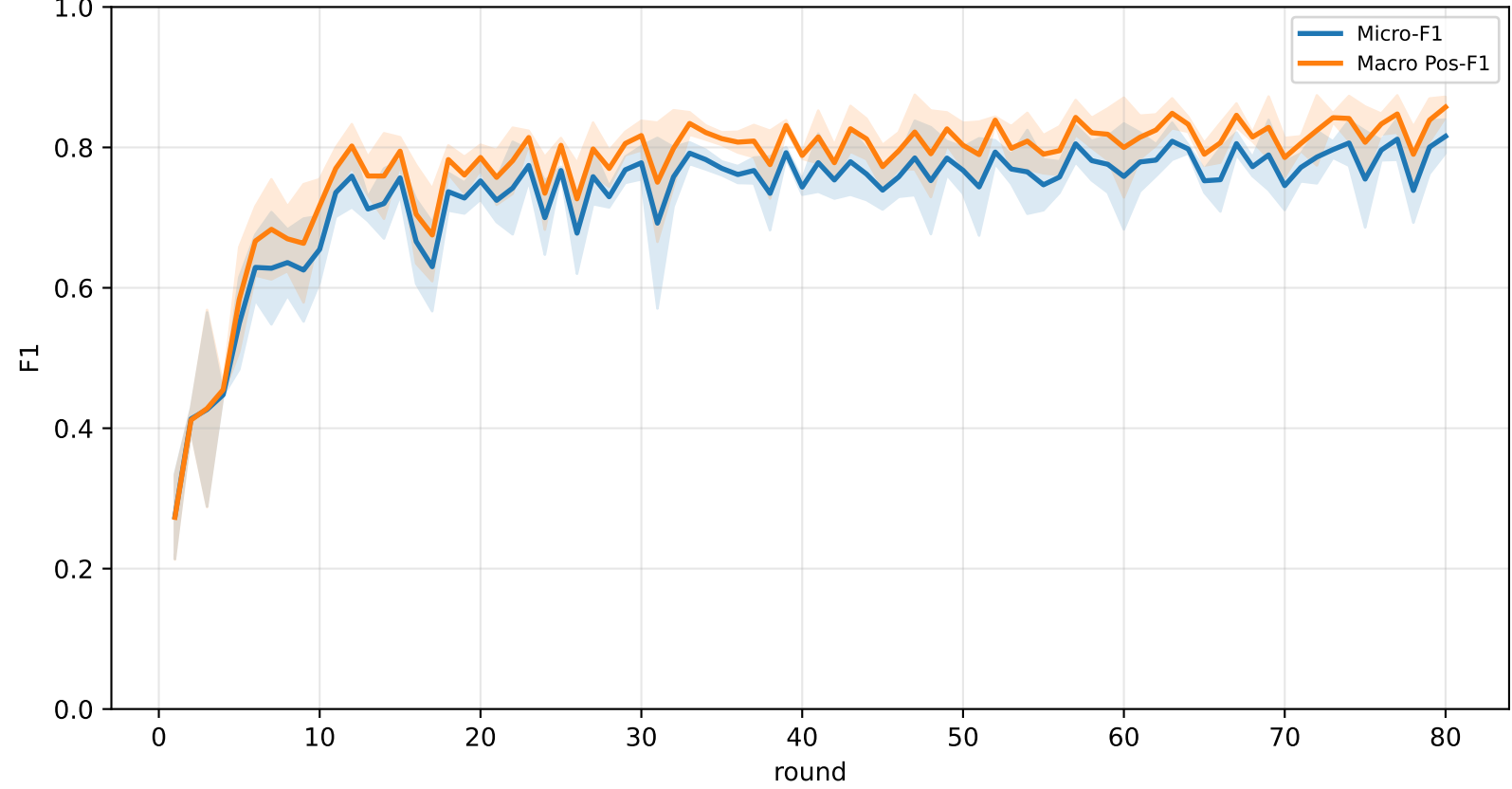
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 82.3% ± 1.4% | 85.7% ± 1.2% | 97.7% ± 1.1%  | 96.2% ± 0.1%  | 89.8% ± 1.6%  | 80.9% ± 4.0%  | 63.8% ± 1.5%  |
| FedAvg            | 78.7% ± 2.2% | 81.3% ± 1.3% | 94.8% ± 2.3%  | 91.9% ± 1.2%  | 79.7% ± 0.4%  | 74.3% ± 3.1%  | 65.7% ± 4.8%  |

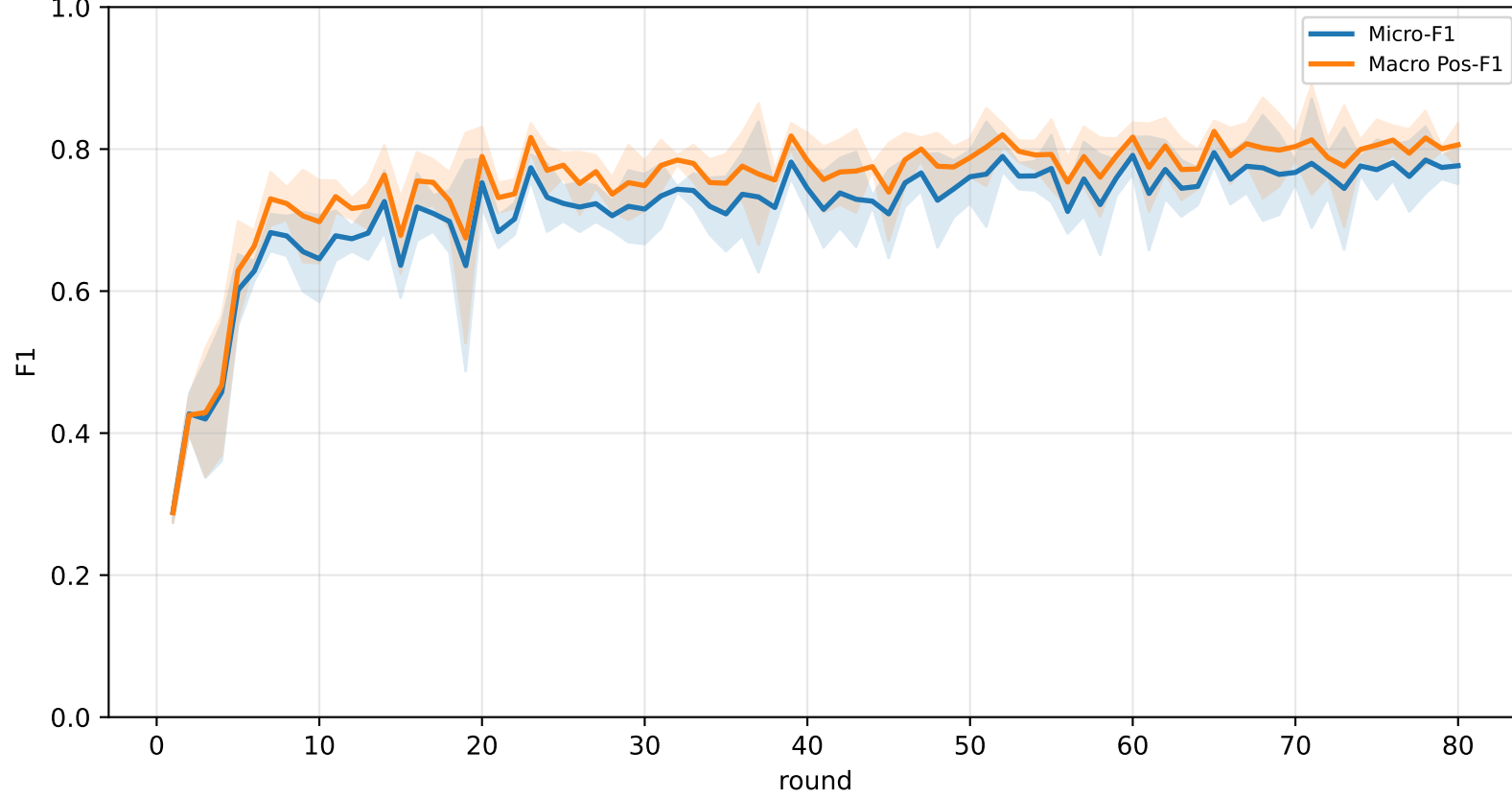
Validation loss comparison



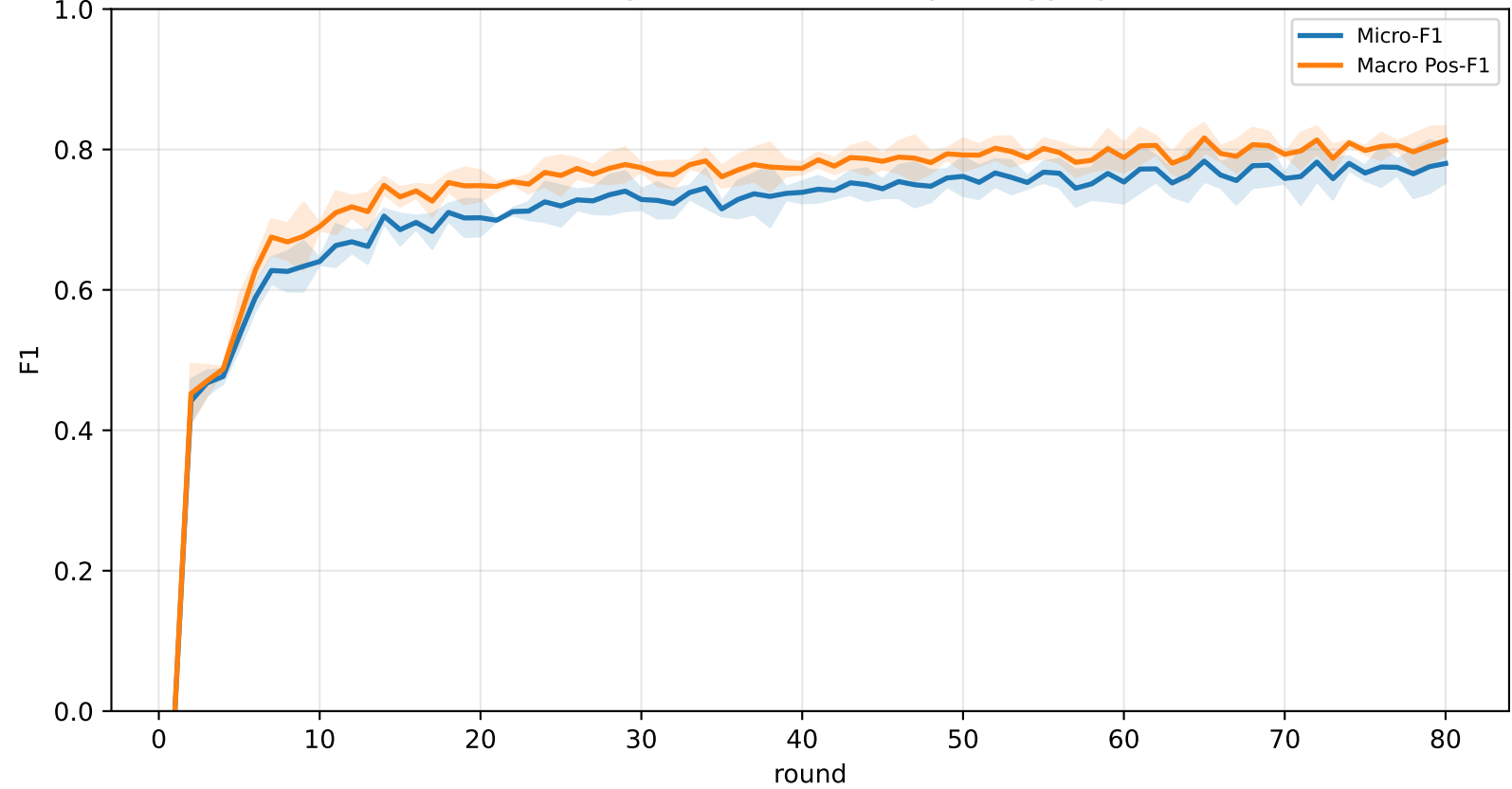
Pooled local centralized micro / macro F1



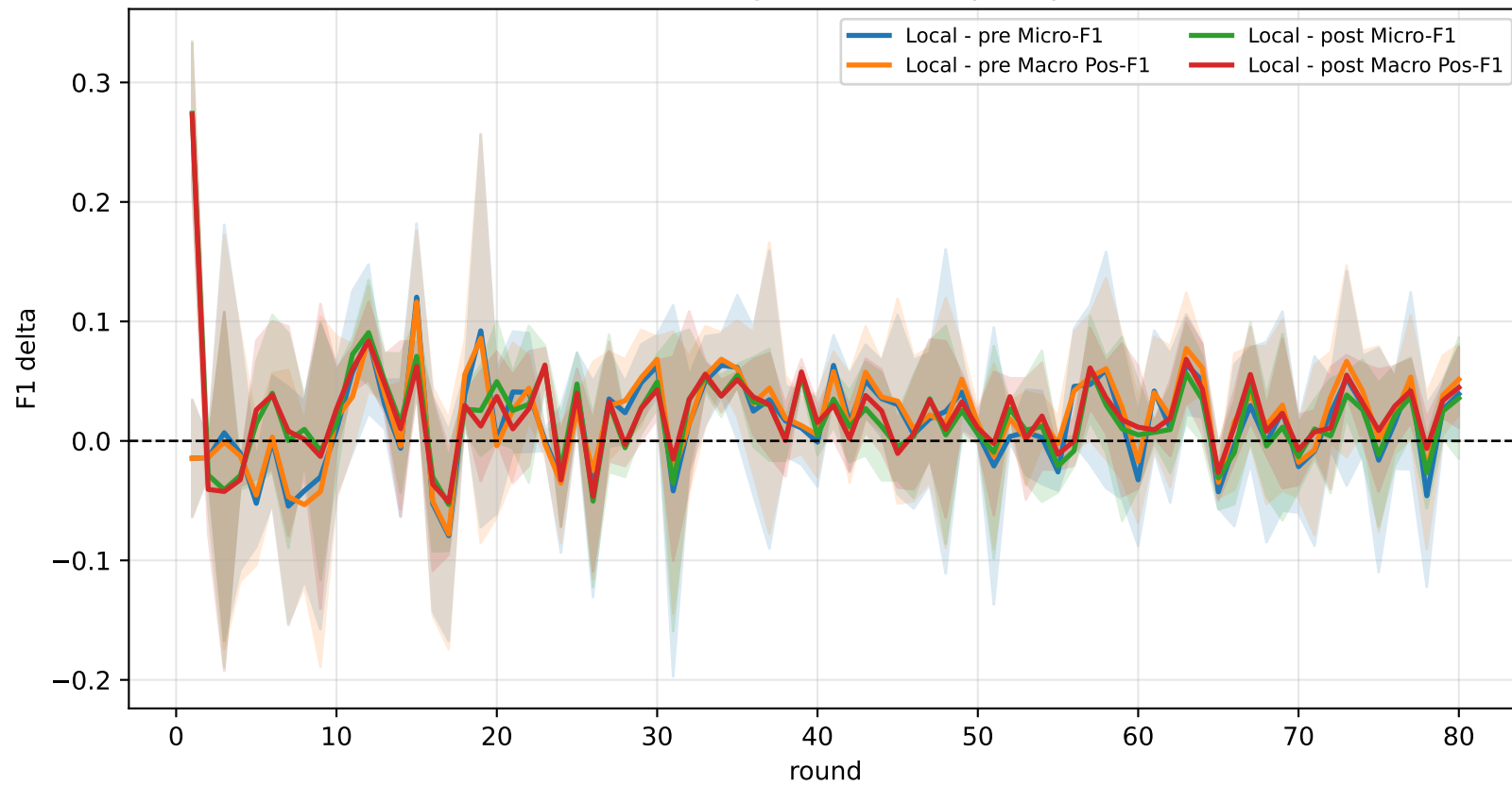
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)

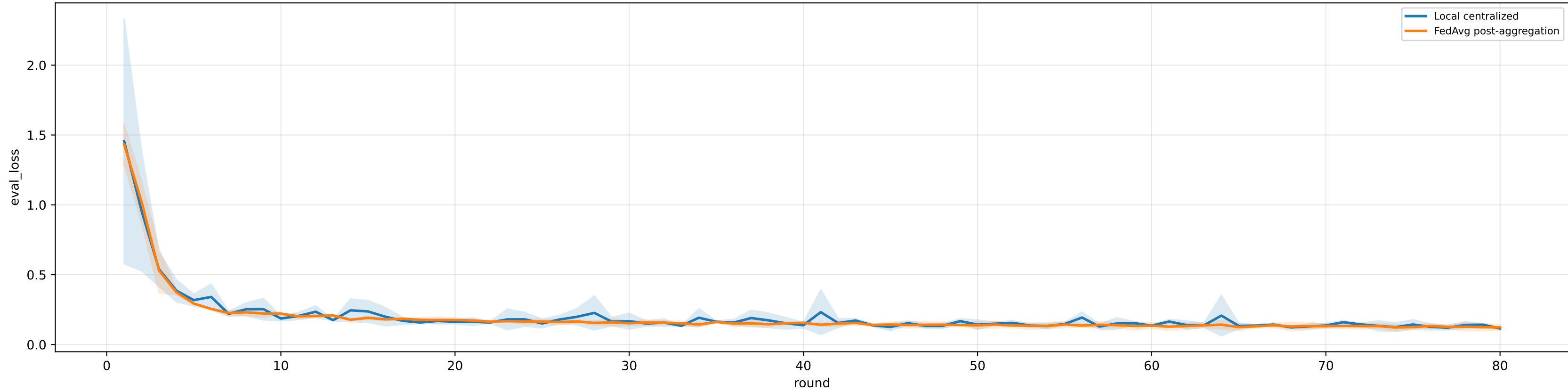


family=specialized\_cycle5 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% | specialization=mid | specialization\_fraction=0.50 | global\_visible\_support=0.60 | task\_jsd\_mean=0.028 | task\_jsd\_median=0.028 | task\_jsd\_max=0.029 | family\_centroid\_jsd\_mean=0.028

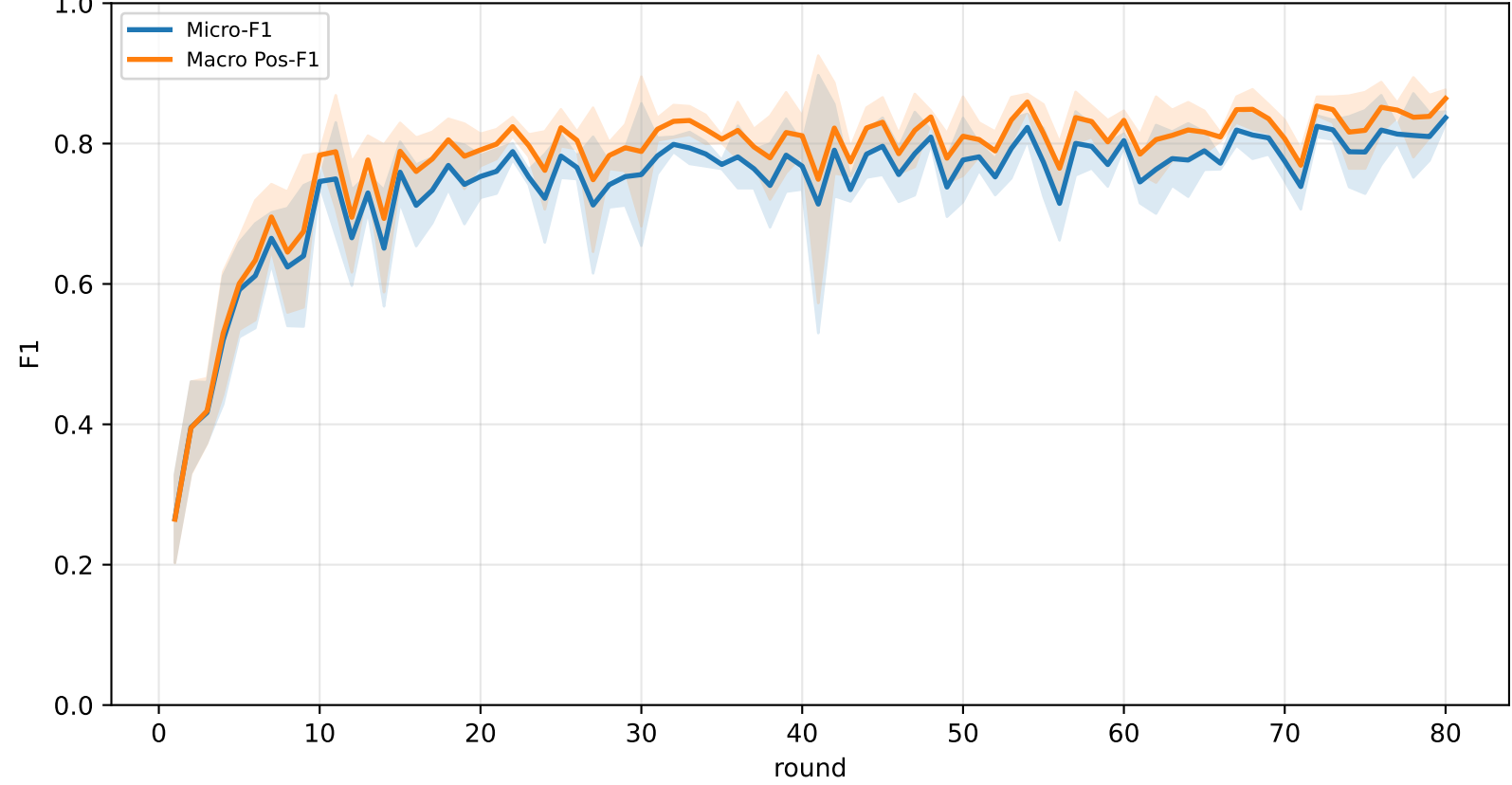
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 82.2% ± 0.8% | 84.9% ± 0.9% | 95.7% ± 1.8%  | 95.1% ± 0.8%  | 86.8% ± 3.3%  | 80.6% ± 1.1%  | 66.1% ± 1.3%  |
| FedAvg            | 78.2% ± 2.1% | 80.7% ± 1.3% | 93.4% ± 1.5%  | 91.9% ± 1.8%  | 78.5% ± 0.4%  | 74.5% ± 3.9%  | 65.4% ± 4.6%  |

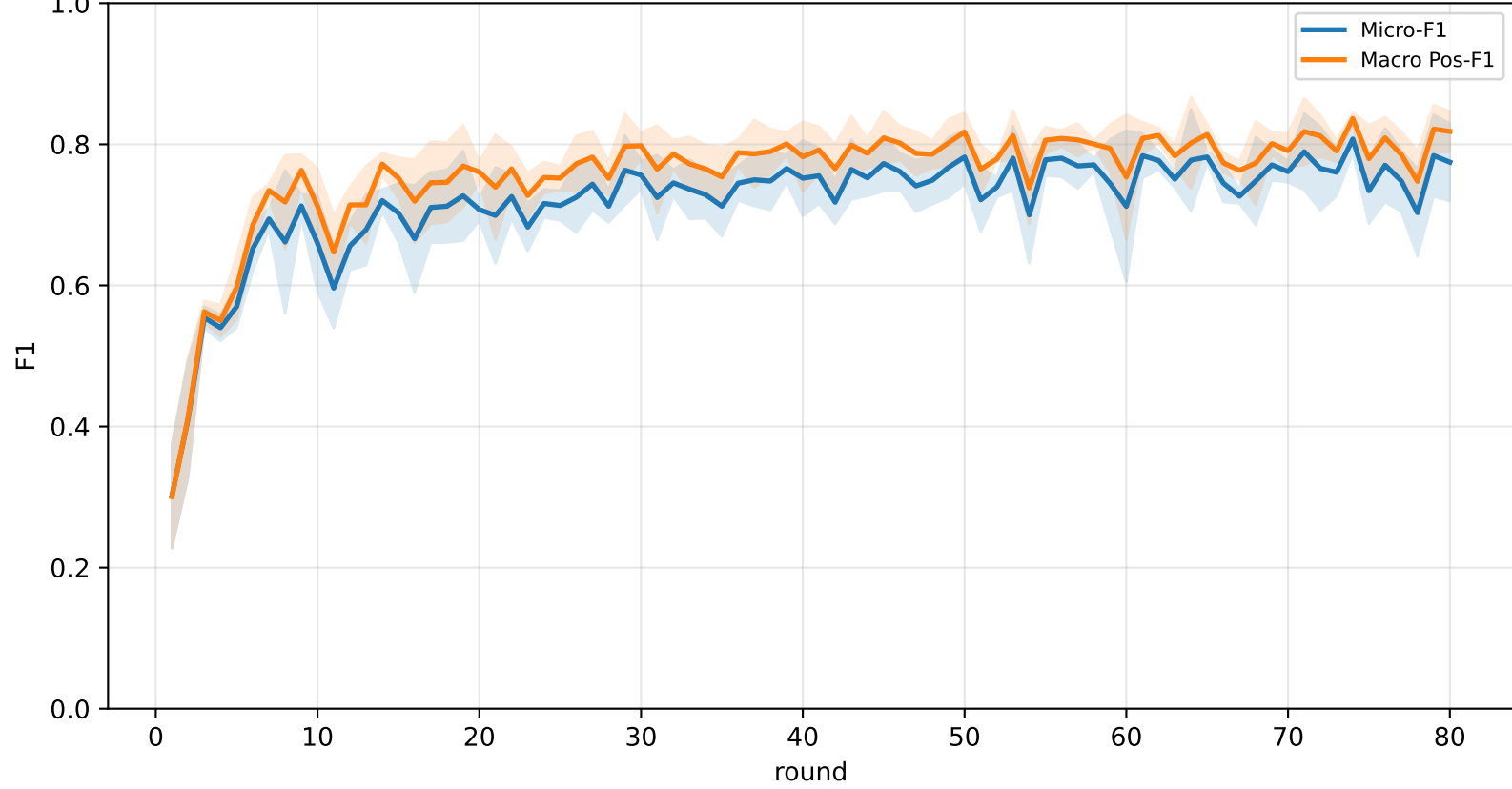
Validation loss comparison



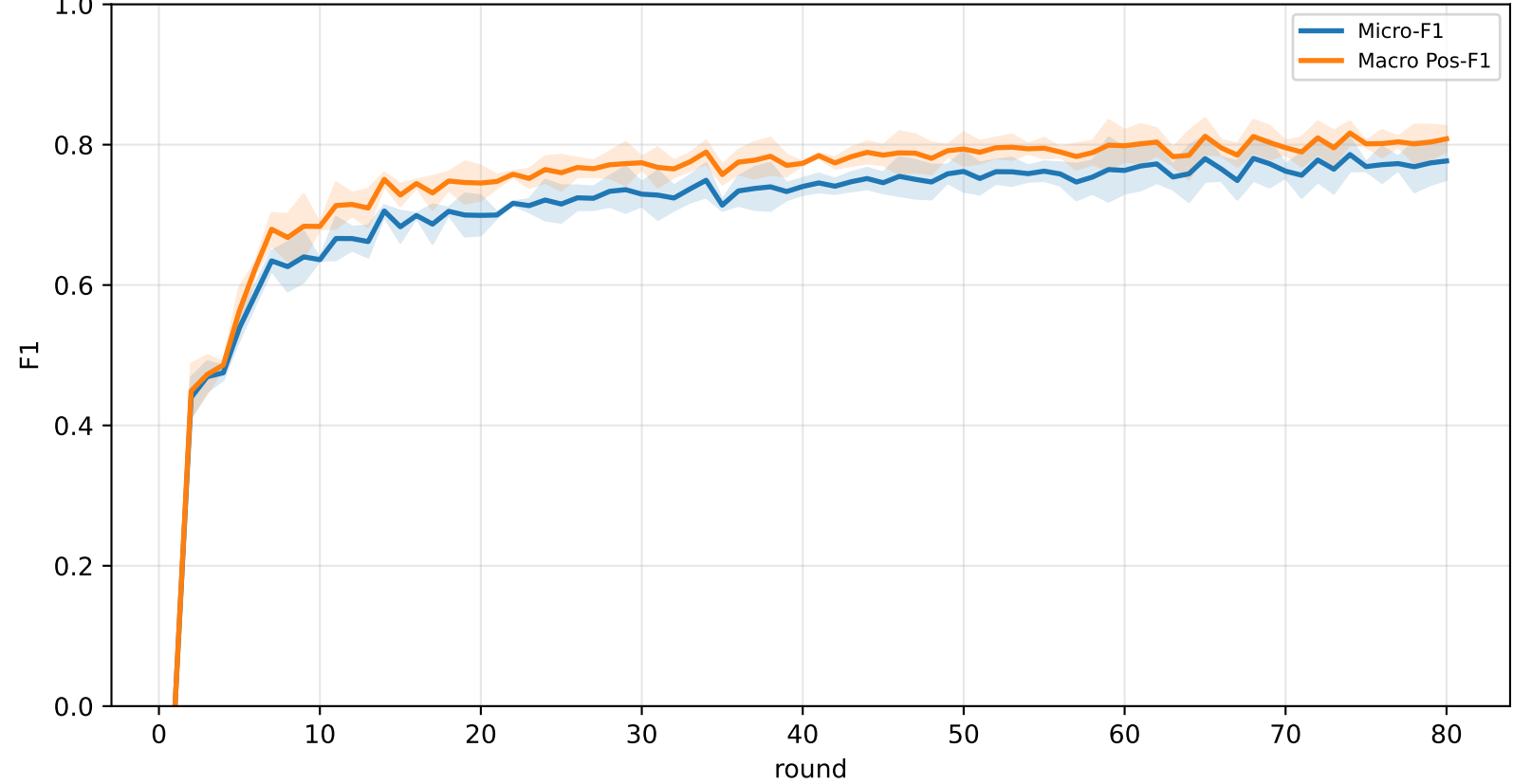
Pooled local centralized micro / macro F1



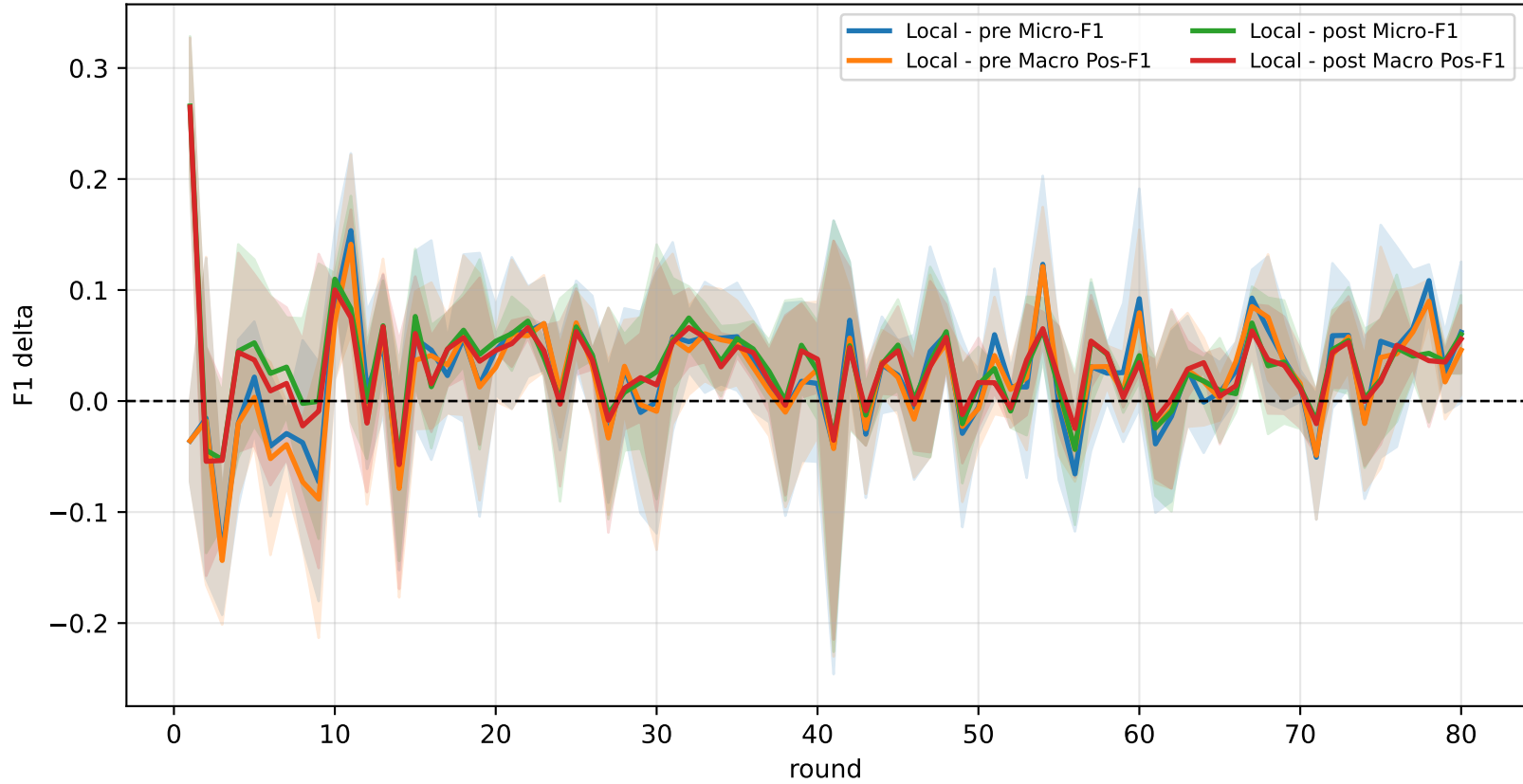
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)



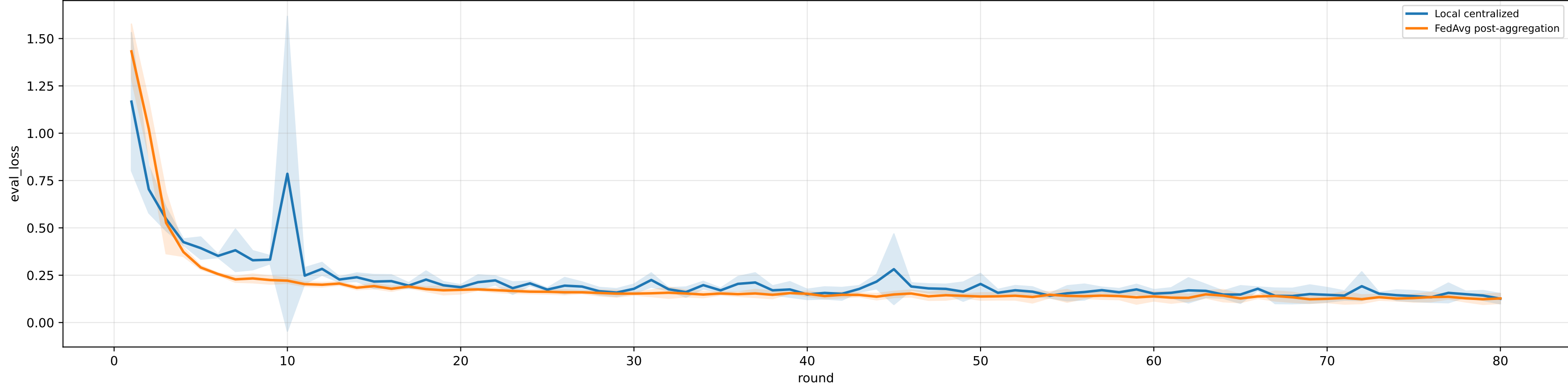


family=specialized\_cycle6 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3% | specialization=mid | specialization\_fraction=0.50 | global\_visible\_support=0.60 | task\_jsd\_mean=0.028 | task\_jsd\_median=0.028 | task\_jsd\_max=0.029 | family\_centroid\_jsd\_mean=0.028

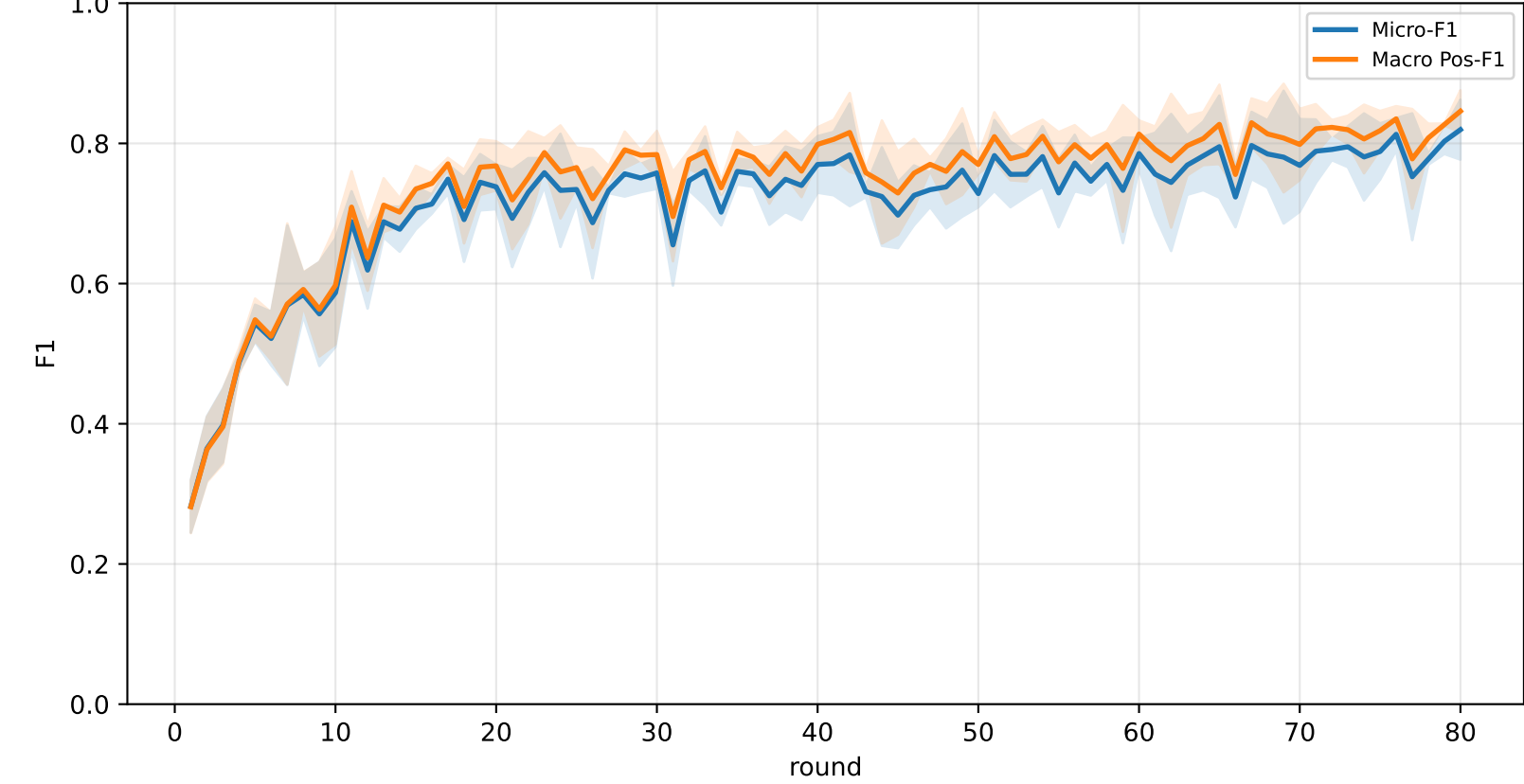
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 81.3% ± 2.7% | 83.7% ± 2.0% | 95.6% ± 1.4%  | 93.8% ± 1.7%  | 84.2% ± 3.5%  | 76.6% ± 4.7%  | 68.5% ± 6.6%  |
| FedAvg            | 79.3% ± 2.6% | 81.9% ± 1.9% | 94.6% ± 1.4%  | 93.3% ± 2.5%  | 80.0% ± 1.1%  | 75.2% ± 3.1%  | 66.1% ± 4.7%  |

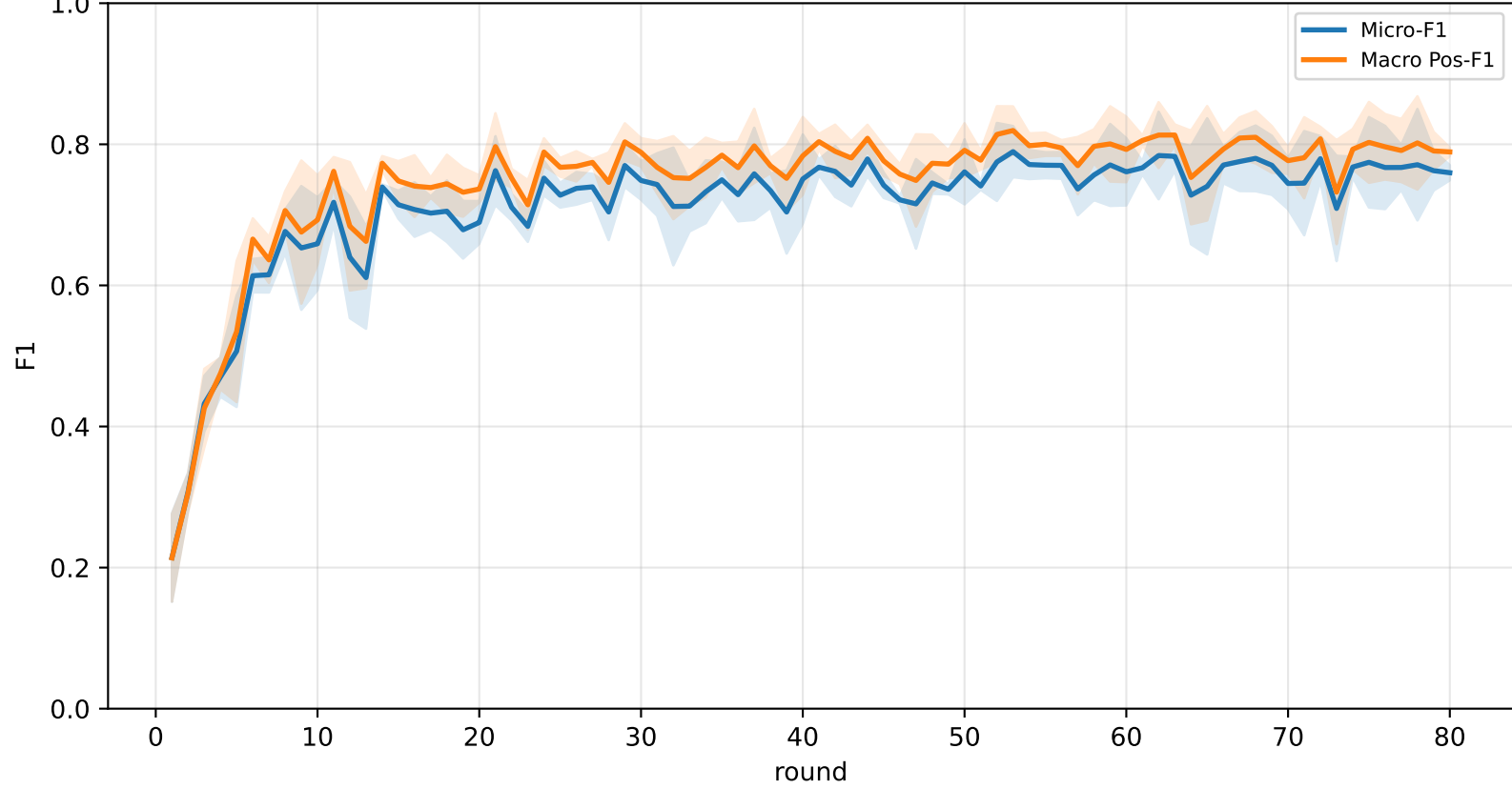
Validation loss comparison



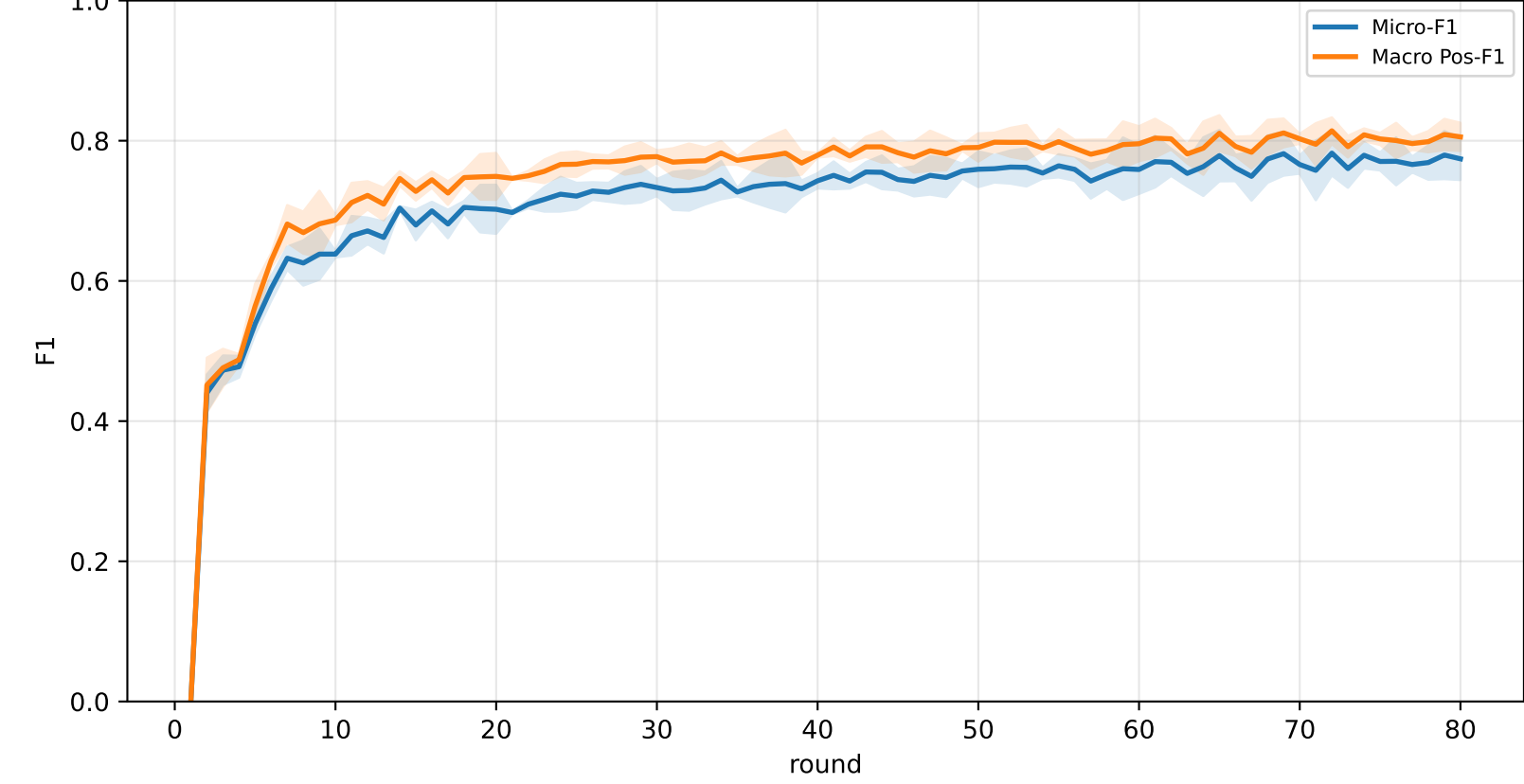
Pooled local centralized micro / macro F1



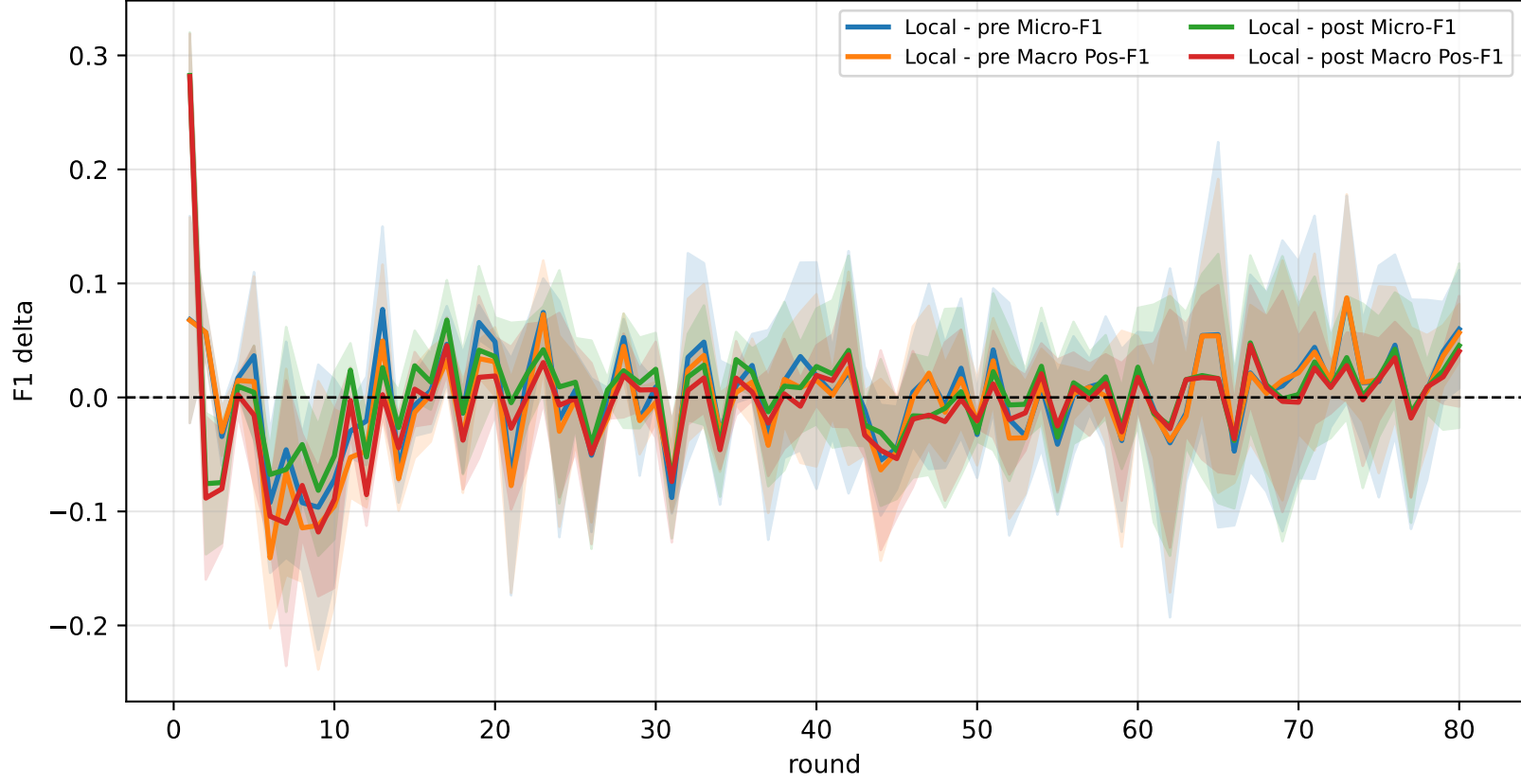
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)





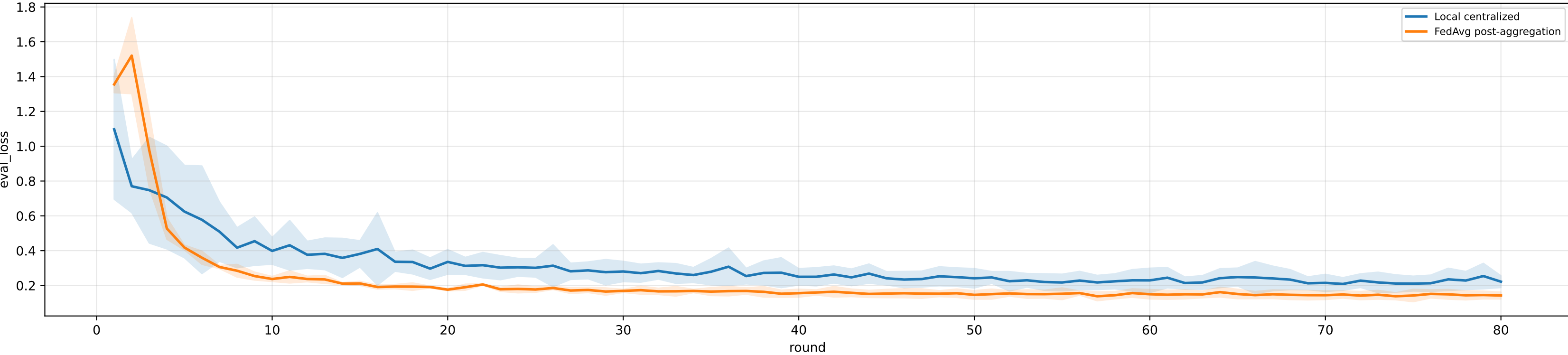
Controlled task-specialization level: High | FedAvg diagnostics | aggregated / pooled over 5 clients | model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64

family=specialized\_cycle2 | n\_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=specialized\_cycle3 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=specialized\_cycle4 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=specialized\_cycle5 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=specialized\_cycle6 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%  
specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.36 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 | family\_centroid\_jsd\_mean=0.162

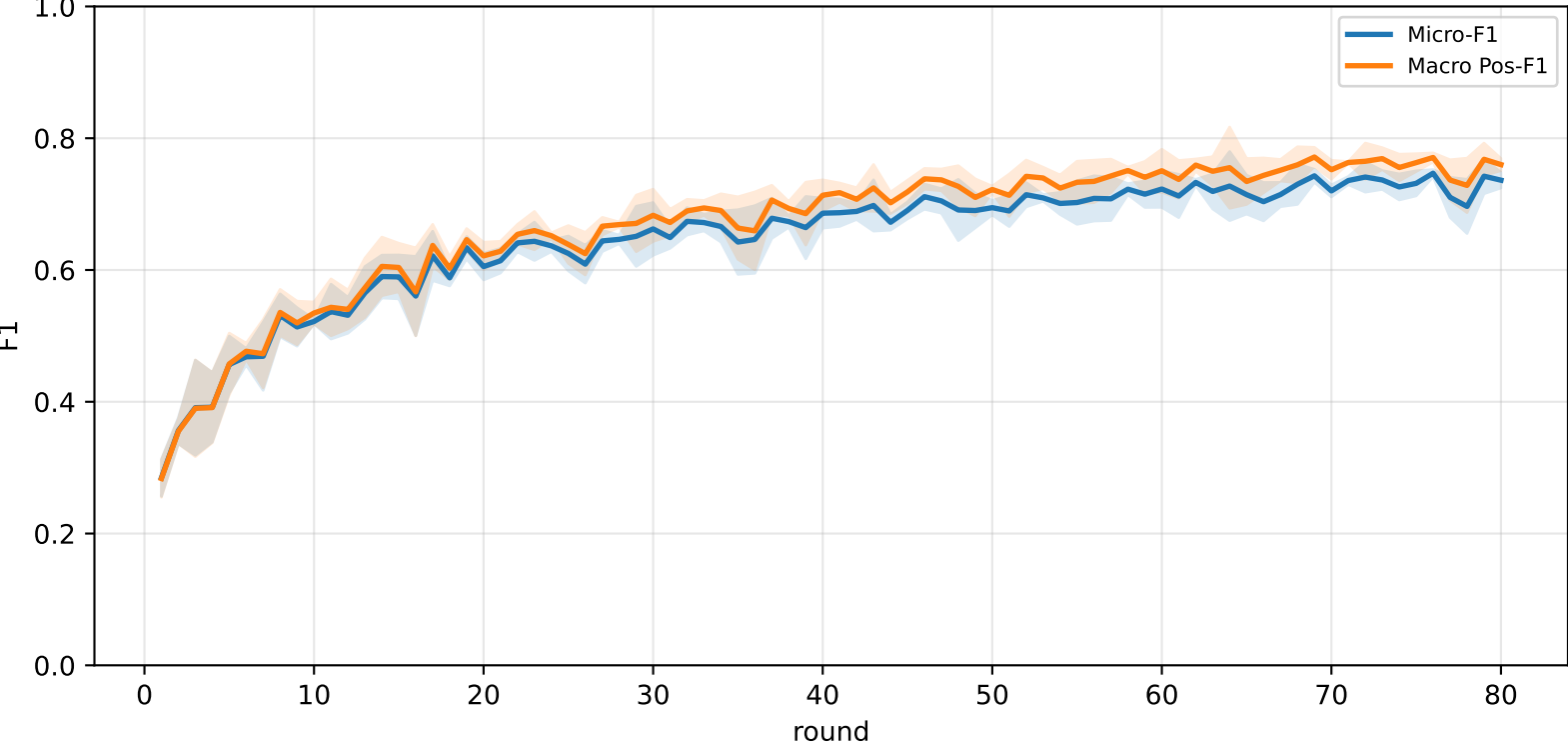
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 74.7% ± 1.4% | 78.7% ± 1.0% | 91.5% ± 2.6%  | 89.1% ± 0.9%  | 80.9% ± 1.9%  | 68.6% ± 3.9%  | 58.8% ± 1.7%  |
| FedAvg            | 76.8% ± 1.8% | 80.5% ± 1.9% | 95.1% ± 1.3%  | 92.3% ± 3.1%  | 81.6% ± 1.4%  | 74.6% ± 3.1%  | 59.0% ± 1.0%  |

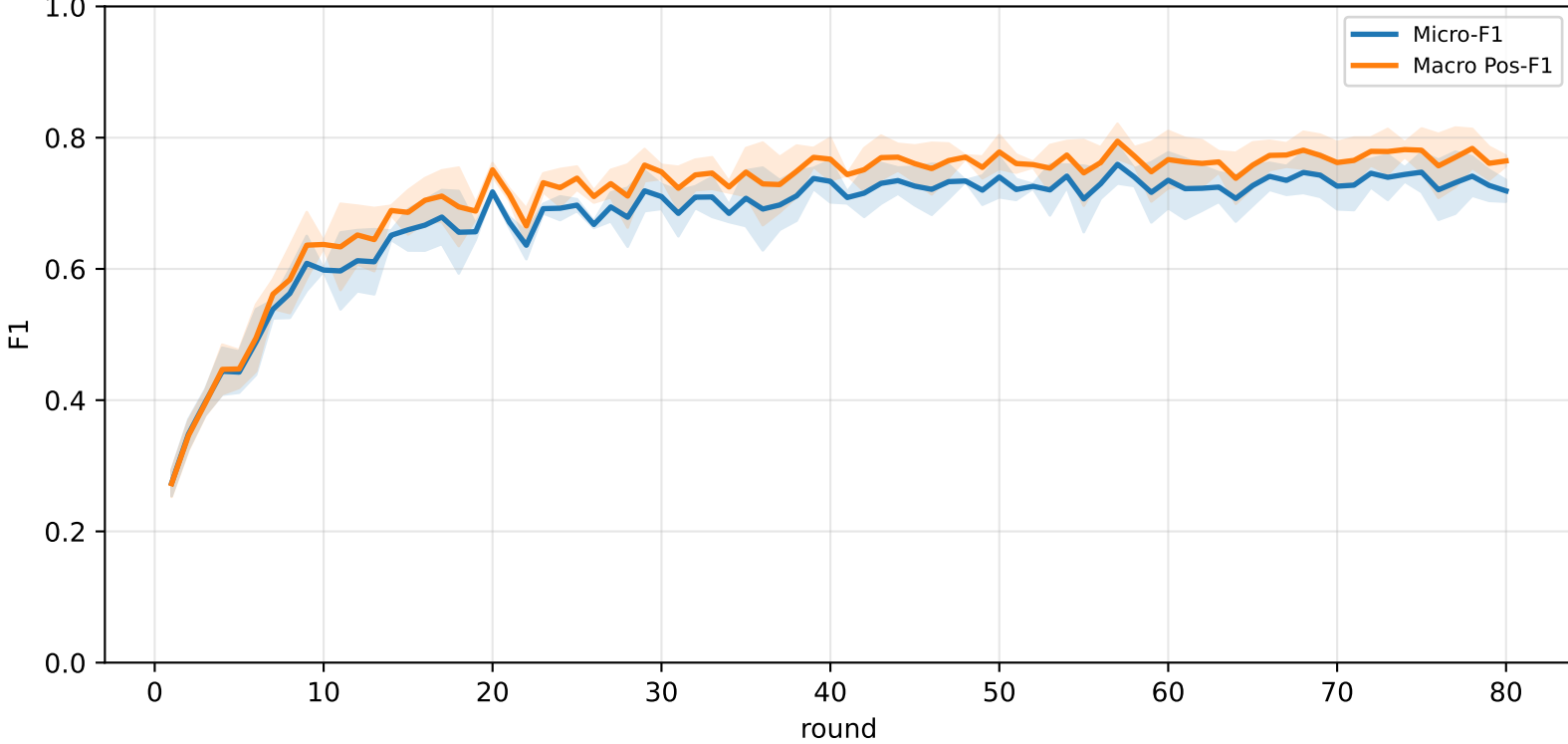
Validation loss comparison



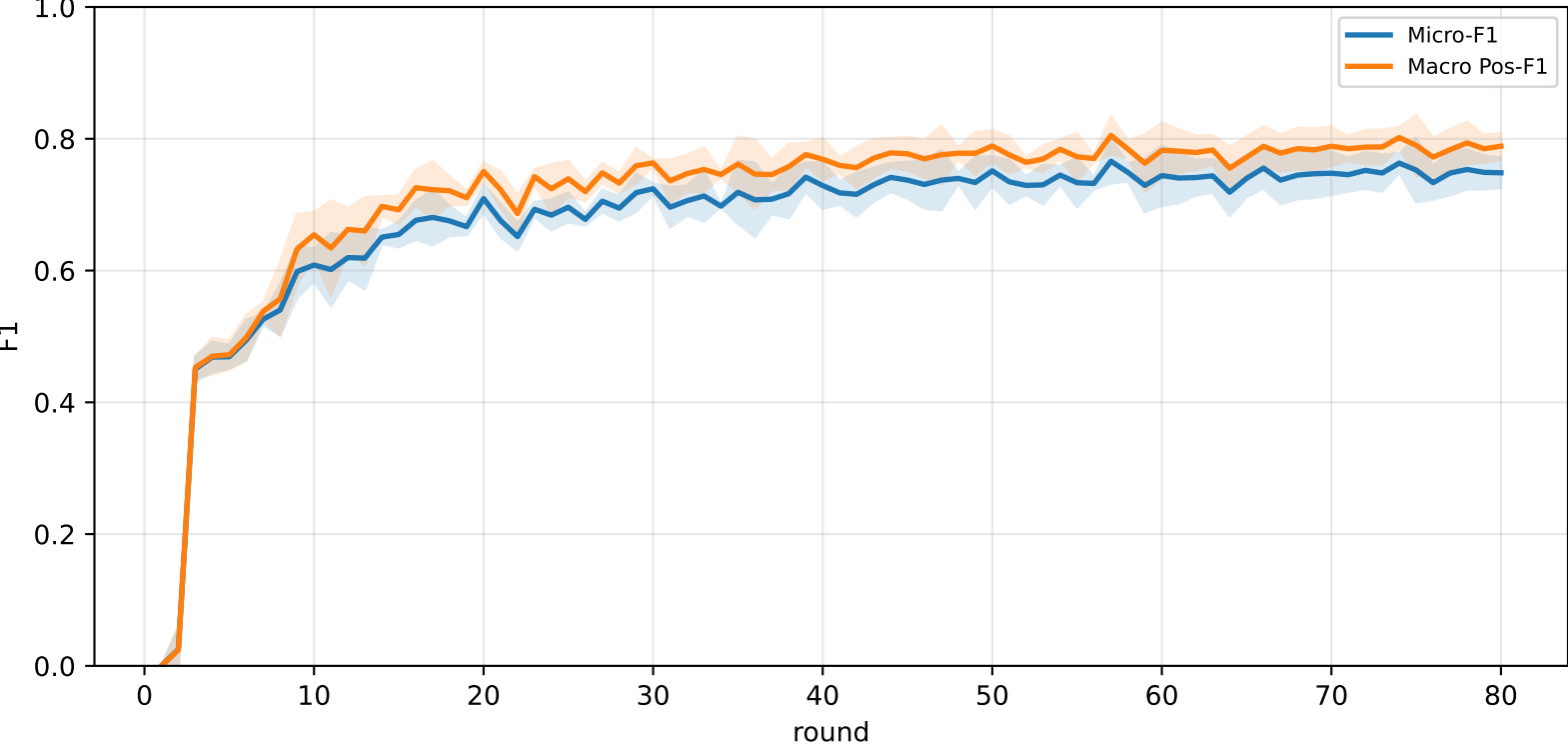
Pooled local centralized micro / macro F1



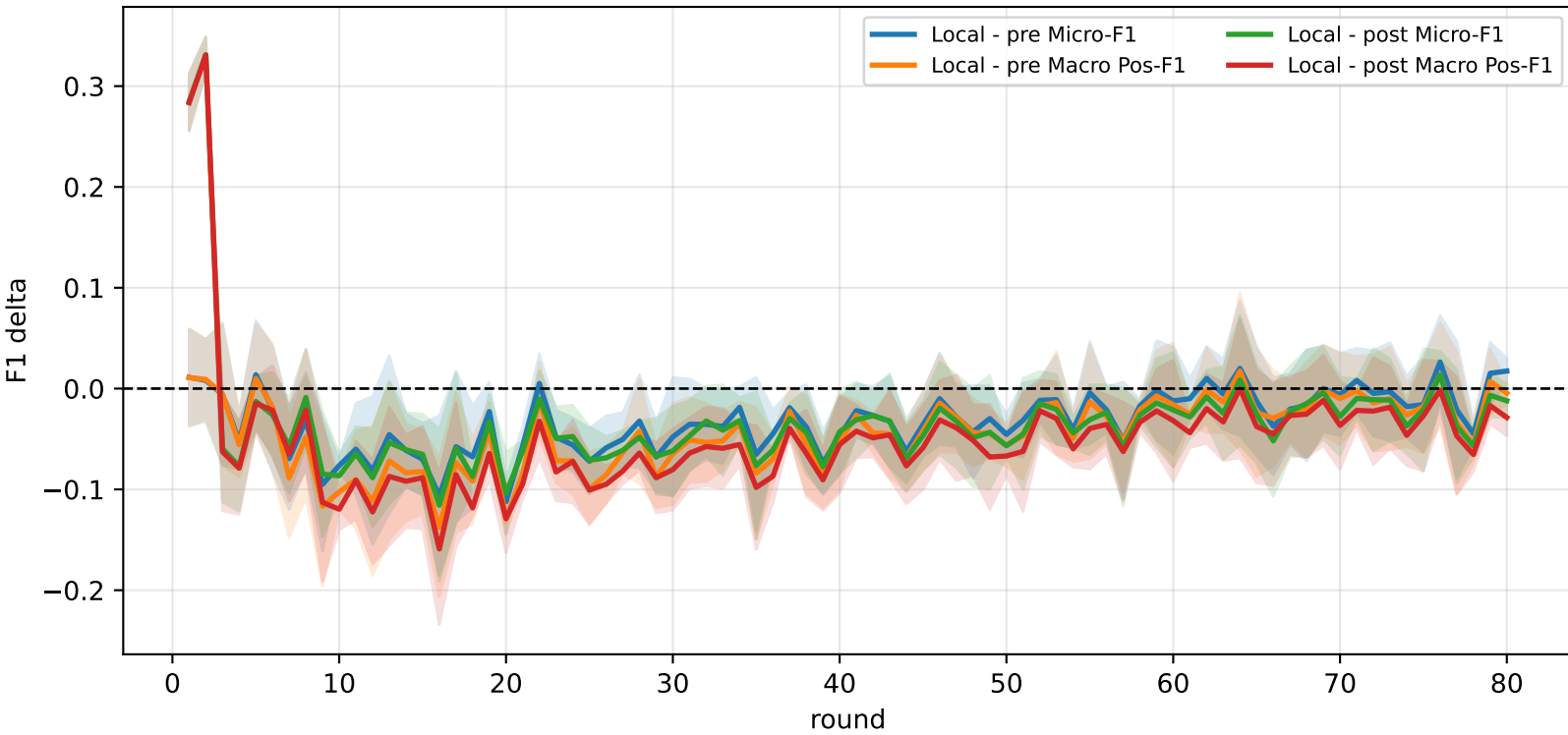
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)

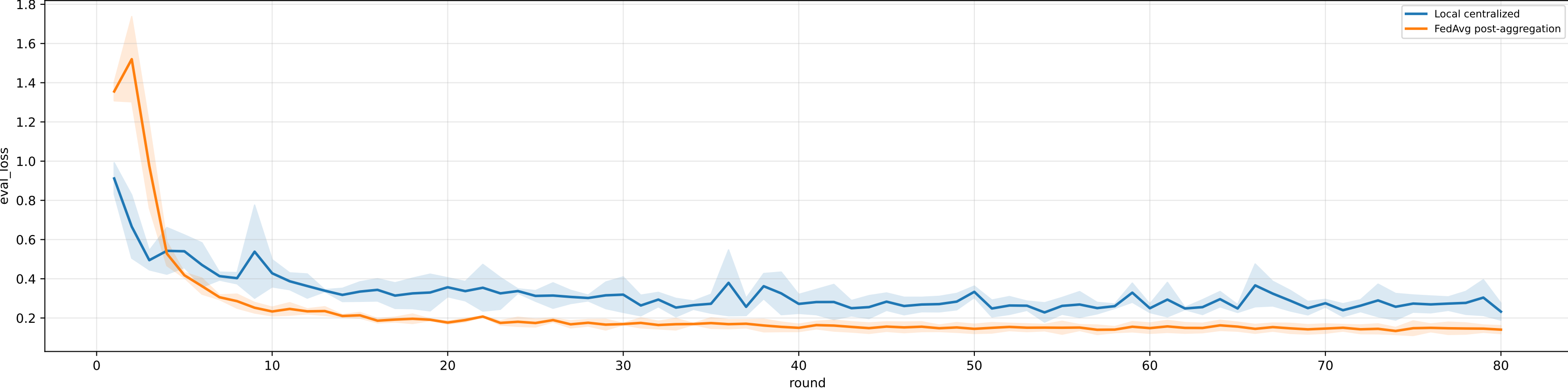


family=specialized\_cycle2 | n\_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.36 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 | family\_centroid\_jsd\_mean=0.162

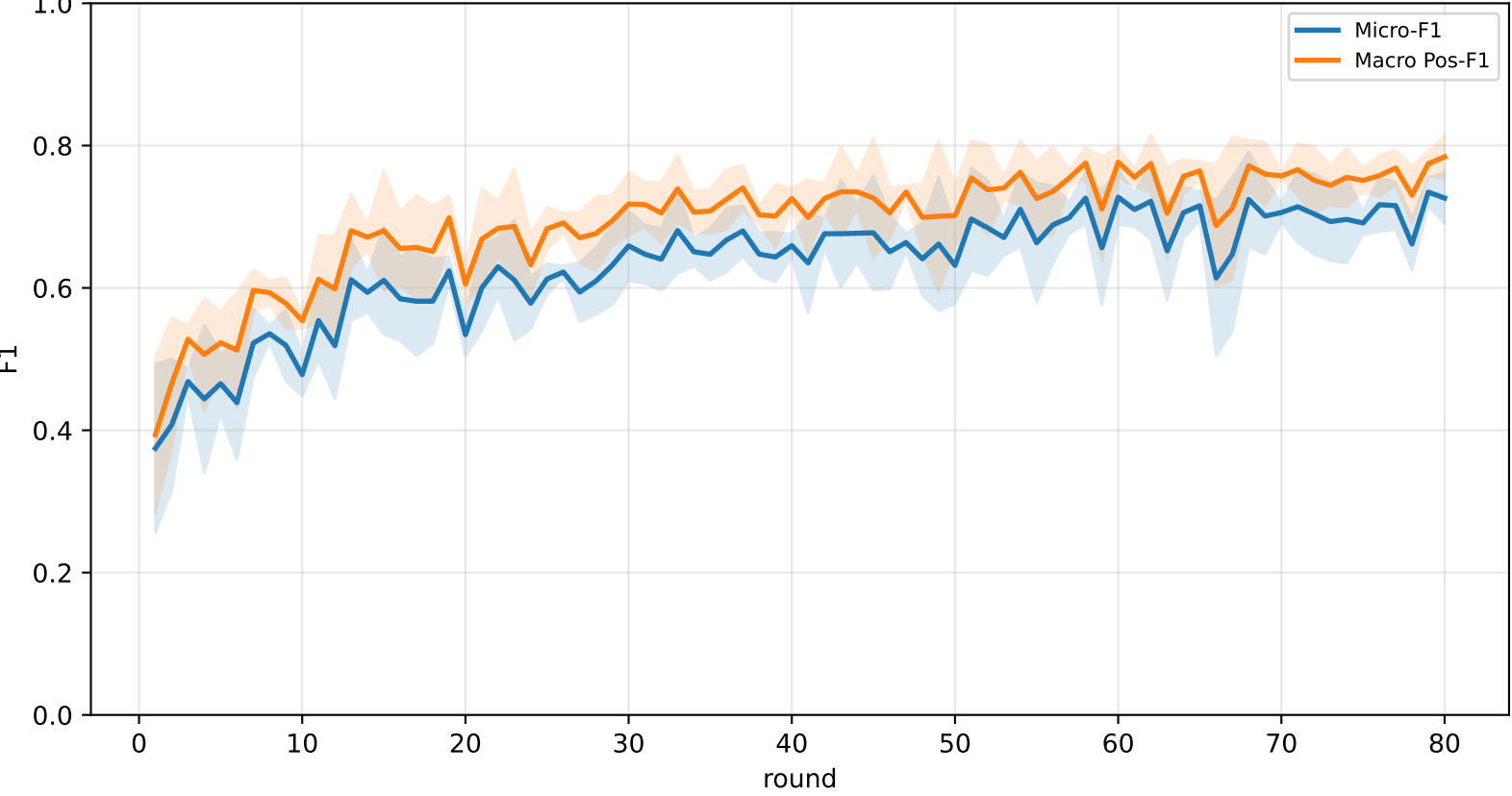
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 71.8% ± 2.2% | 76.9% ± 0.8% | 98.1% ± 0.8%  | 90.4% ± 2.4%  | 76.9% ± 4.5%  | 65.2% ± 5.3%  | 53.7% ± 4.1%  |
| FedAvg            | 77.0% ± 1.8% | 80.7% ± 1.8% | 95.6% ± 0.6%  | 92.2% ± 3.4%  | 81.6% ± 1.2%  | 74.7% ± 2.9%  | 59.3% ± 1.1%  |

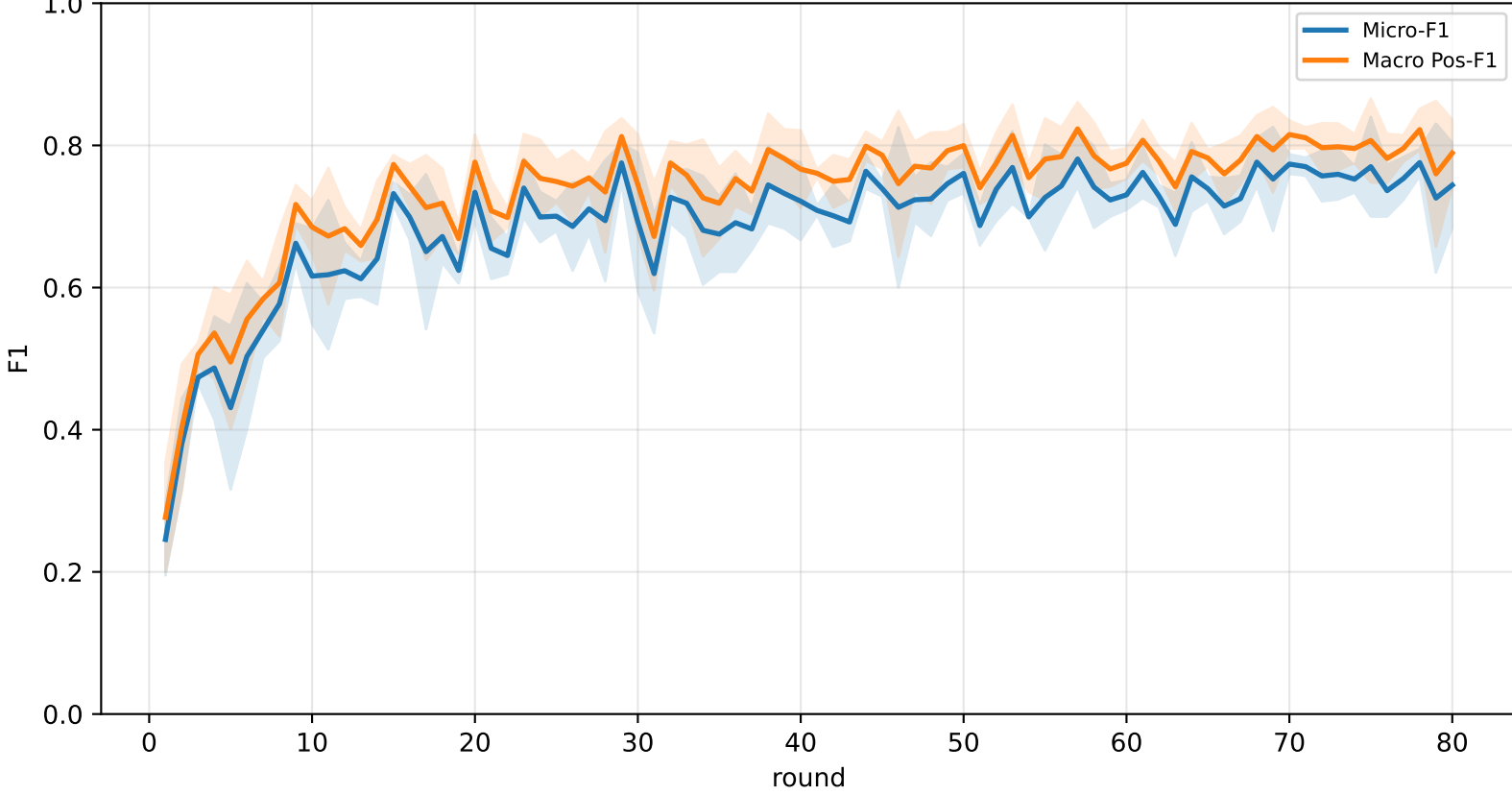
Validation loss comparison



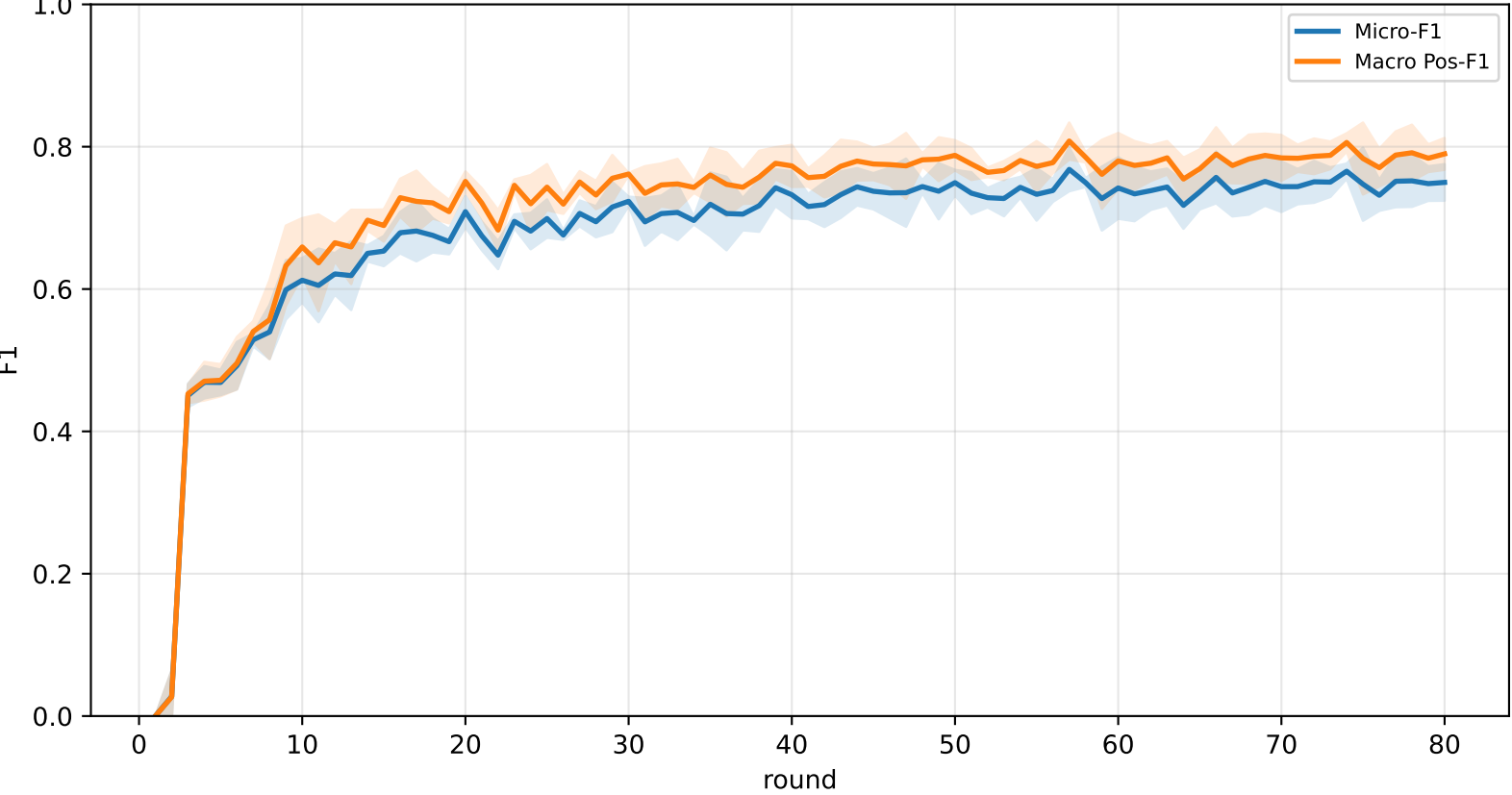
Pooled local centralized micro / macro F1



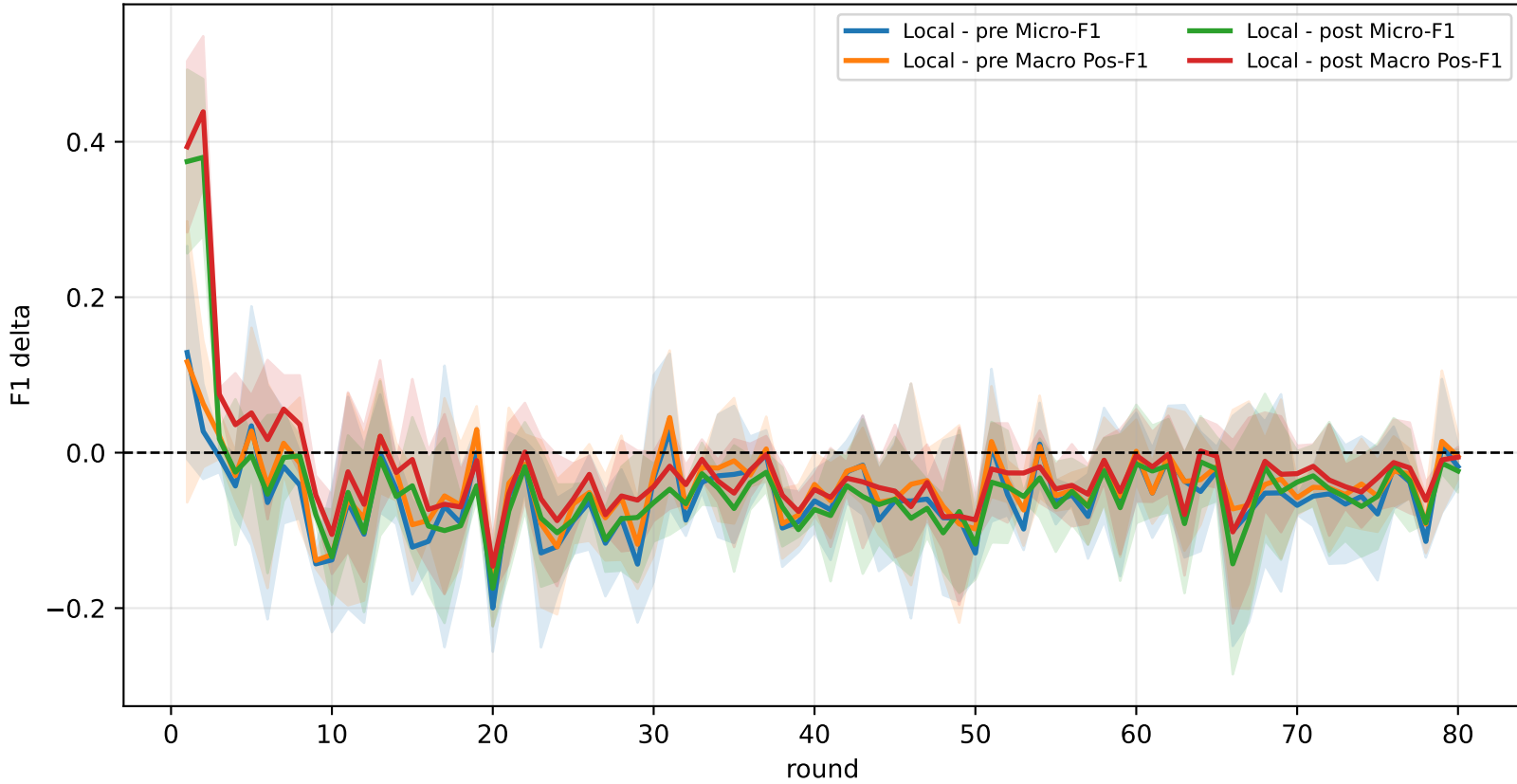
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)

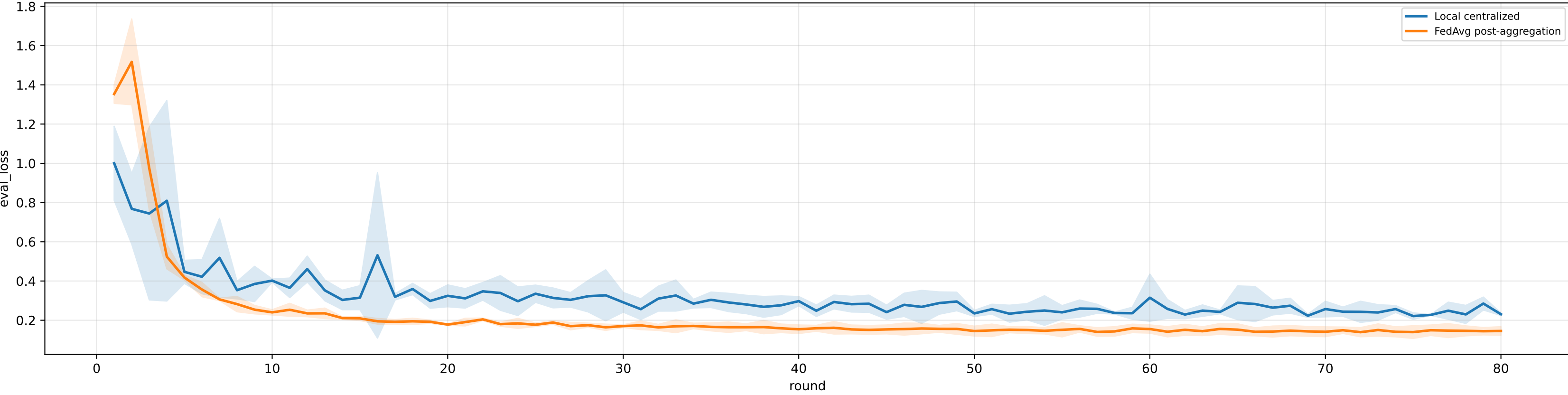


family=specialized\_cycle3 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.36 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 |  
family\_centroid\_jsd\_mean=0.162

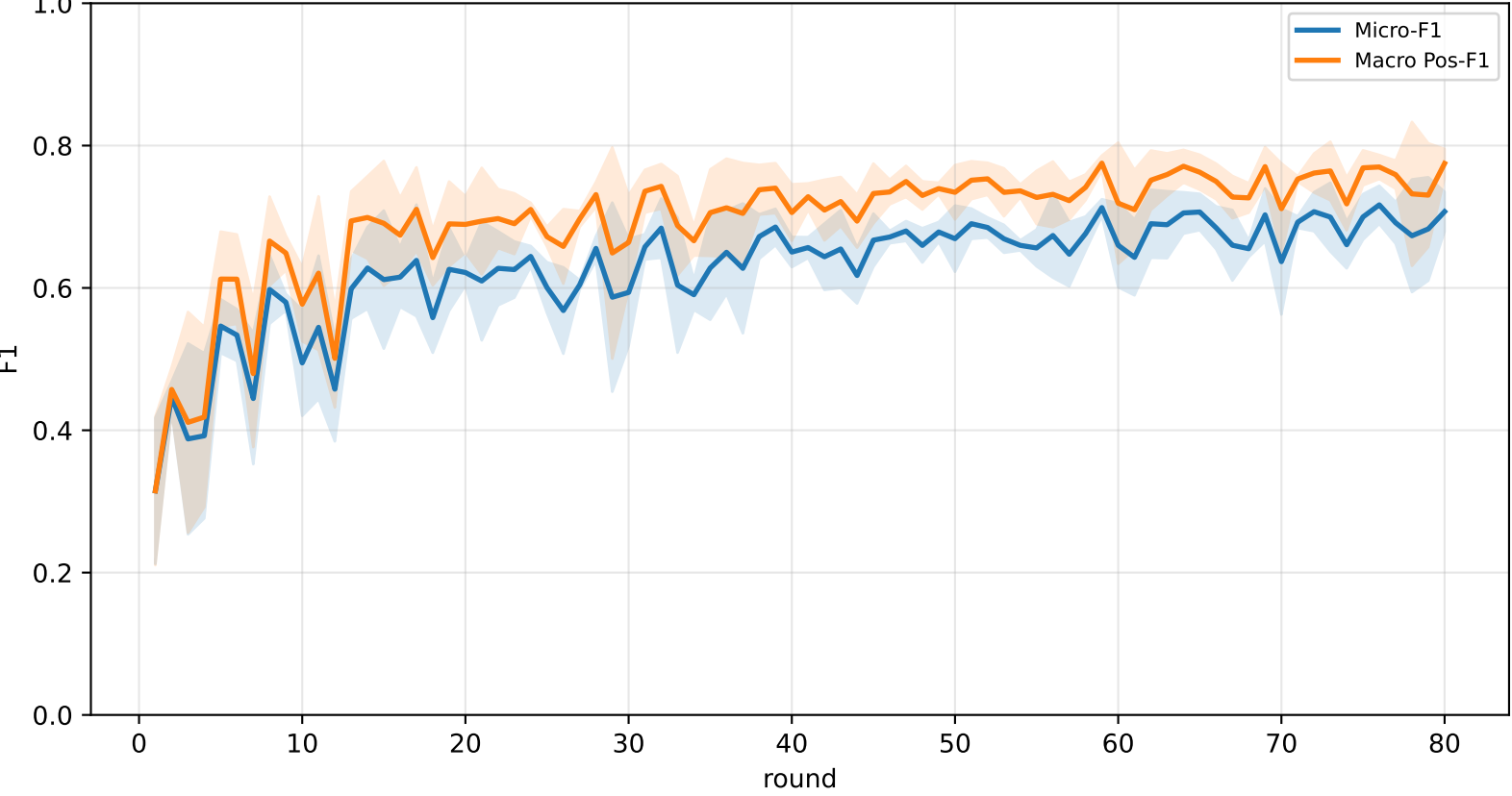
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 69.6% ± 3.7% | 76.6% ± 3.0% | 96.4% ± 1.1%  | 94.7% ± 1.9%  | 86.7% ± 5.4%  | 52.7% ± 5.2%  | 52.7% ± 4.1%  |
| FedAvg            | 76.9% ± 1.3% | 80.7% ± 1.4% | 94.9% ± 1.8%  | 93.0% ± 2.3%  | 82.0% ± 0.5%  | 74.4% ± 2.1%  | 59.0% ± 1.0%  |

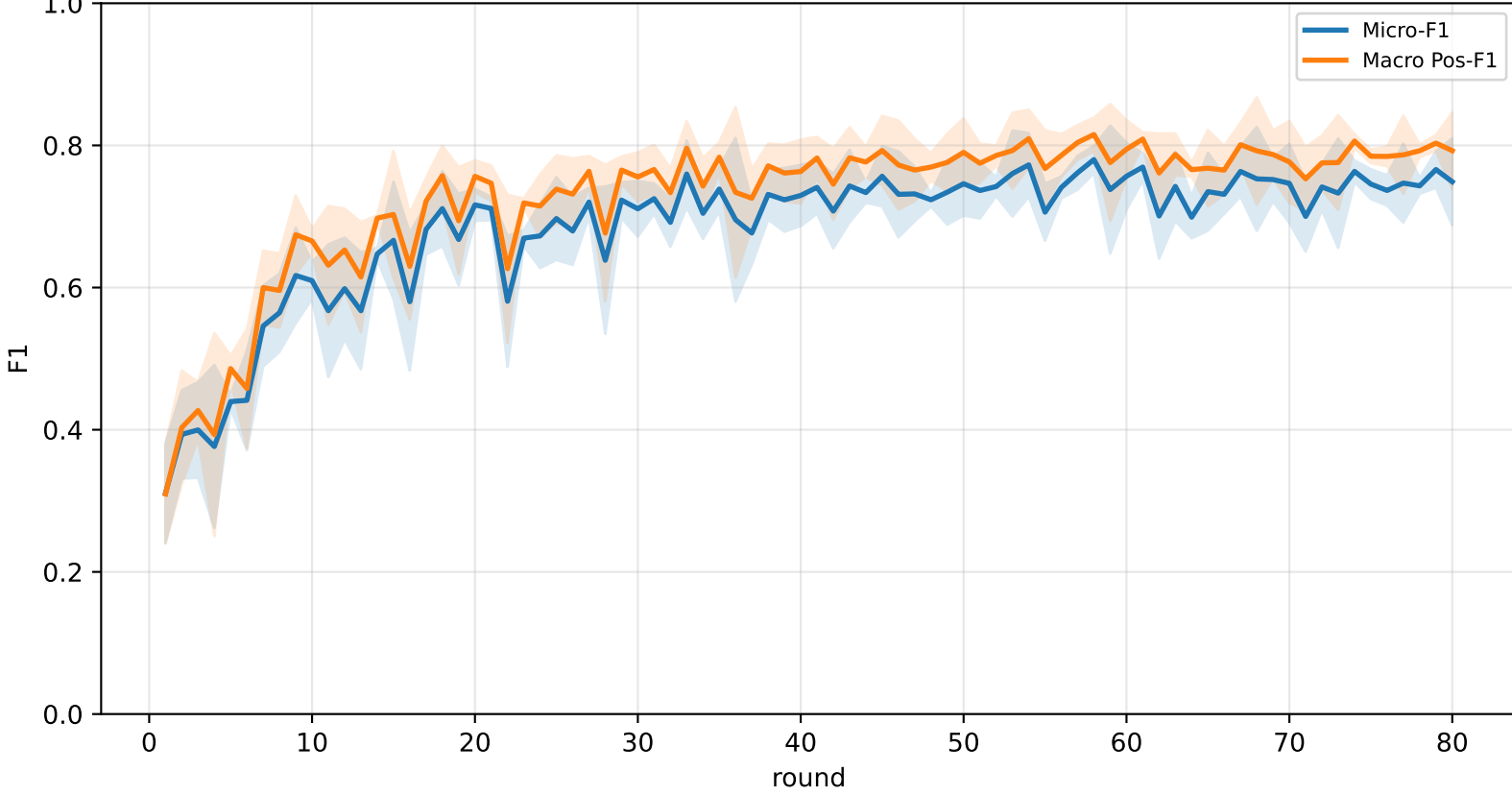
Validation loss comparison



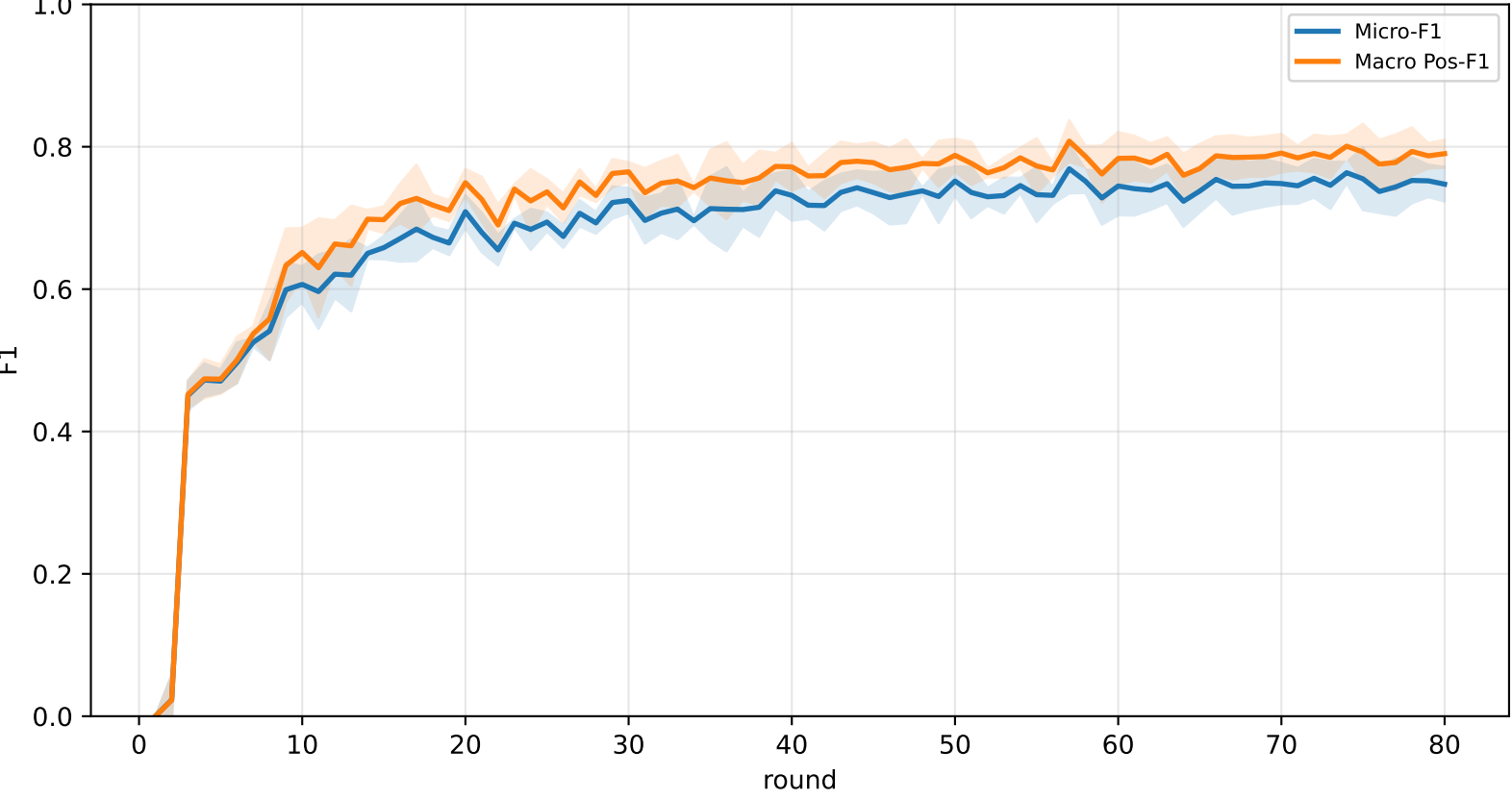
Pooled local centralized micro / macro F1



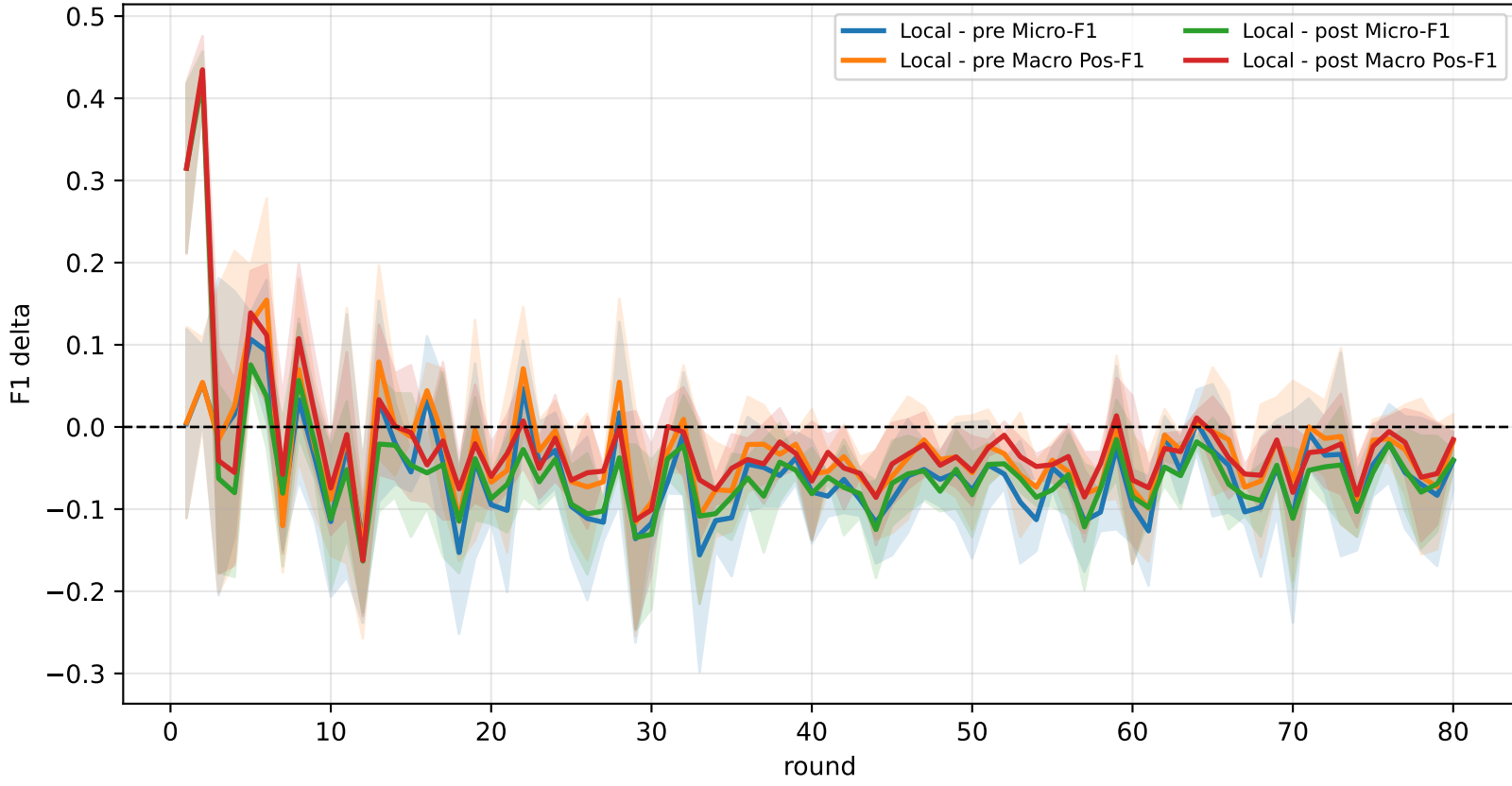
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)



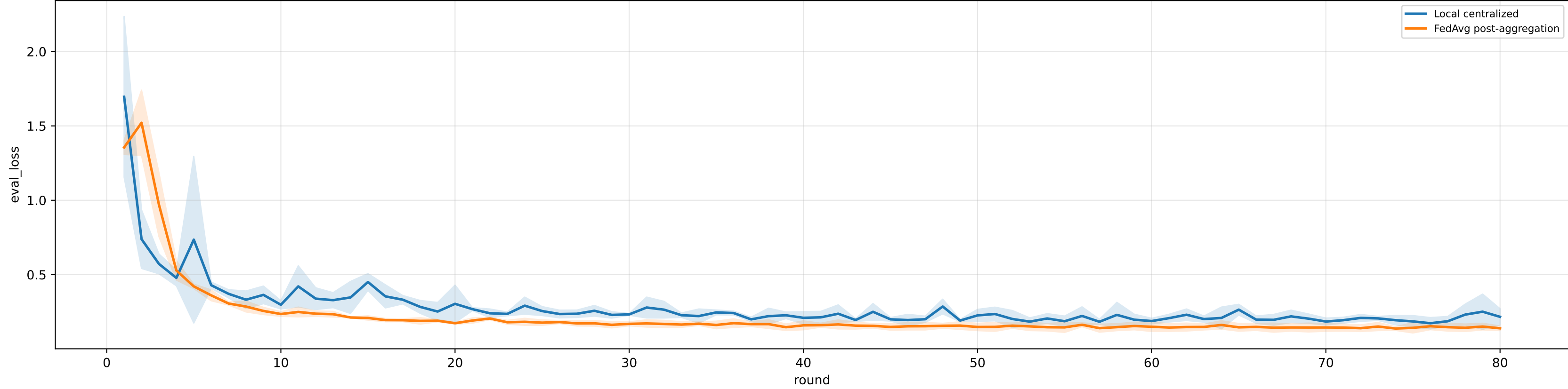


family=specialized\_cycle4 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.36 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 |  
family\_centroid\_jsd\_mean=0.162

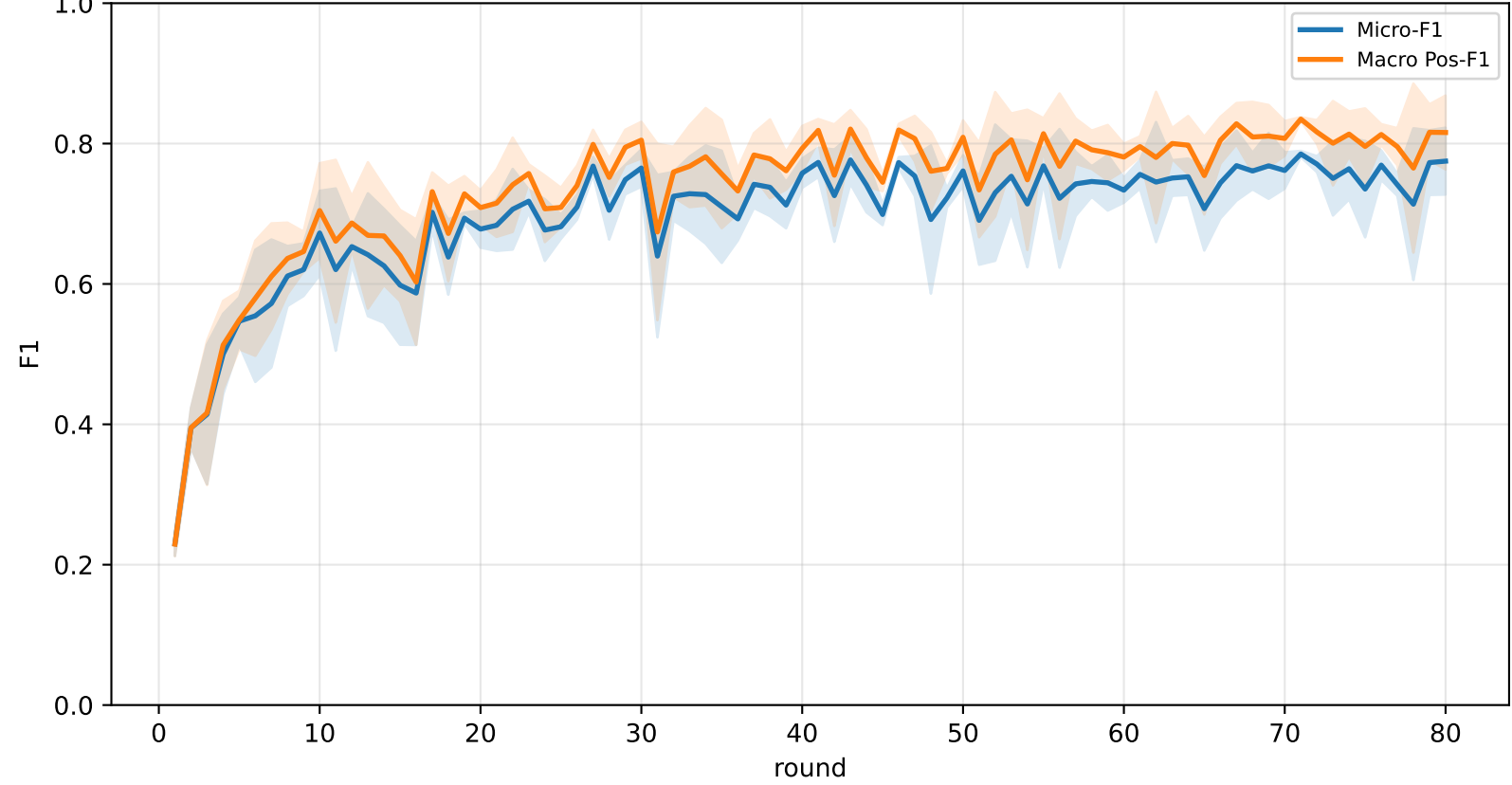
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 78.0% ± 1.1% | 82.8% ± 1.1% | 95.4% ± 2.6%  | 95.6% ± 1.0%  | 91.0% ± 2.8%  | 76.7% ± 2.4%  | 55.2% ± 0.7%  |
| FedAvg            | 76.7% ± 1.9% | 80.5% ± 1.9% | 95.2% ± 0.1%  | 92.3% ± 3.6%  | 81.3% ± 1.8%  | 74.7% ± 3.4%  | 59.0% ± 1.3%  |

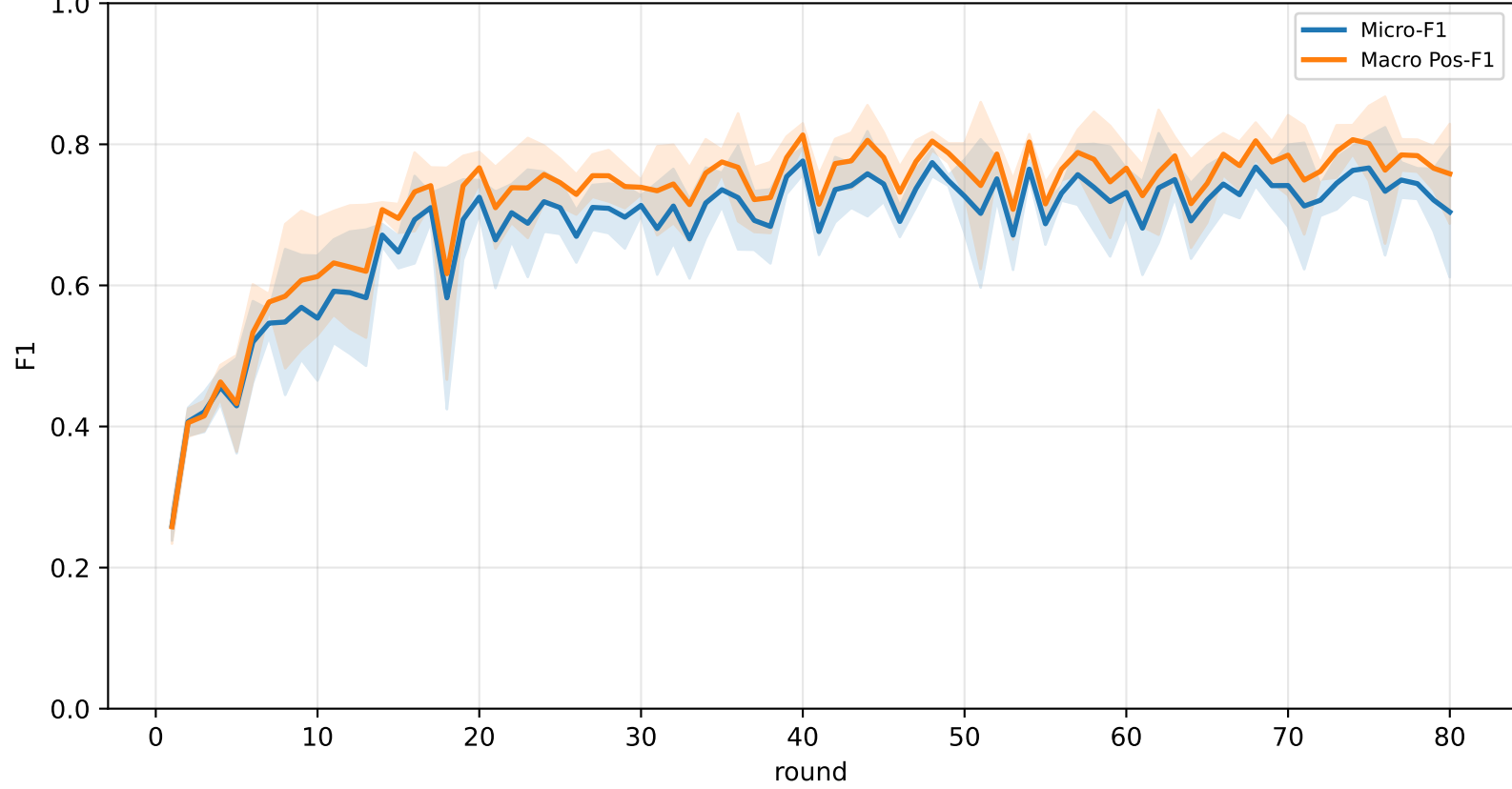
Validation loss comparison



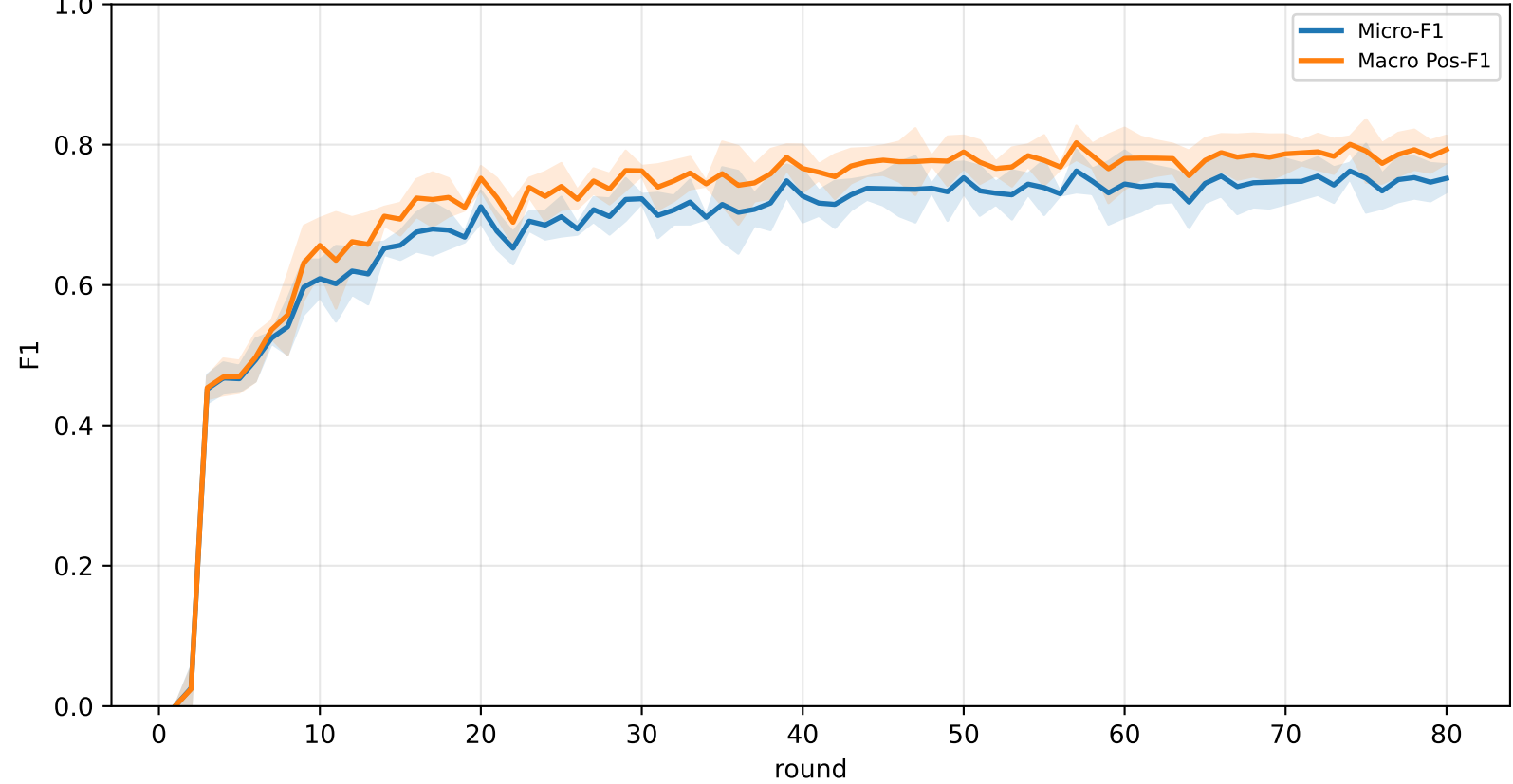
Pooled local centralized micro / macro F1



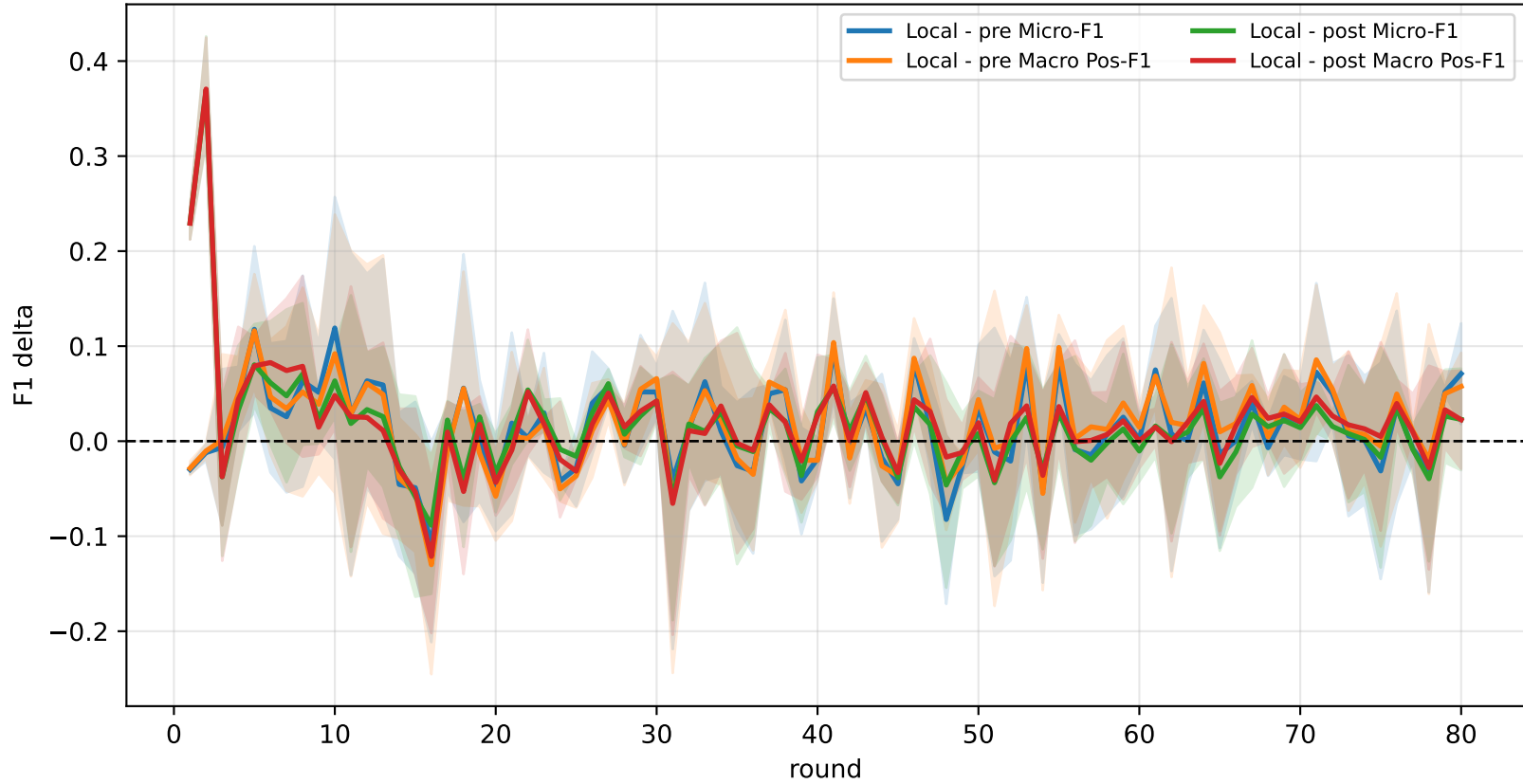
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)



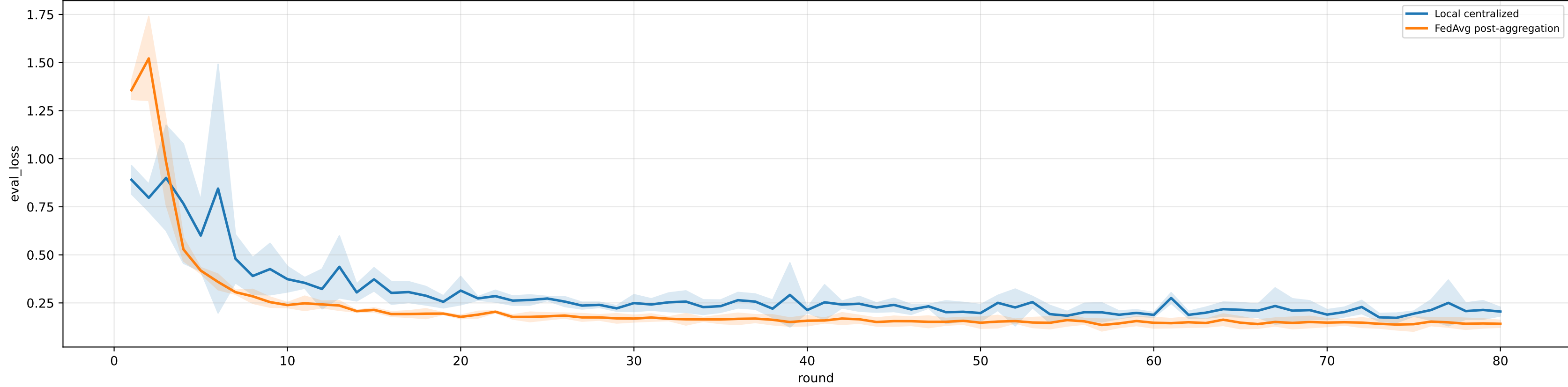


family=specialized\_cycle5 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.36 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 | family\_centroid\_jsd\_mean=0.162

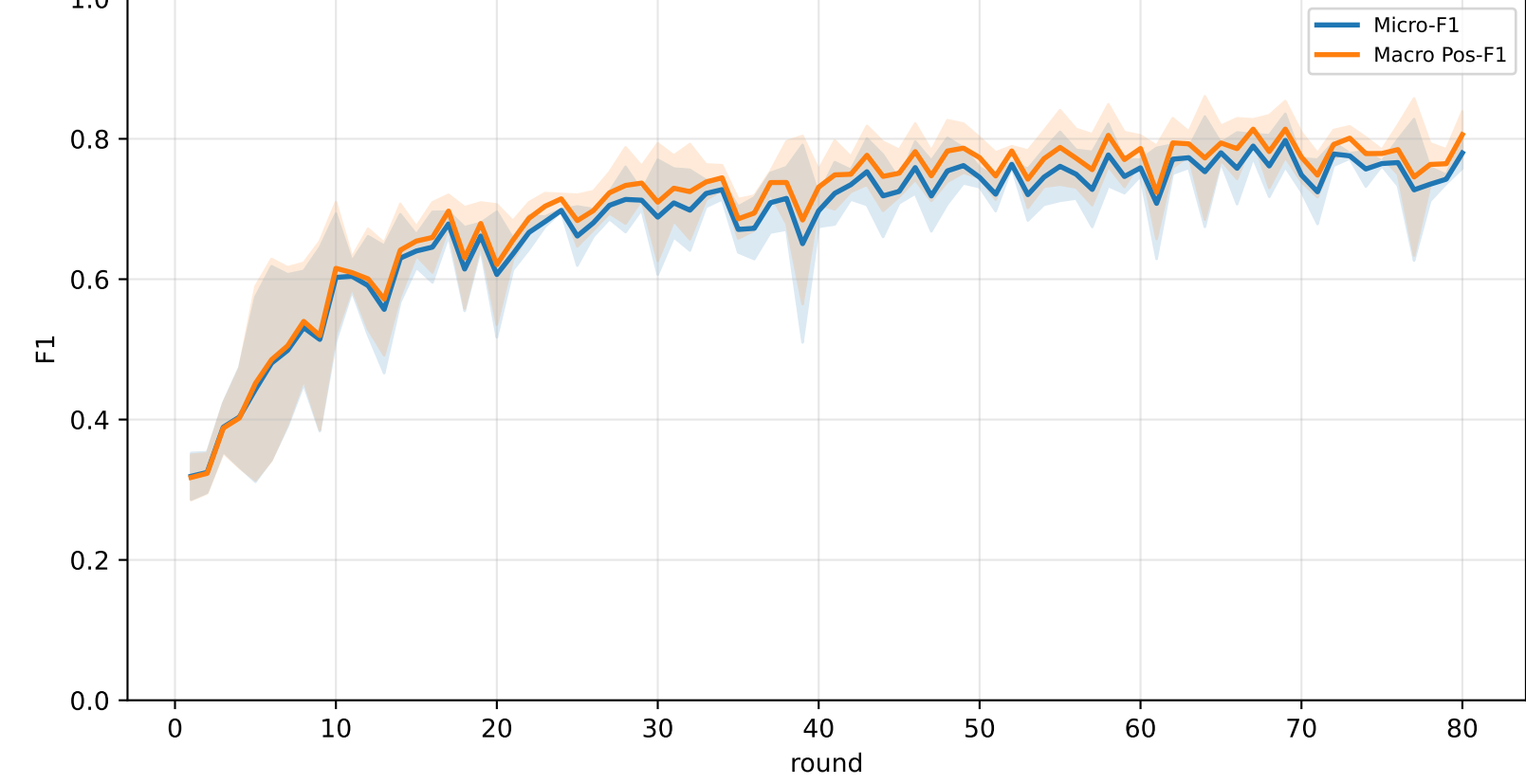
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 75.8% ± 0.9% | 78.3% ± 1.5% | 87.0% ± 1.3%  | 87.4% ± 3.9%  | 77.0% ± 3.6%  | 79.2% ± 5.6%  | 60.7% ± 3.5%  |
| FedAvg            | 76.5% ± 1.9% | 80.2% ± 2.0% | 94.7% ± 0.9%  | 91.3% ± 3.3%  | 81.3% ± 1.5%  | 74.4% ± 3.6%  | 59.0% ± 1.0%  |

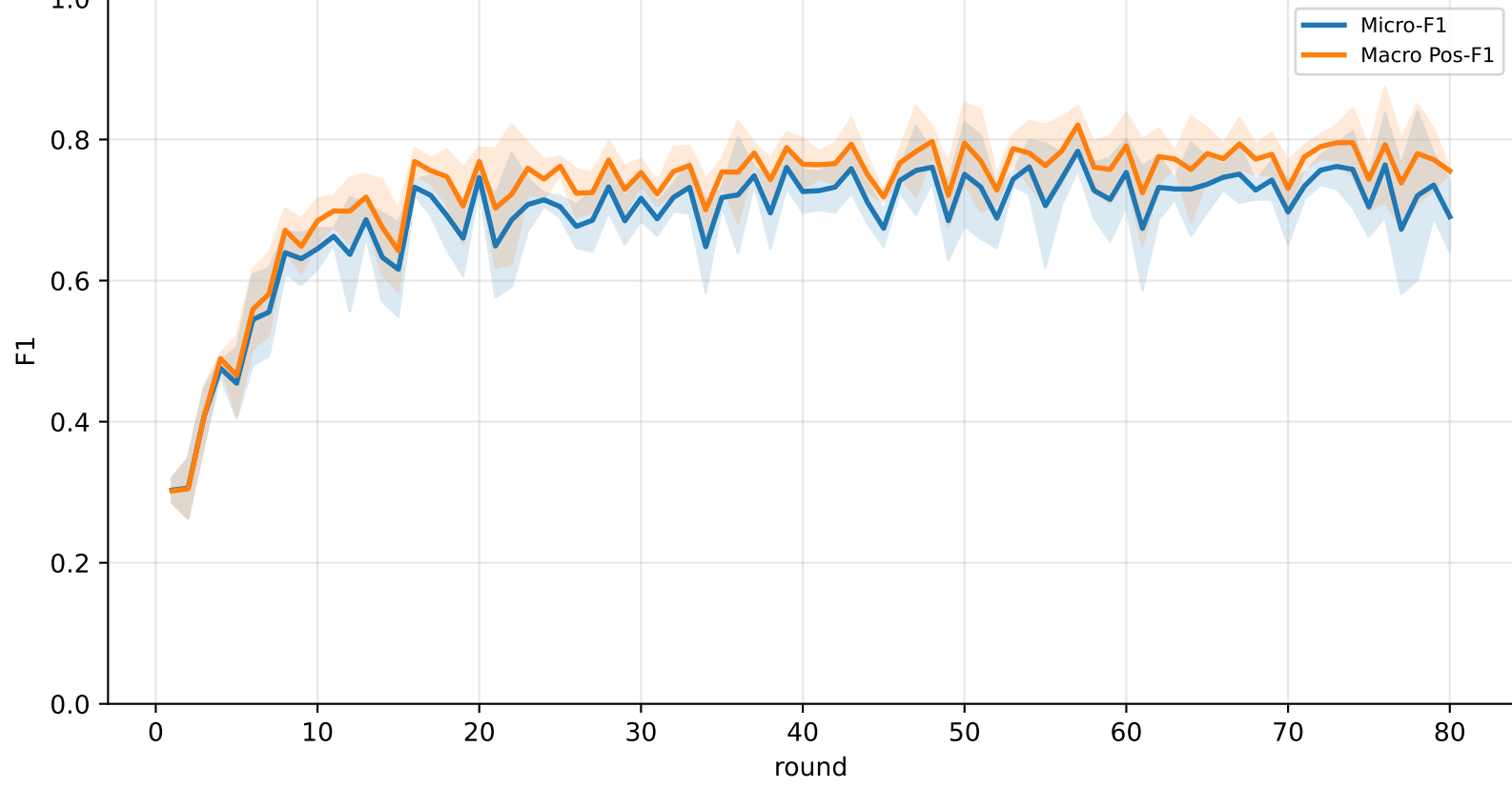
Validation loss comparison



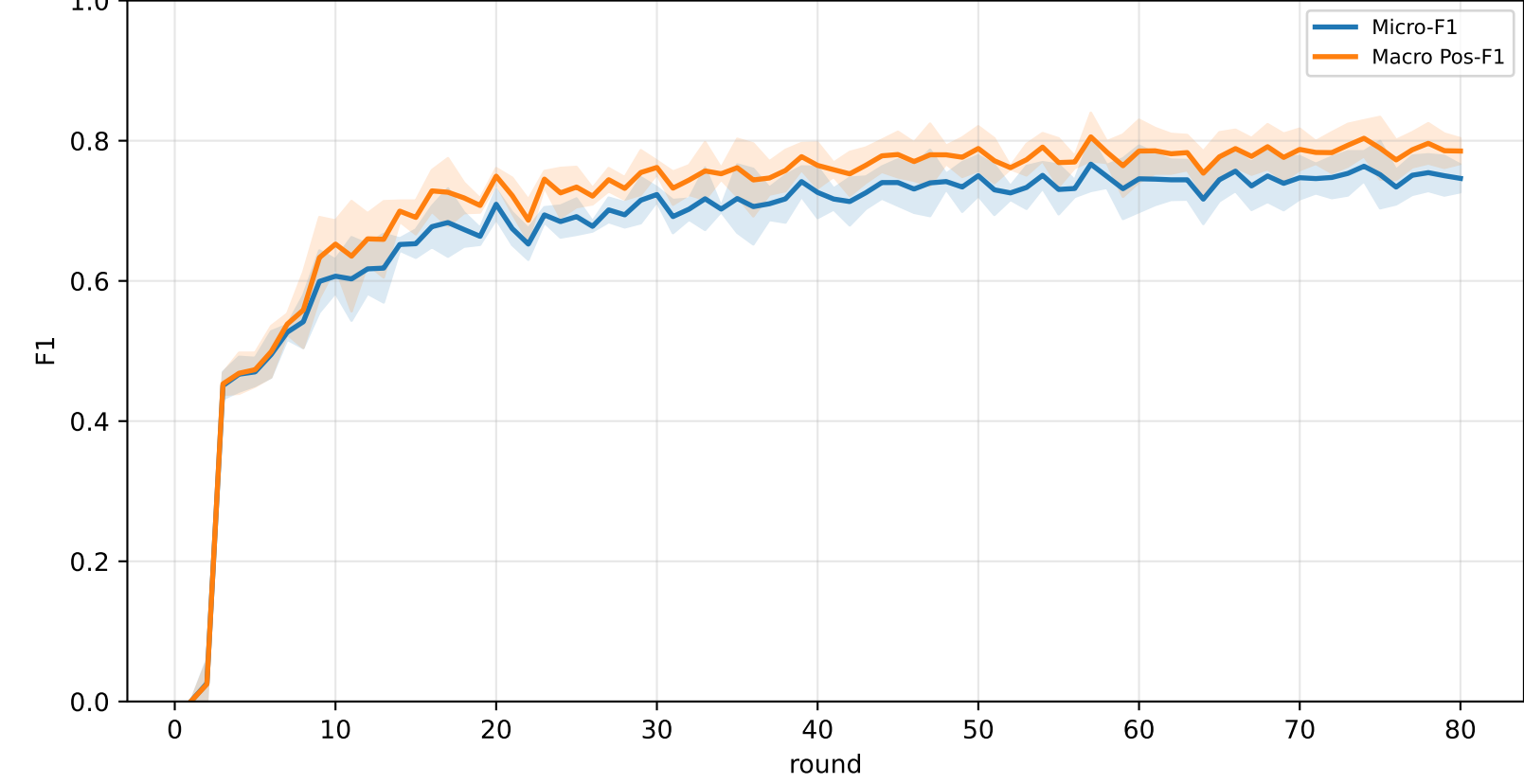
Pooled local centralized micro / macro F1



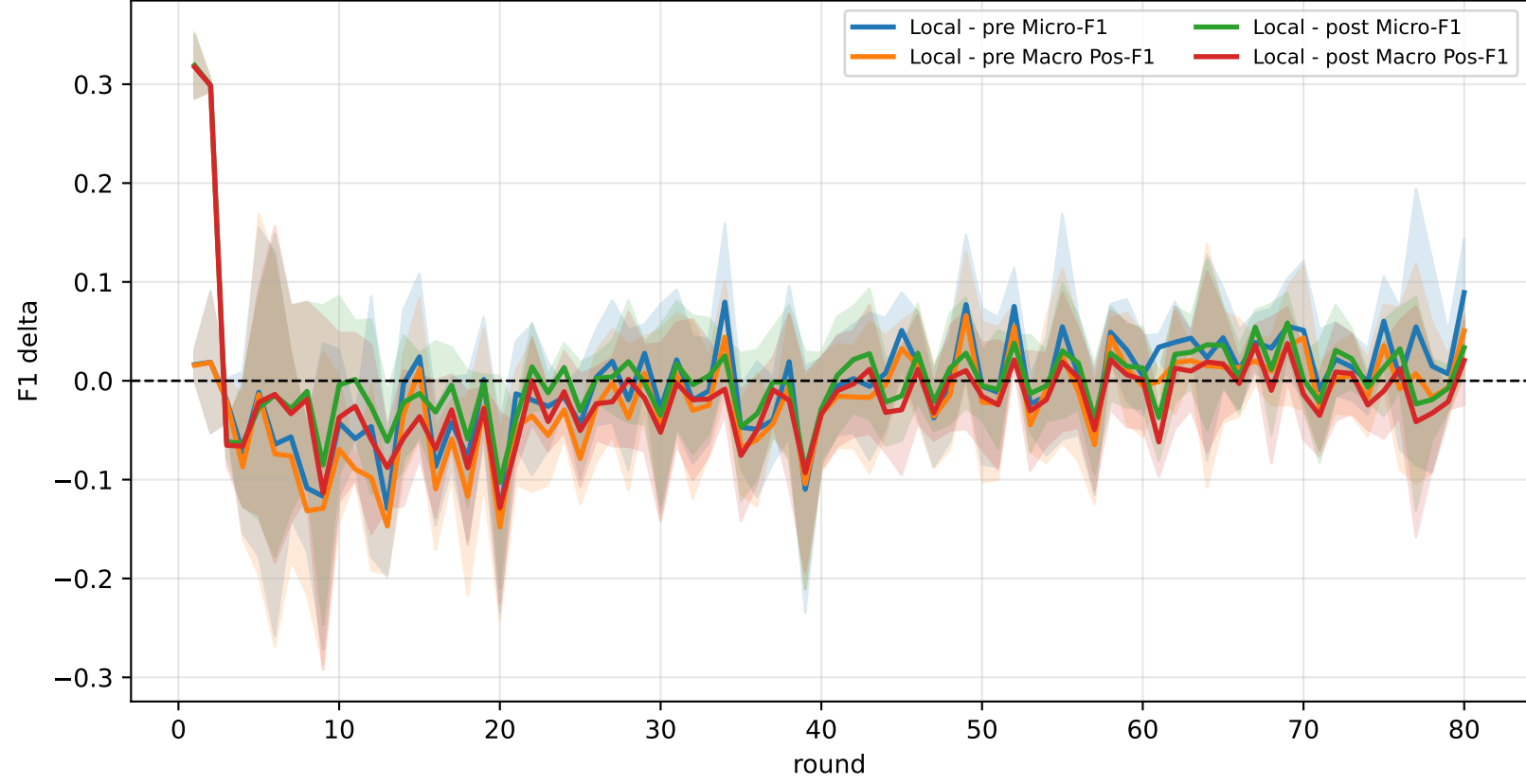
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)

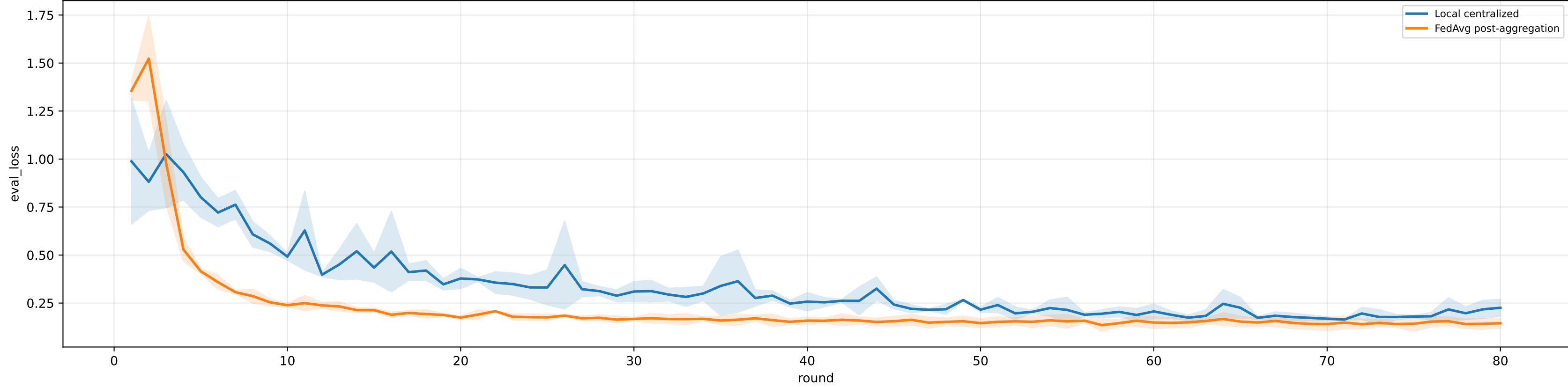


family=specialized\_cycle6 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%  
specialization=high | specialization\_fraction=0.80 | global\_visible\_support=0.36 | task\_jsd\_mean=0.162 | task\_jsd\_median=0.162 | task\_jsd\_max=0.165 |  
family\_centroid\_jsd\_mean=0.162

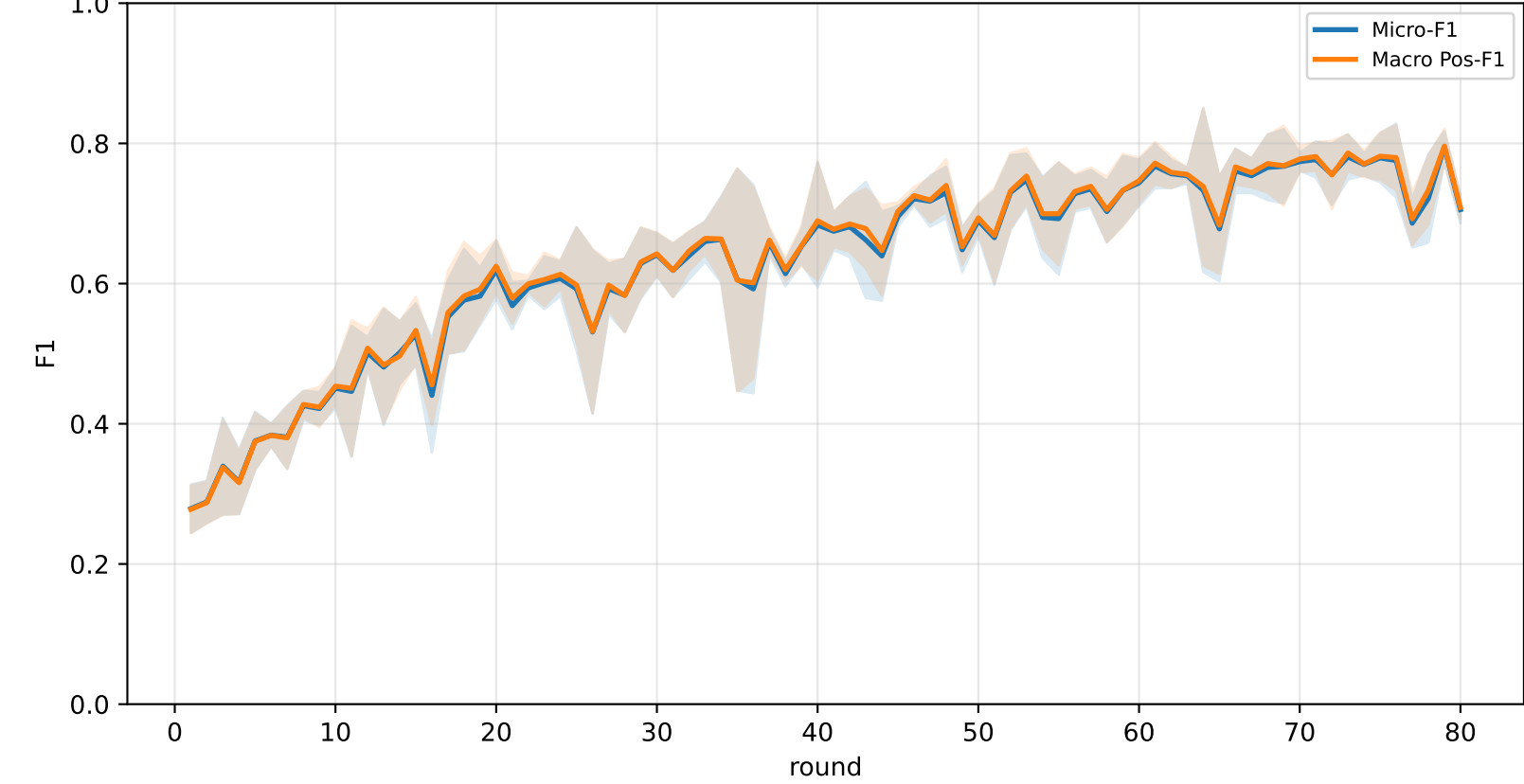
Best logged test performance (FedAvg vs Local centralized)

| Algorithm         | Micro-F1     | Macro Pos-F1 | cycle2 pos-F1 | cycle3 pos-F1 | cycle4 pos-F1 | cycle5 pos-F1 | cycle6 pos-F1 |
|-------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|
| Local centralized | 78.5% ± 3.5% | 79.1% ± 2.8% | 83.3% ± 8.5%  | 79.7% ± 5.3%  | 76.3% ± 4.2%  | 77.7% ± 6.0%  | 78.4% ± 6.9%  |
| FedAvg            | 77.0% ± 2.2% | 80.7% ± 2.4% | 95.2% ± 3.0%  | 92.8% ± 2.9%  | 81.7% ± 2.4%  | 75.0% ± 3.6%  | 58.9% ± 1.0%  |

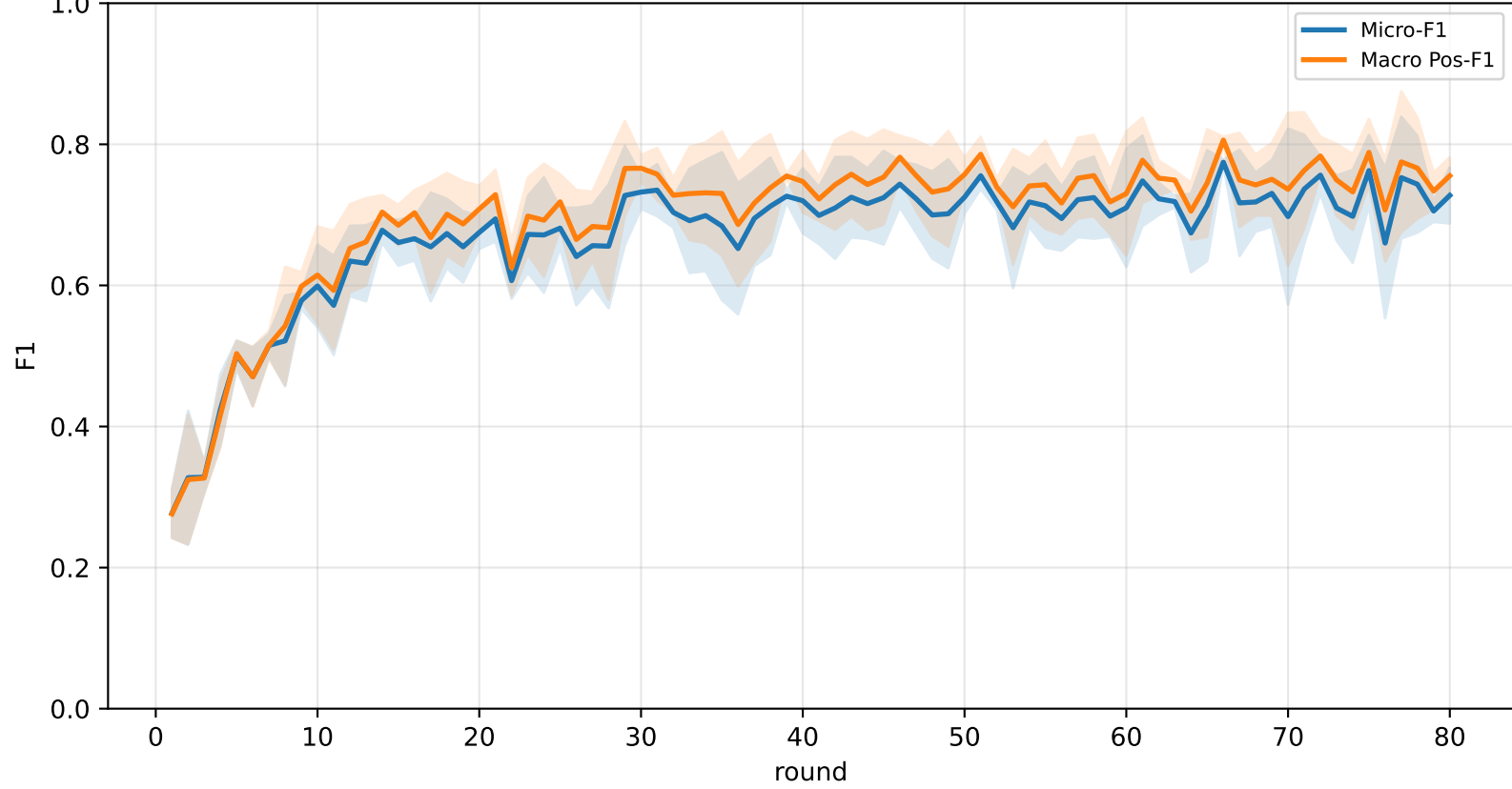
Validation loss comparison



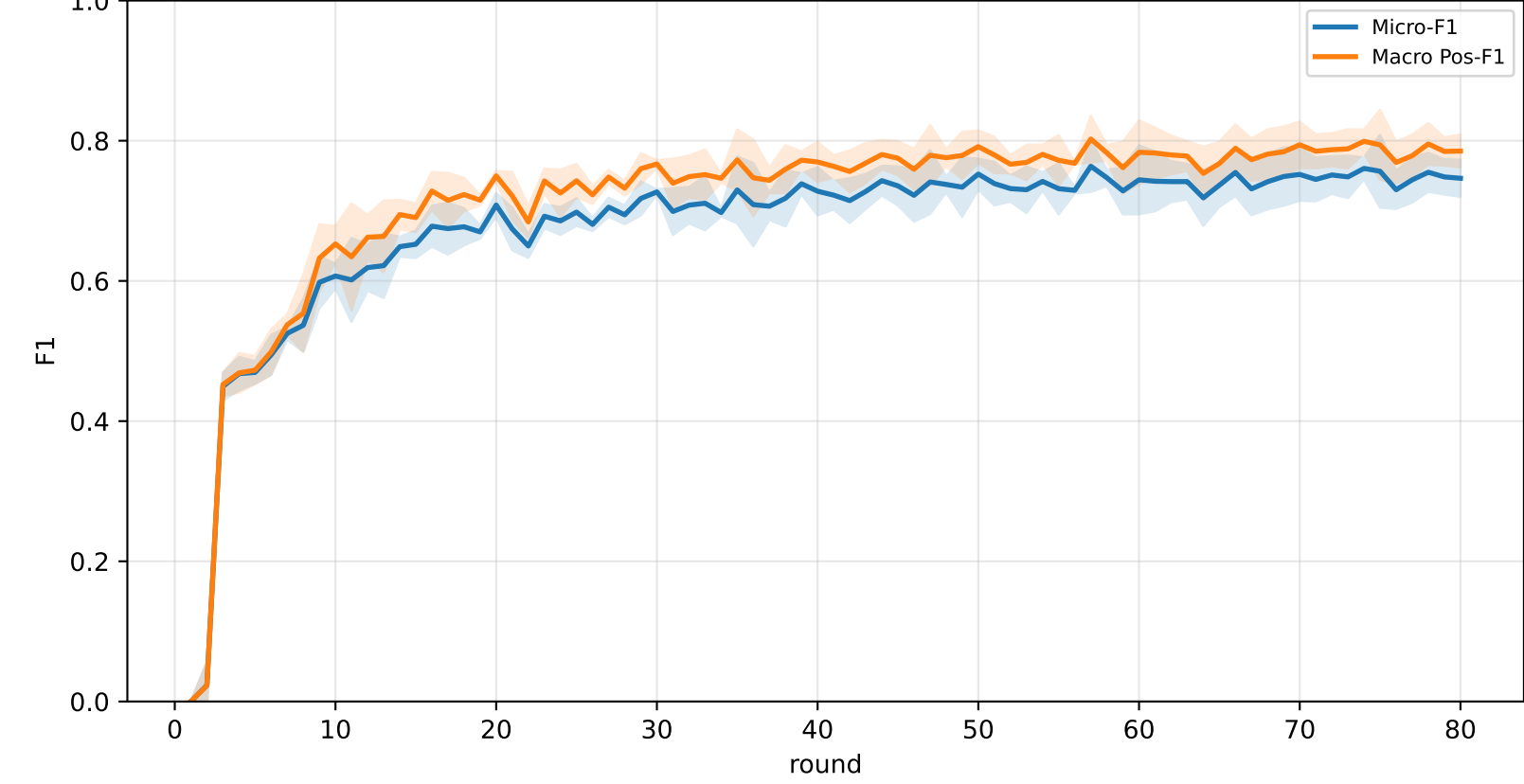
Pooled local centralized micro / macro F1



Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta vs FedAvg (local minus pre / post)

